



SRI SHANMUGHA COLLEGE OF ENGINEERING AND TECHNOLOGY
(Approved by AICTE, Accredited by NAAC and Affiliated to Anna University)
Pullipalayam, Morur (P.O), Sankari (T.k), Salem (D.T)

CS8711 – Cloud Computing Lab Manual

Dept / Year : CSE/ IV

Semester : VII

Department of Computer Science & Engineering

OBJECTIVES:***The student should be made to:***

- Υ Be exposed to tool kits for grid and cloud environment.
- Υ Be familiar with developing web services/Applications in grid framework
- Υ Learn to run virtual machines of different configuration.
- Υ Learn to use Hadoop

LIST OF EXPERIMENTS:***Course Objective:***

- Υ To develop web applications in cloud
- Υ To learn the design and development process involved in creating a cloud based application
- Υ To learn to implement and use parallel programming using Hadoop

Exercises:

1. Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of windows 7 or 8.
2. Install a C compiler in the virtual machine created using virtual box and execute Simple Programs
3. Install Google App Engine. Create hello world app and other simple web applications using python/java.
4. Use GAE launcher to launch the web applications.
5. Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim.
6. Find a procedure to transfer the files from one virtual machine to another virtual machine.
7. Find a procedure to launch virtual machine using trystack (Online Openstack Demo Version)
8. Install Hadoop single node cluster and run simple applications like wordcount.

Course Outcome:

On completion of this course, the students will be able to:

- Υ Configure various virtualization tools such as Virtual Box, VMware workstation.
- Υ Design and deploy a web application in a PaaS environment.
- Υ Learn how to simulate a cloud environment to implement new schedulers.
- Υ Install and use a generic cloud environment that can be used as a private cloud.
- Υ Manipulate large data sets in a parallel environment.

TOTAL: 45 PERIODS



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TECHNOLOGY**

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ANNAUNIVERSITY AND ACCREDITED BY NAAC & NBA(ECE,CSE,MECH)



Tiruchengode-Sankari main road, Pullipalayam, Morur
(Po), Sankari (Tk), Salem (Dt) Pin: 637 304

RECORD NOTE BOOK

**CS8711-CLOUD COMPUTING
LABORATORY**

NAME :

REG NO :

Submitted for the Anna University Practical Examinations



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RECORD NOTE BOOK

REG NO

Certified that this is a bonafide record of Practical work done by
Mr/Ms of the
Semester Branch during the Academic year
in the Laboratory

Staff-in-charge

Head of the Department

Submitted for the University Practical

Examination held on.....

Internal Examiner

External Examiner

CS8711- CLOUD COMPUTING LABORATORY

DEGREE / BRANCH: **B.E / CSE**

YEAR / SEM: IV / VII

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EX.No:1

**Install Virtualbox/VMware Workstation with different
flavours of linux or windows OS on top of windows**

Date:

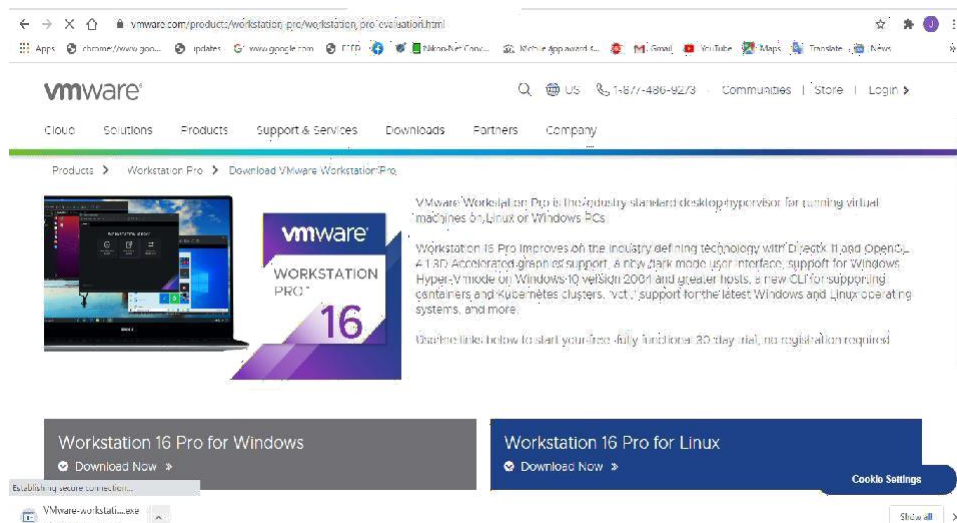
Aim:

To install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of windows7 or 8 or 10.

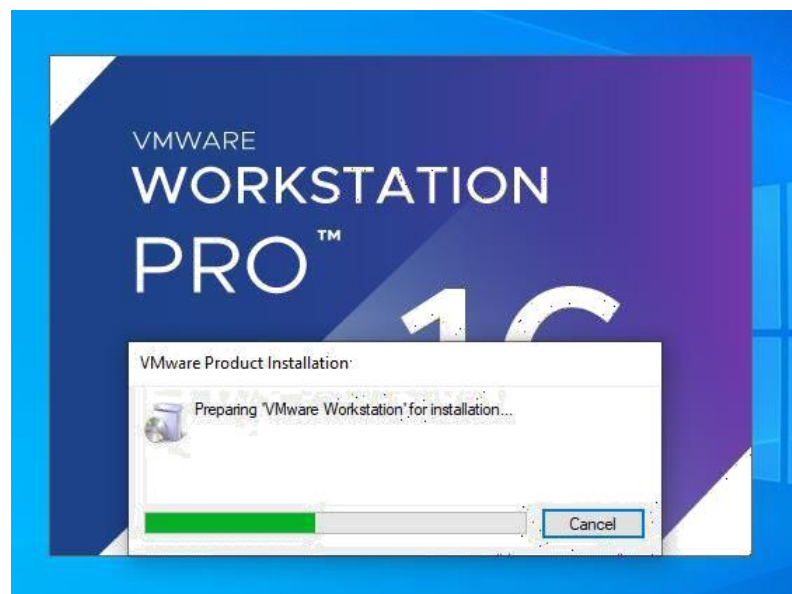
Procedure:

Downloading and installing VMware

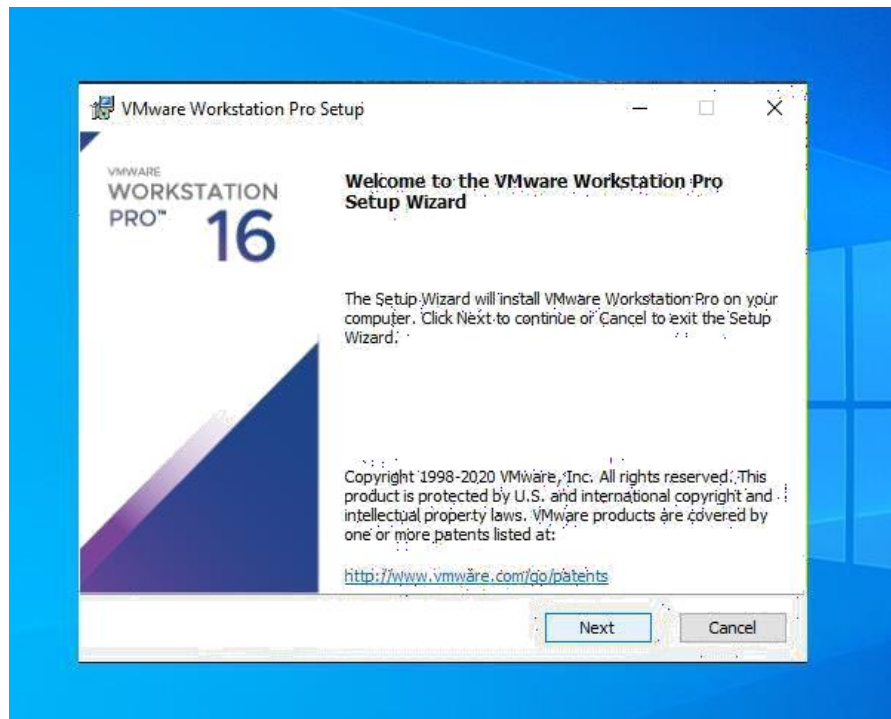
Step 1: Download VMware



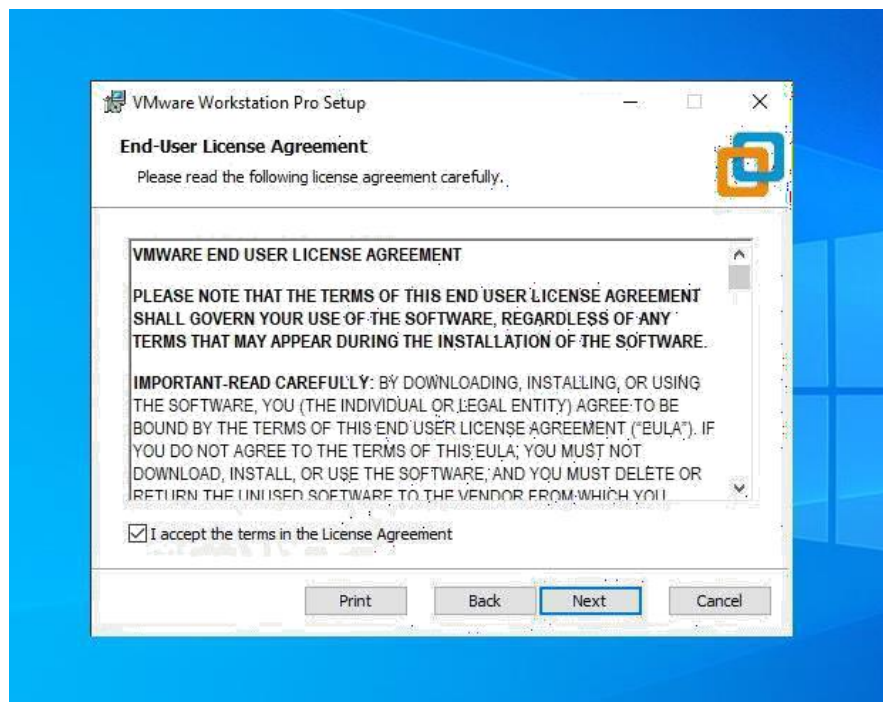
Step 2: Install the VMware Application



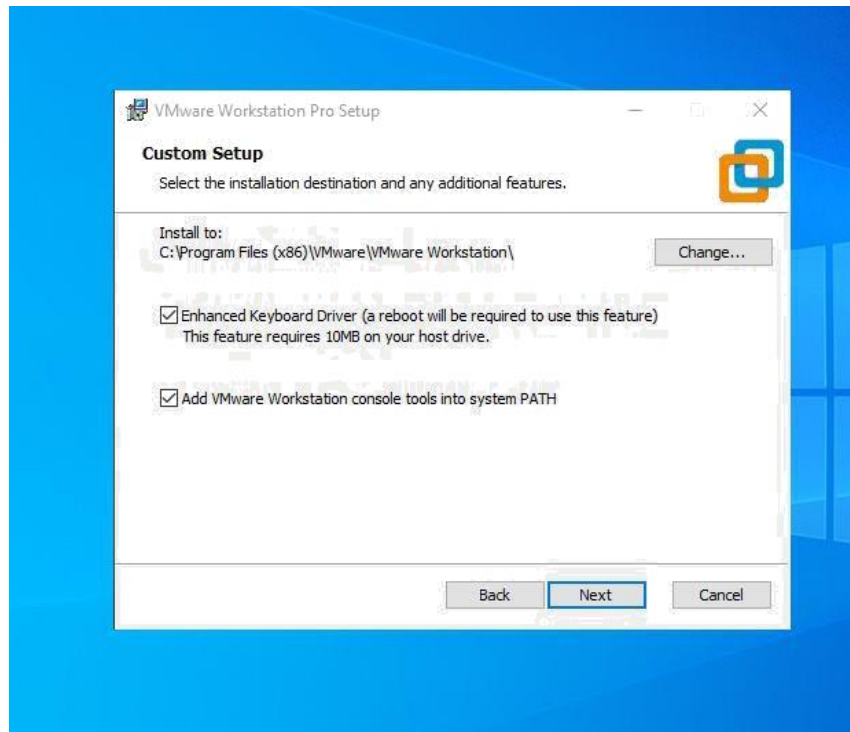
Step 3: Click Next



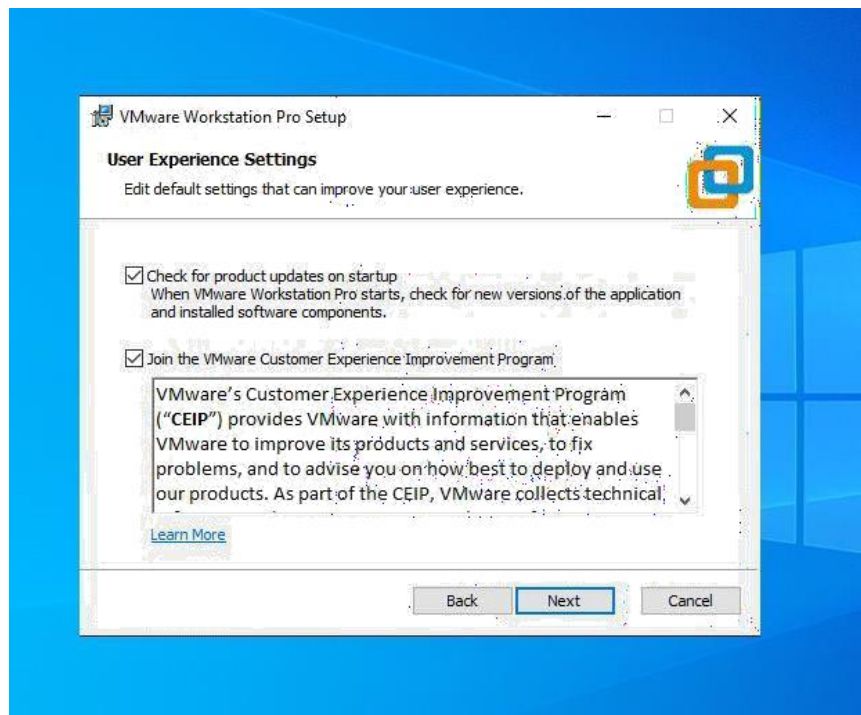
Step 4: Accept and Click Next



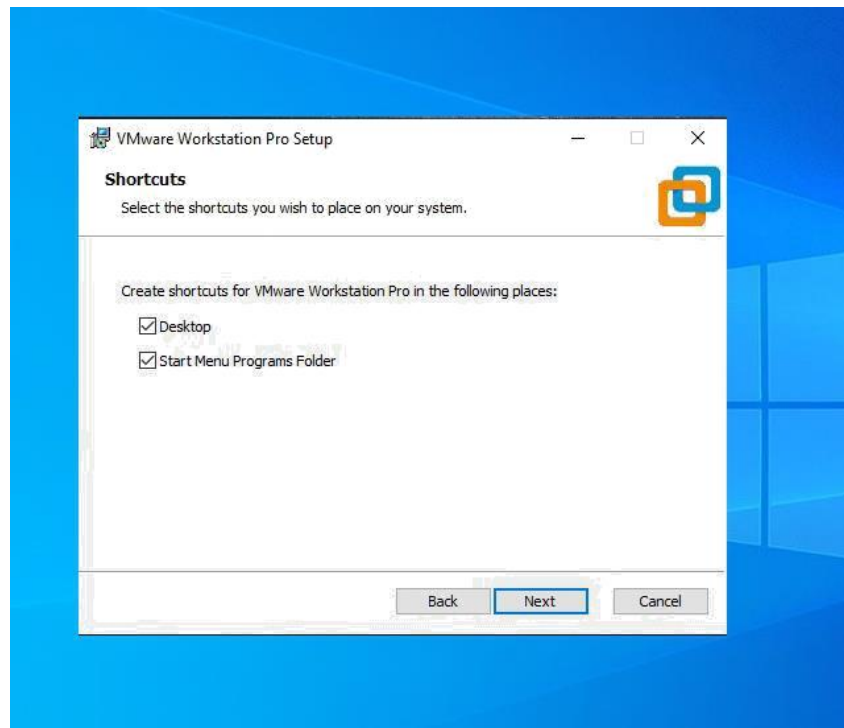
Step 5: Click Next



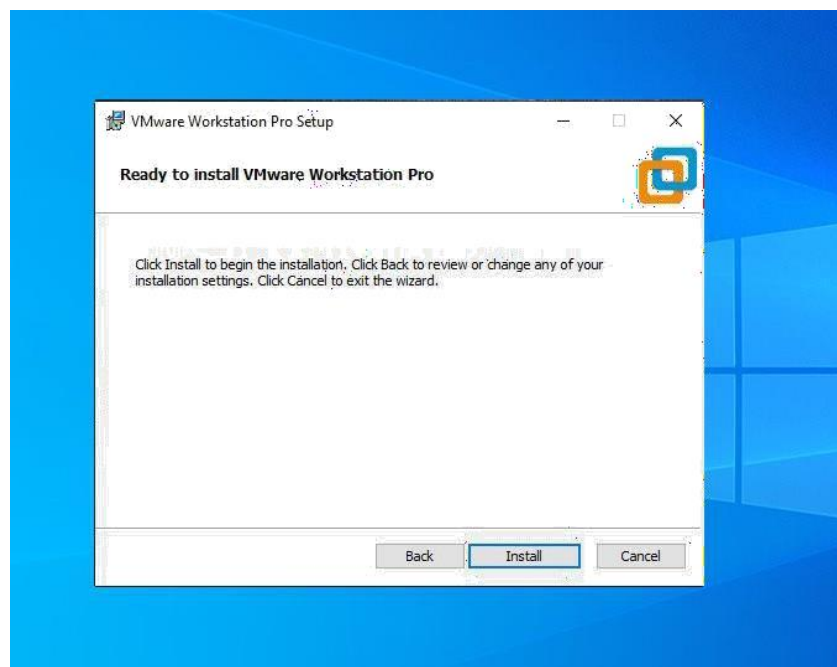
Step 6: Click Next



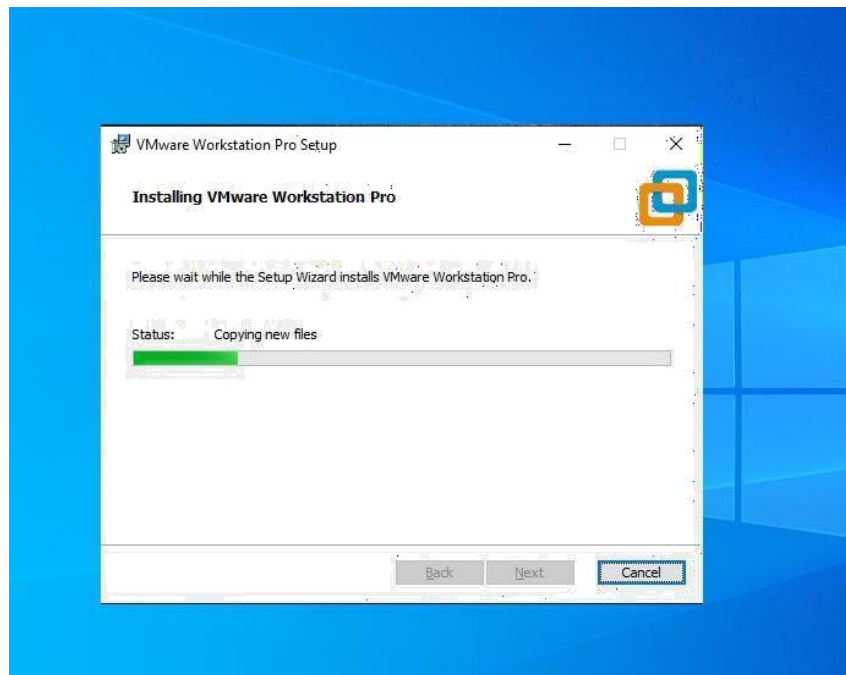
Step 7: Click Next



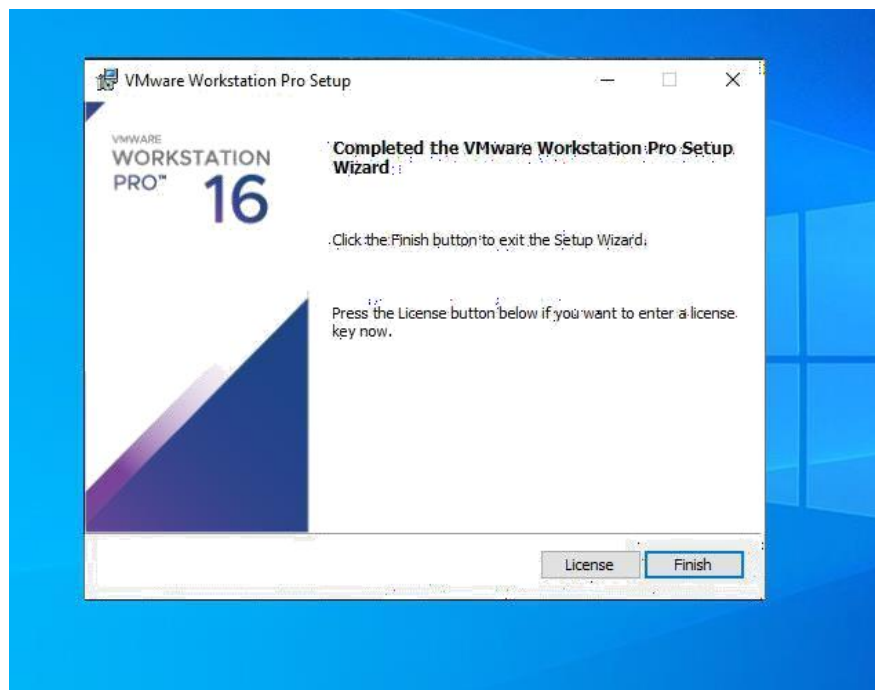
Step 8: Click Install



Step 9: Installing

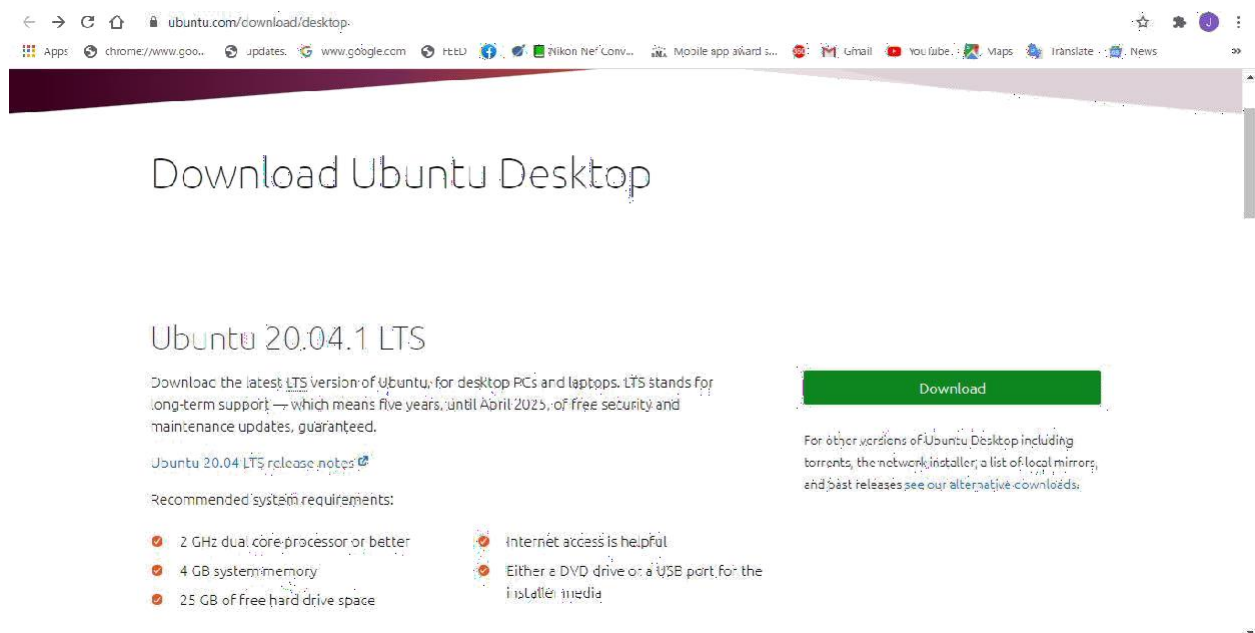


Step 10: Click Finish

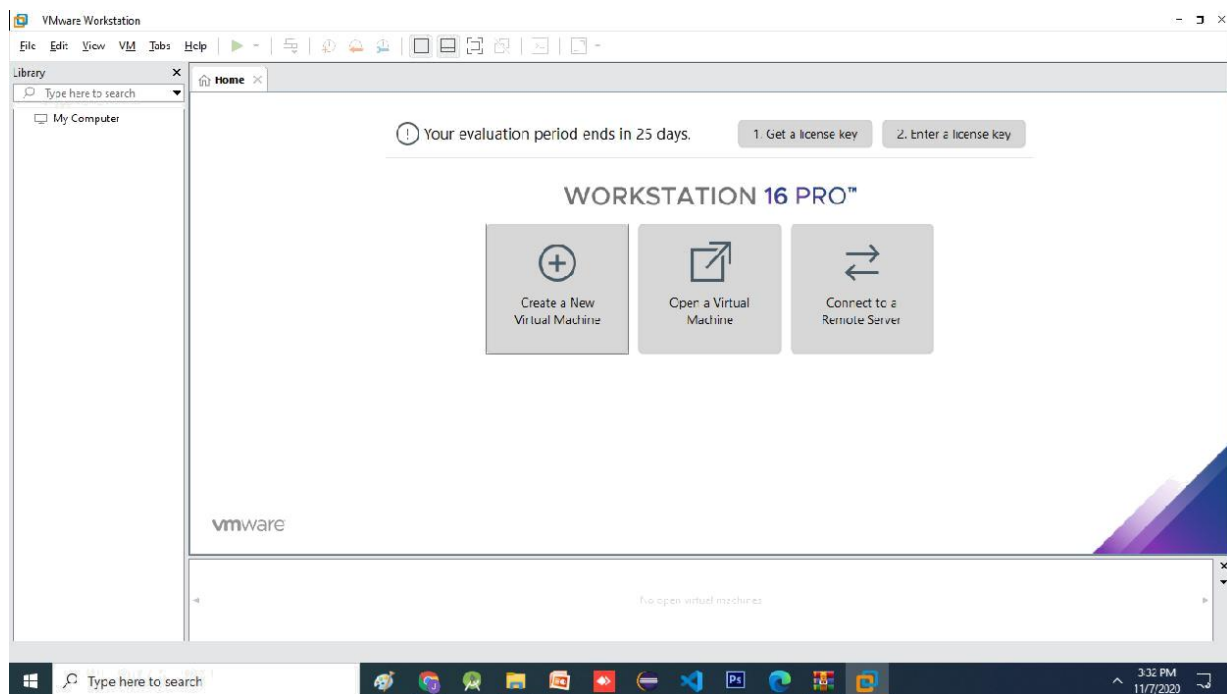


Downloading Ubuntu

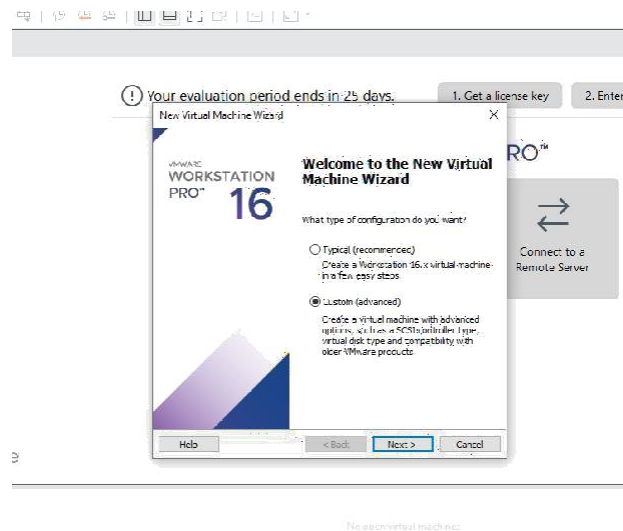
Step 11: Download the Ubuntu OS



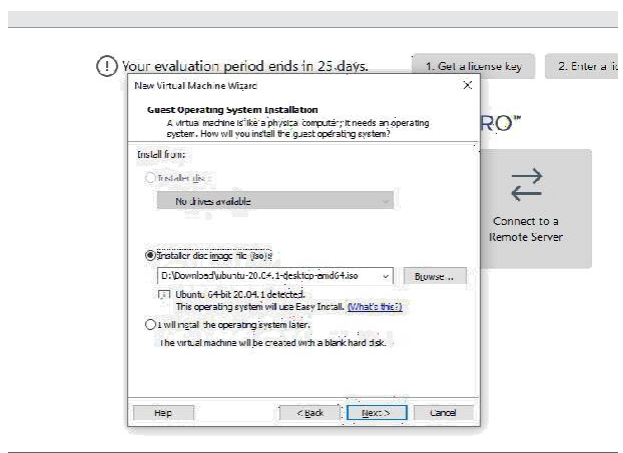
Step 12: Create new virtual machine



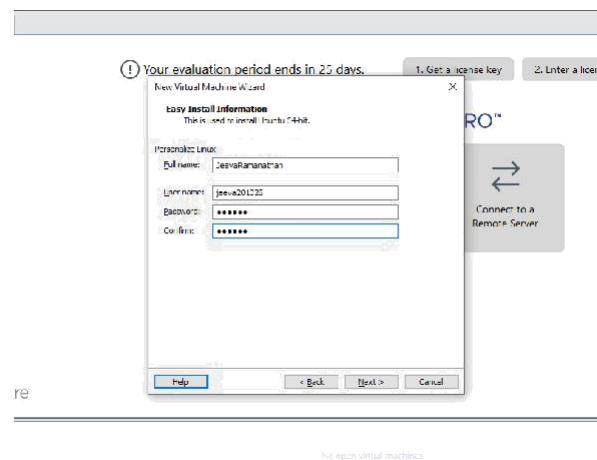
Step 13: Click Next



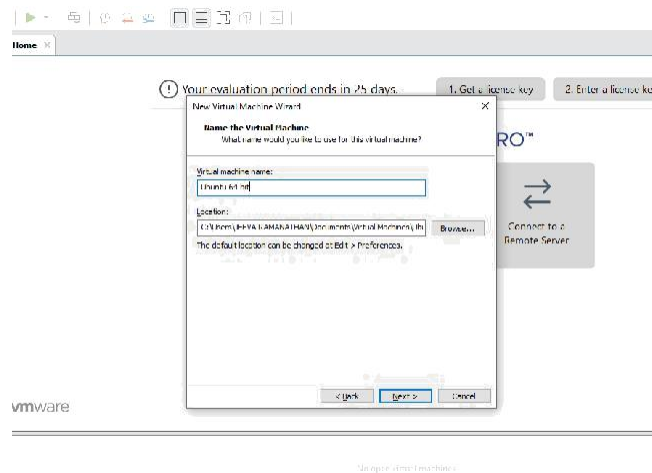
Step 14: Browse the downloaded Ubuntu file and click next



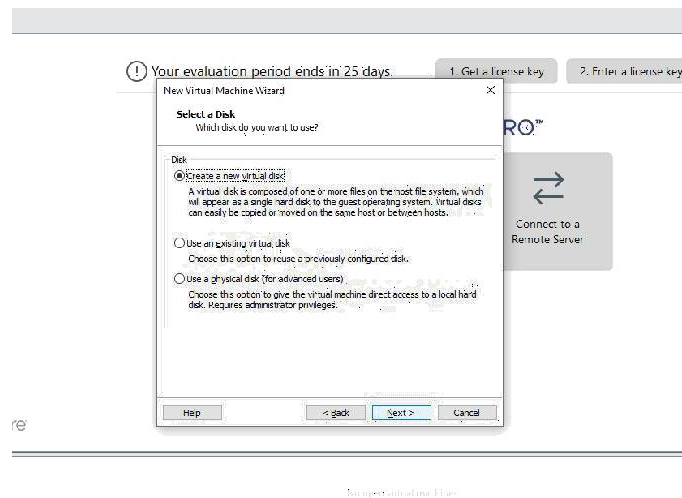
Step 15: Create an username and password and click next



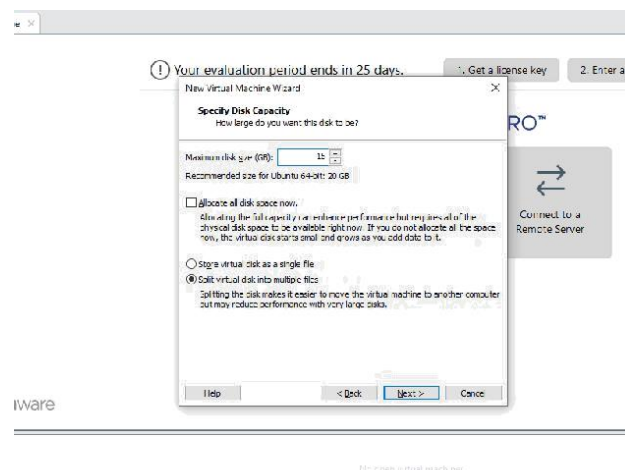
Step 16: Choose the location to use your virtual machine and click next



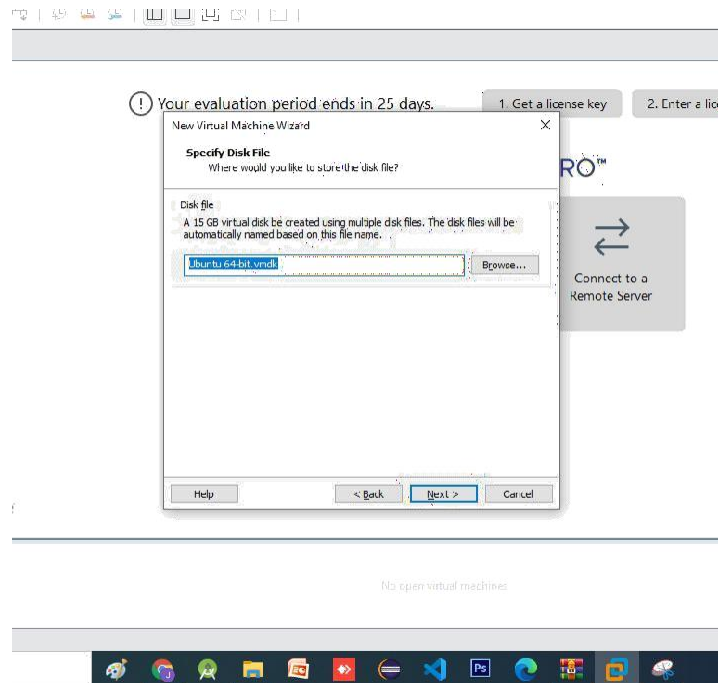
Step 17: Select create a new virtual disk and click next



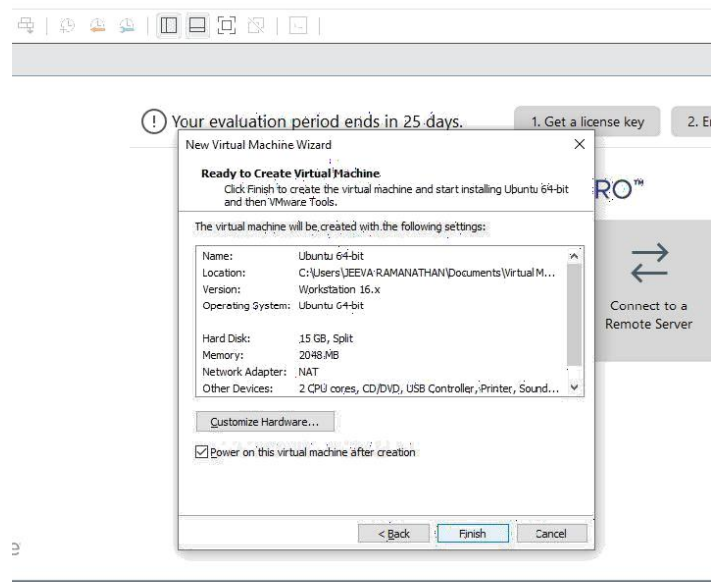
Step 18: Specify the disk size and click next



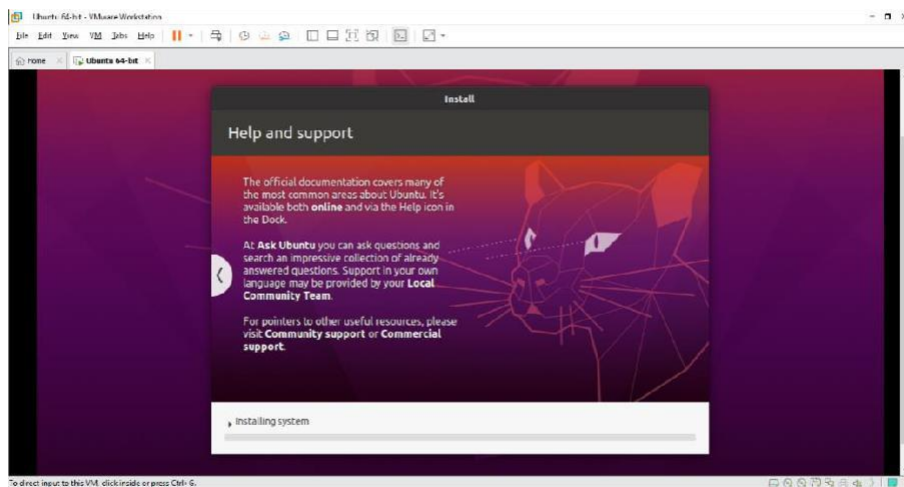
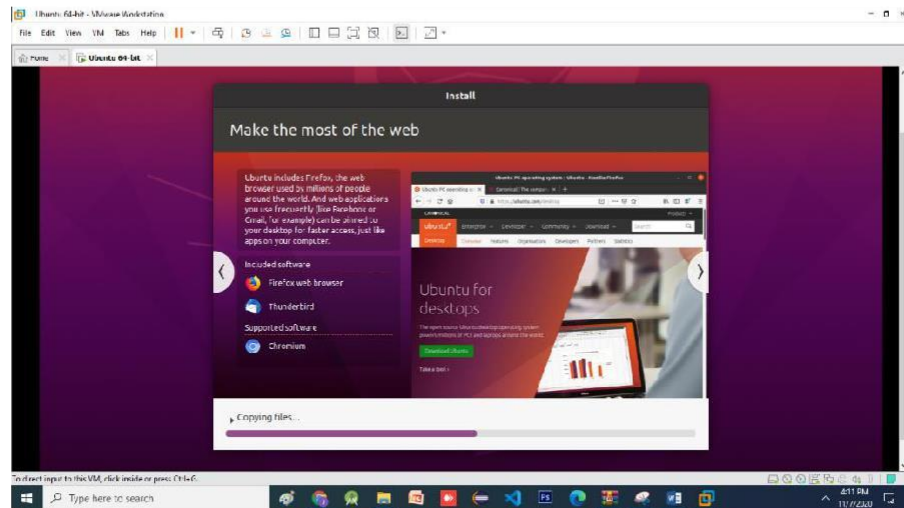
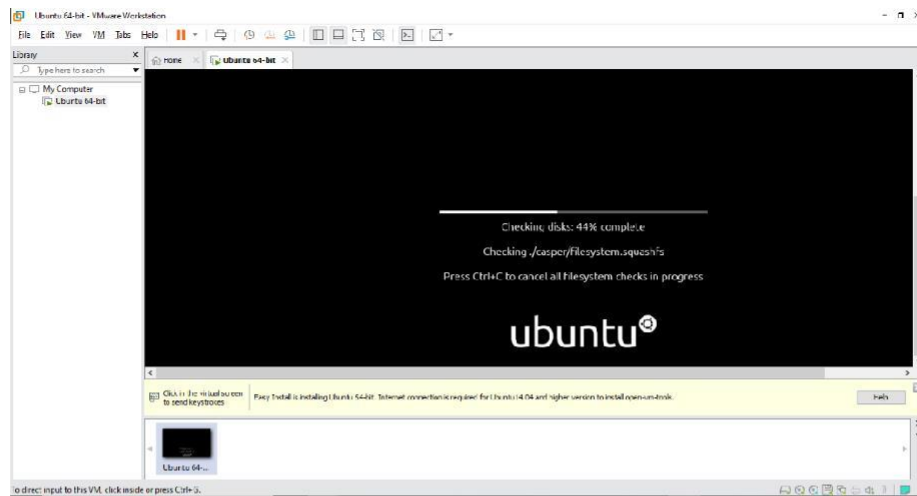
Step 19: Click Next



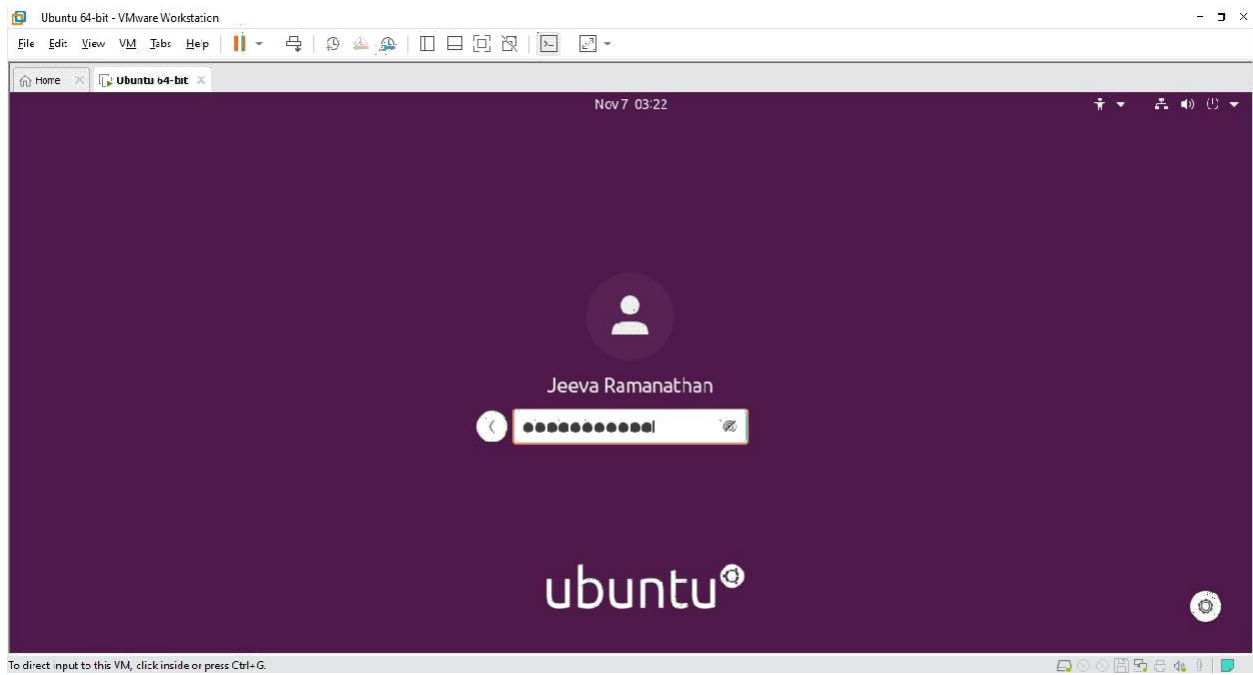
Step 20: Click Finish



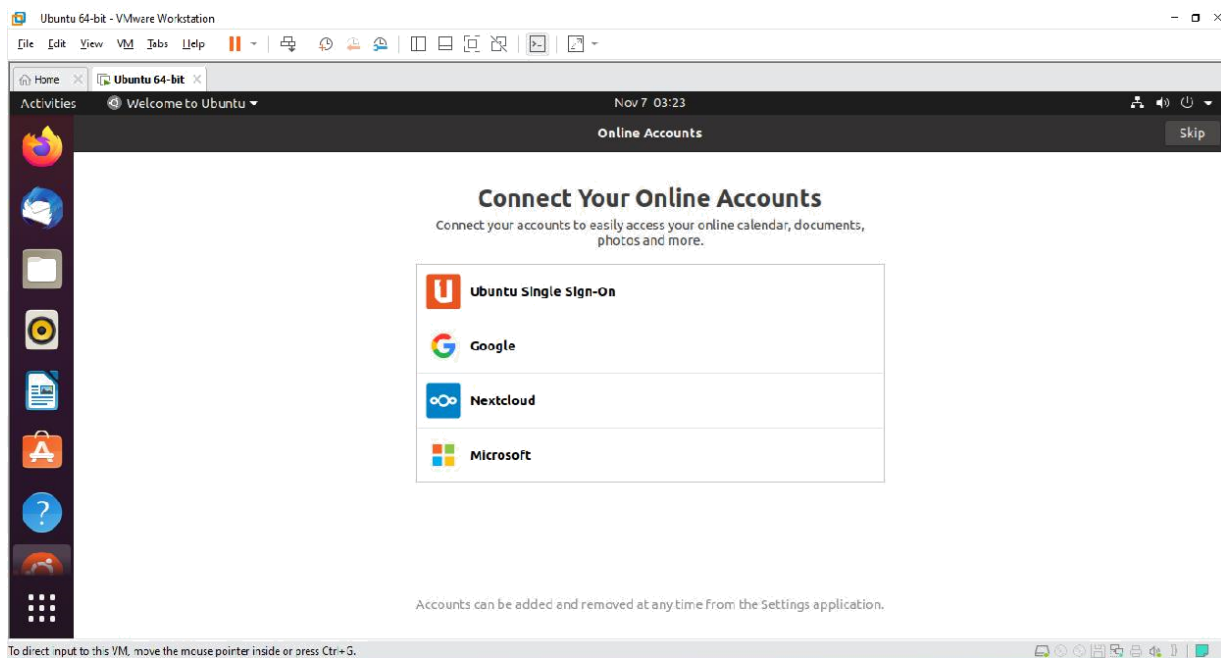
Step 21: Installing Ubuntu on VMware and unzipping files



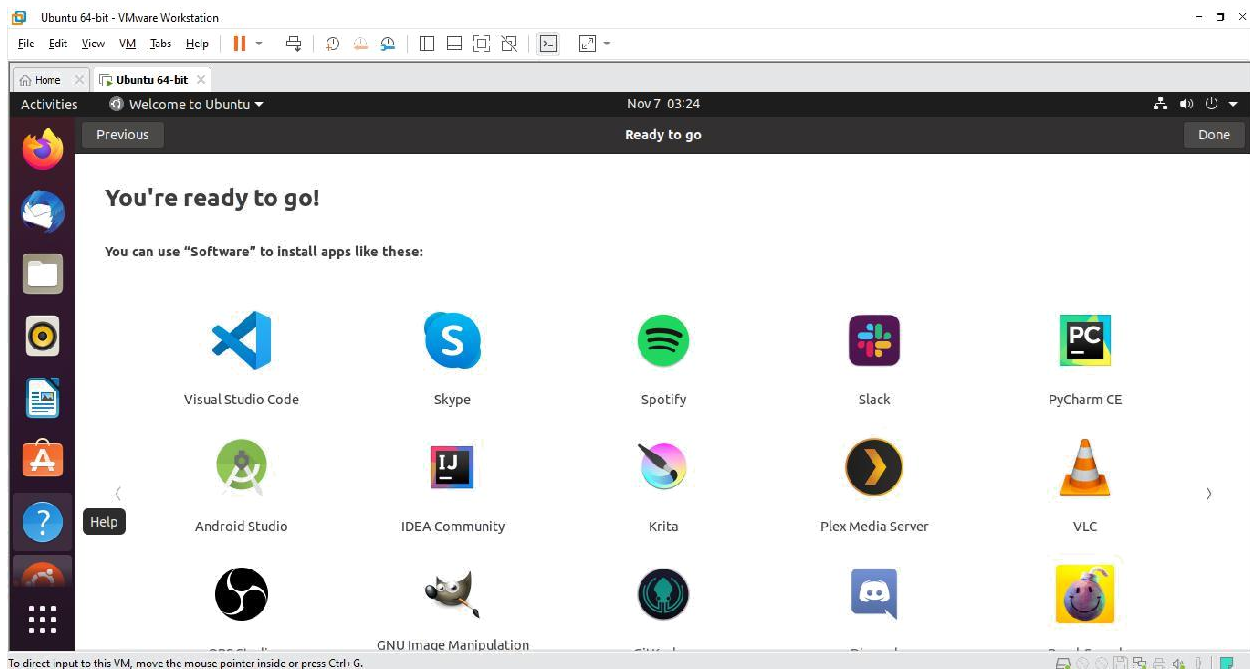
Step 22: Login to Ubuntu



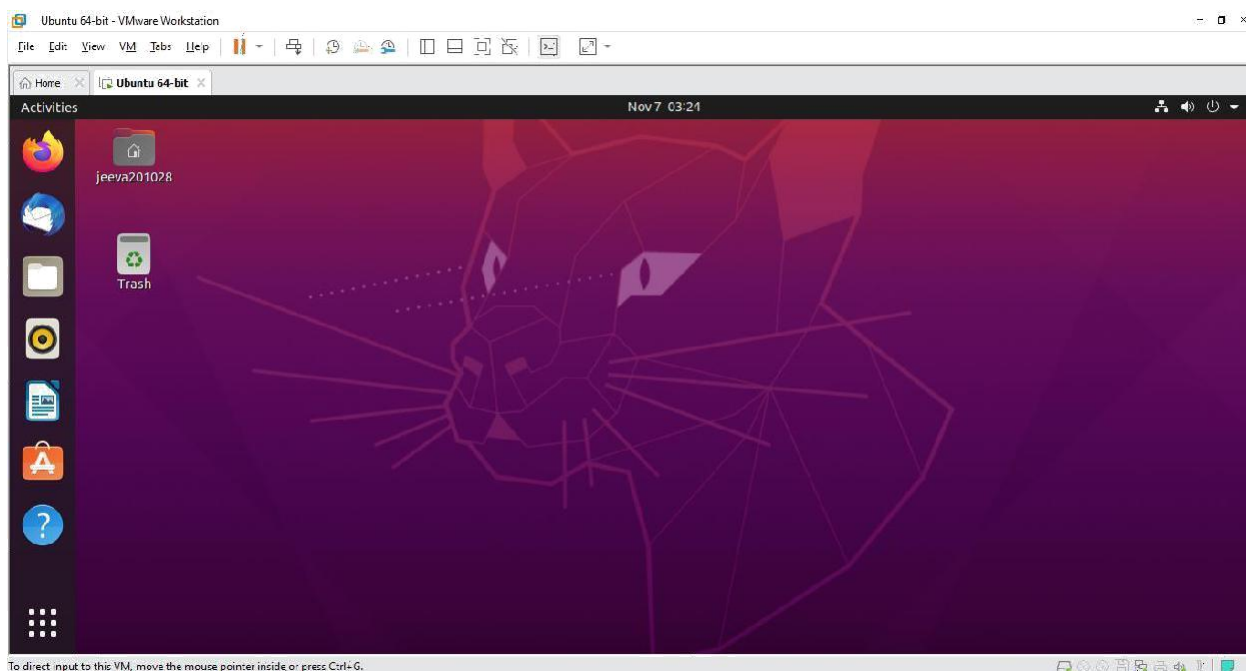
Step 23: Skip everything



Step 24: Click Done



Step 25: Thus we have installed VMware Workstation with different flavours of linux on top of windows



Result: Thus, VMware Workstation with different flavours of linux or windows OS on top of windows7 or 8 or 10.has been successfully installed and executed.

EX.No:2

**Install a C compiler in the virtual machine created using
virtual box and execute a simple program**

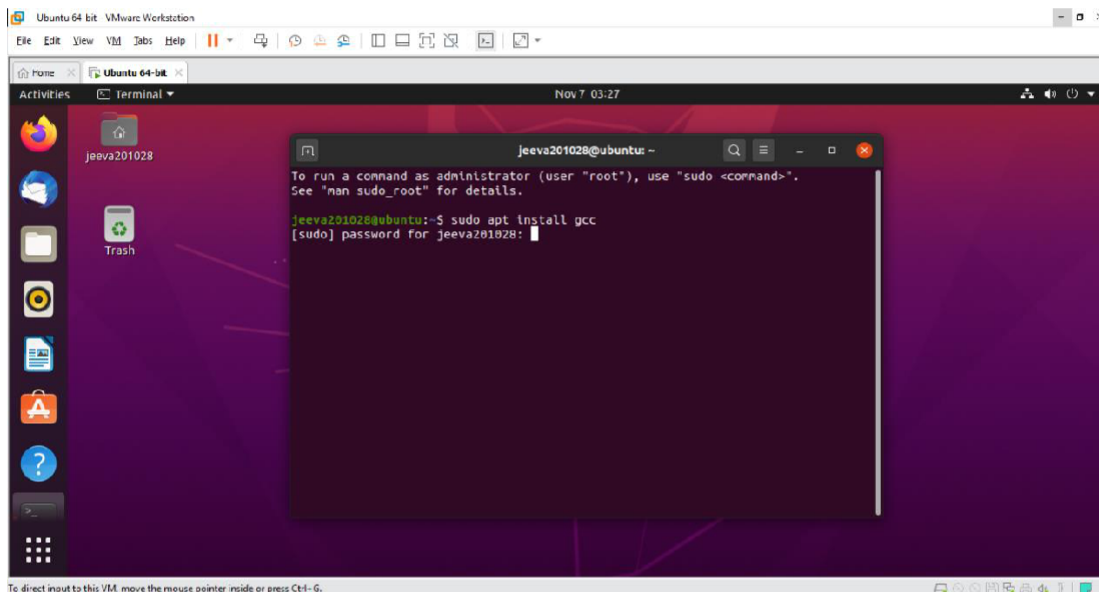
Date:

Aim:

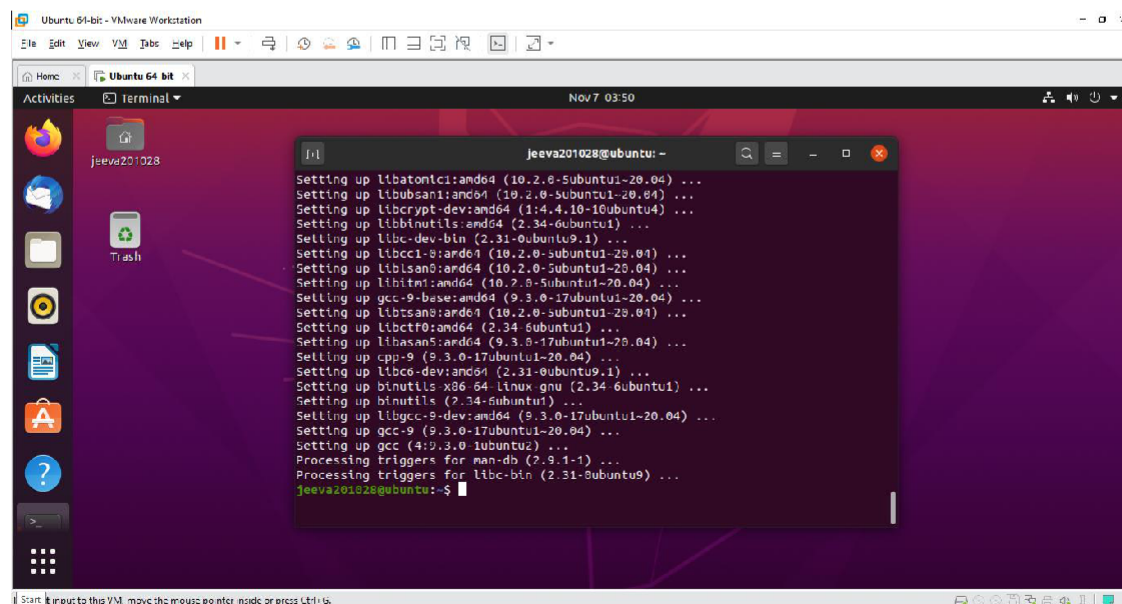
To install a C Compiler in the virtual machine created using virtual box and execute a simple C program.

Procedure:

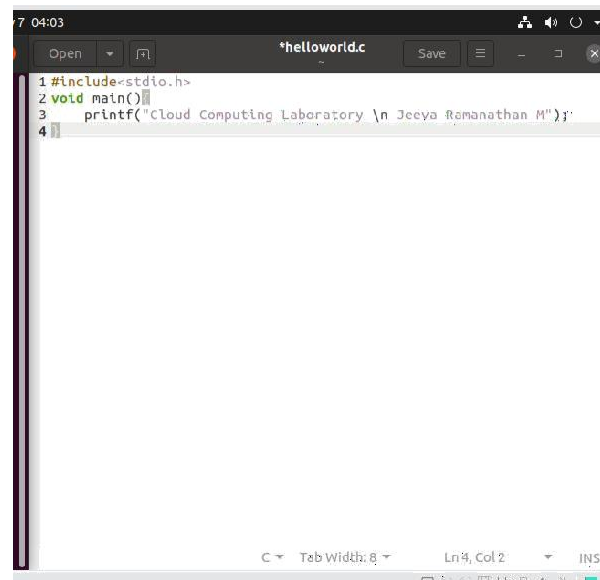
Step 1: Open the terminal on Ubuntu and install C compiler – “sudo apt install gcc”



Step 2: Once the installation finished open the editor

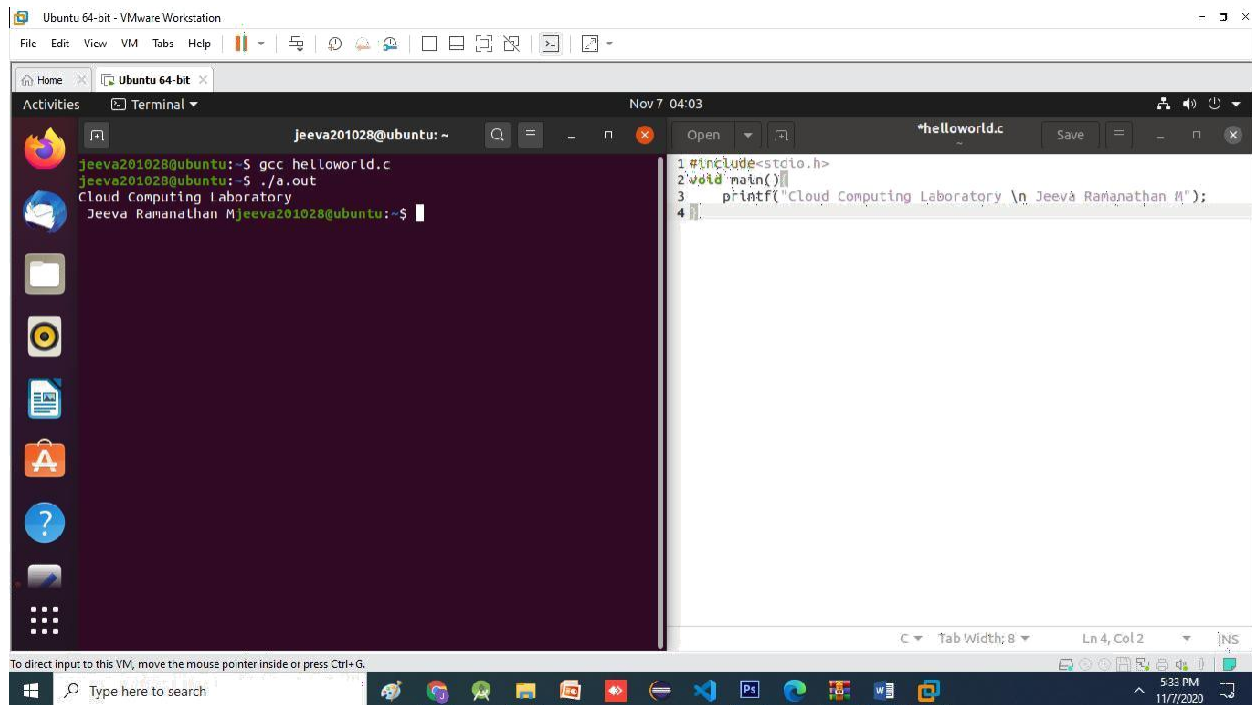


Step 3: Type a simple C program on editor and save it



```
1 #include <stdio.h>
2 void main()
3 {
4     printf("Cloud Computing Laboratory \n Jeeva Ramanathan M");
5 }
```

Step 4: Compile and run the C program



```
Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Activities Terminal Nov 7 04:03
jееva201028@ubuntu: ~
jееva201028@ubuntu:~$ gcc helloworld.c
jееva201028@ubuntu:~$ ./a.out
Cloud Computing Laboratory
Jeeva Ramanathan Mjееva201028@ubuntu:~$

1 #include <stdio.h>
2 void main()
3 {
4     printf("Cloud Computing Laboratory \n Jeeva Ramanathan M");
5 }
```

Result:

Thus a C compiler is installed in the virtual machine and C program was executed and output was obtained successfully.

EX.No:3 **Transfer files from one host machine to another virtual machine**

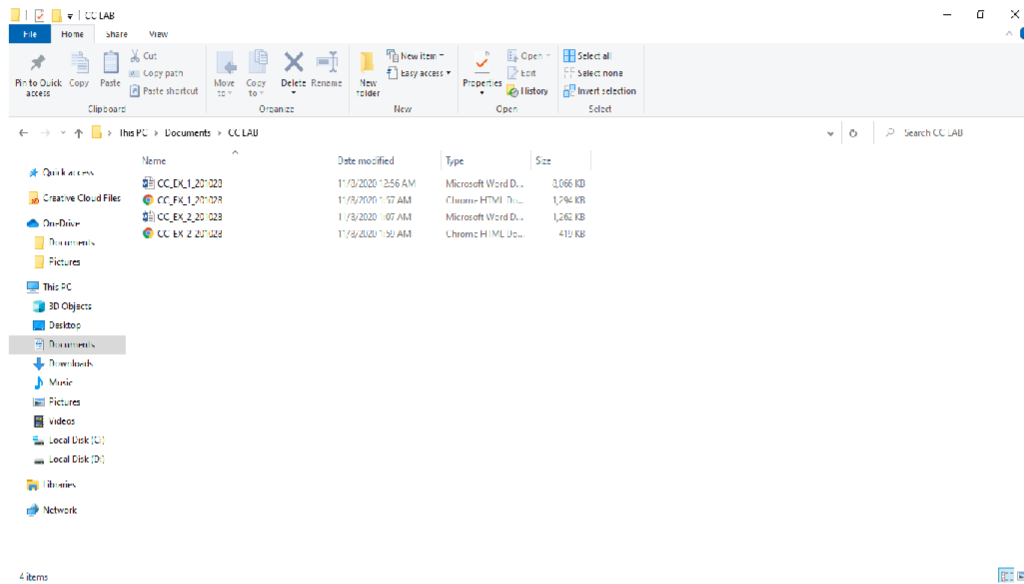
Date:

Aim:

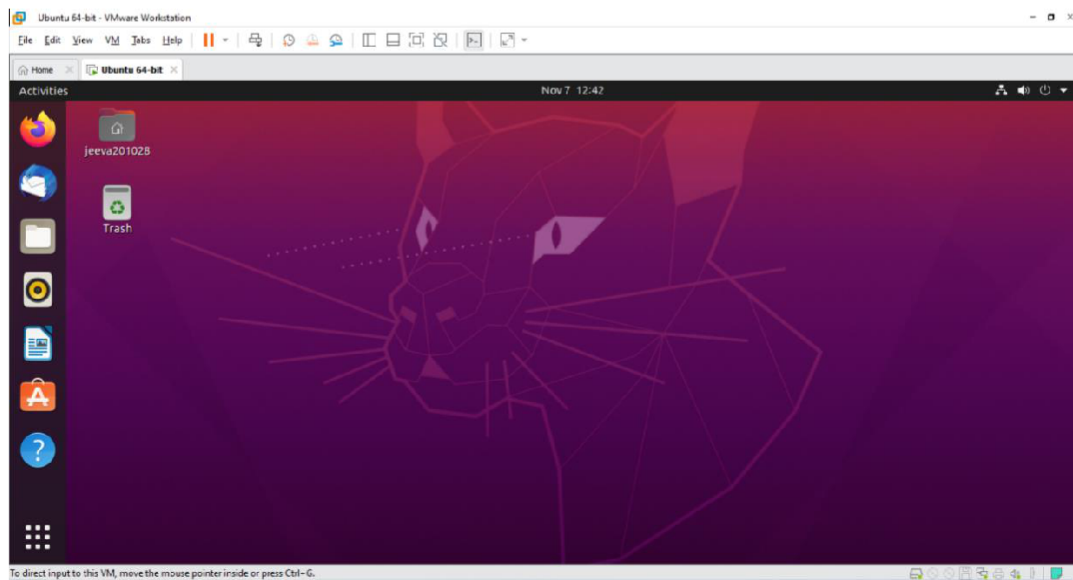
To transfer files/folders from the host machine to the virtual machine.

Procedure:

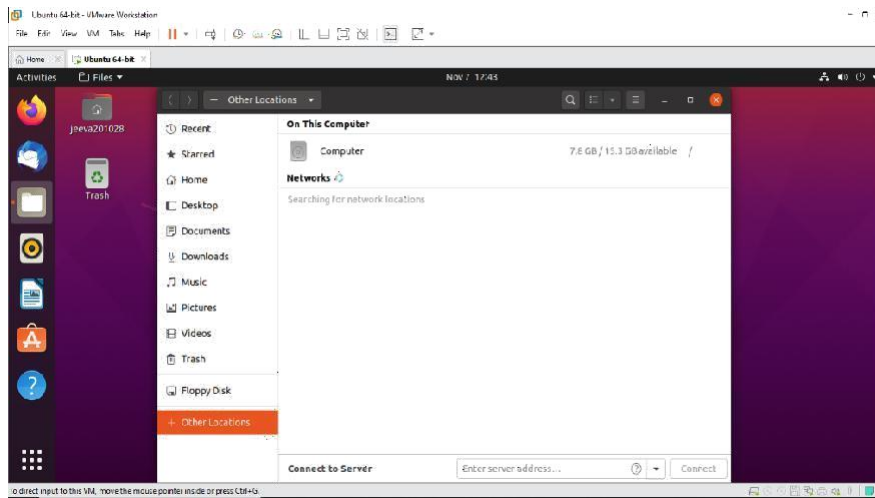
Files that are to be shared to virtual machine



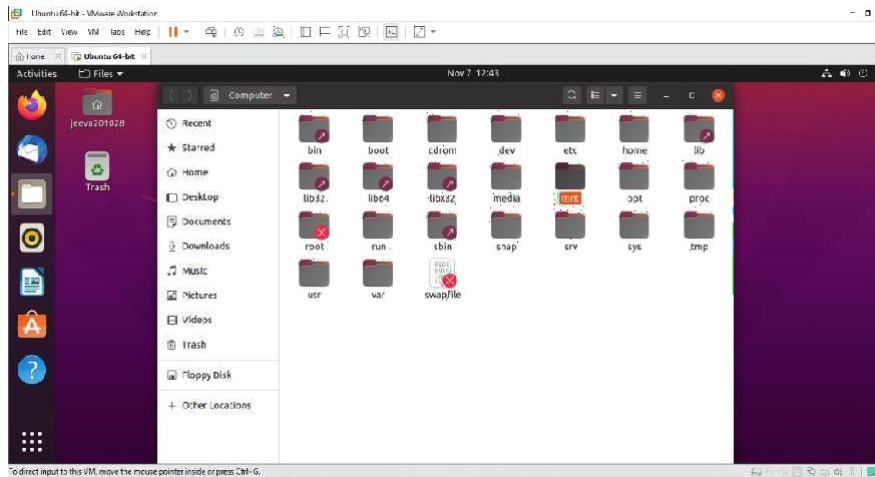
Step 1: Open the ubuntu in VM



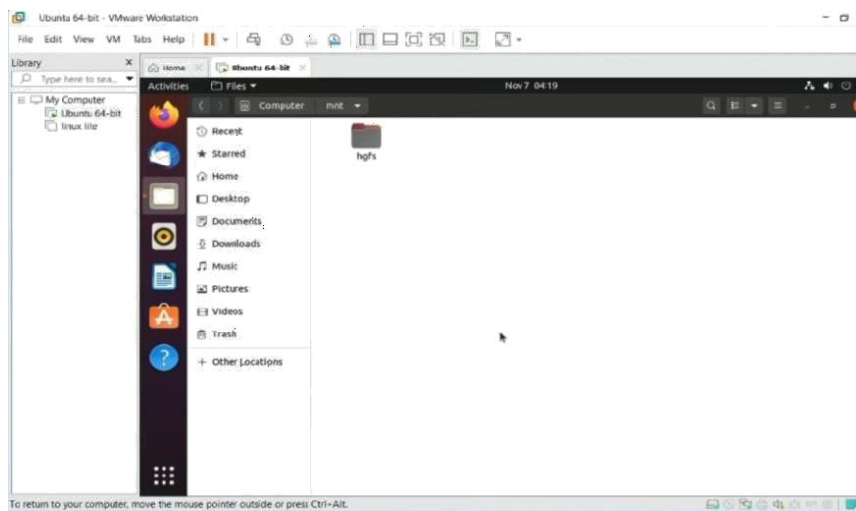
Step 2: Open the file manager and go to Other Locations



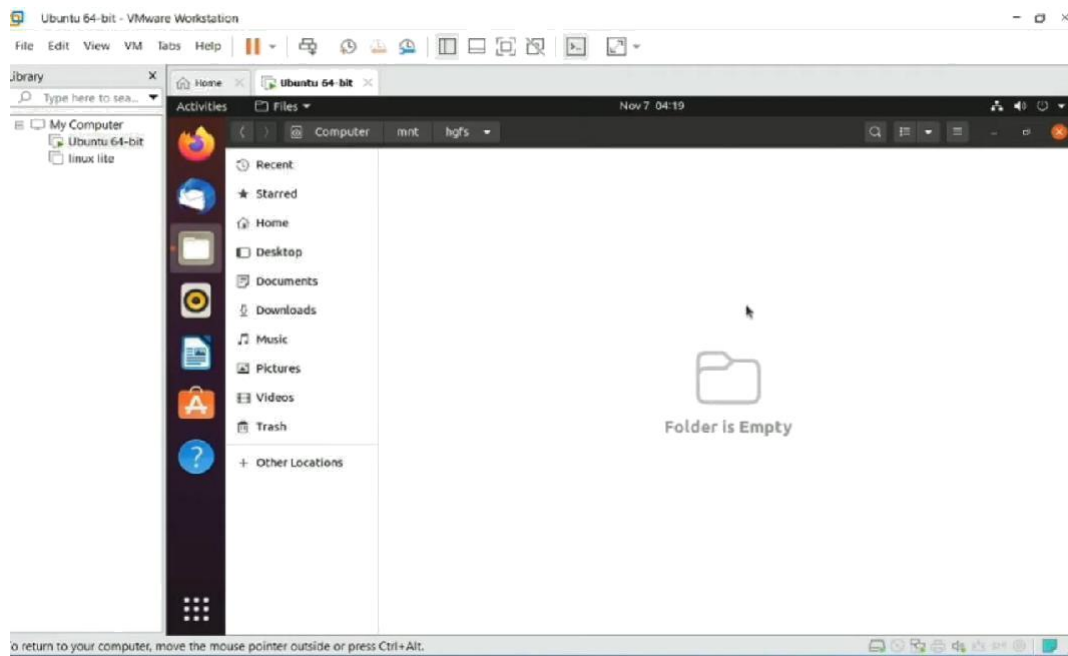
Step 3: Open the folder named “mnt”



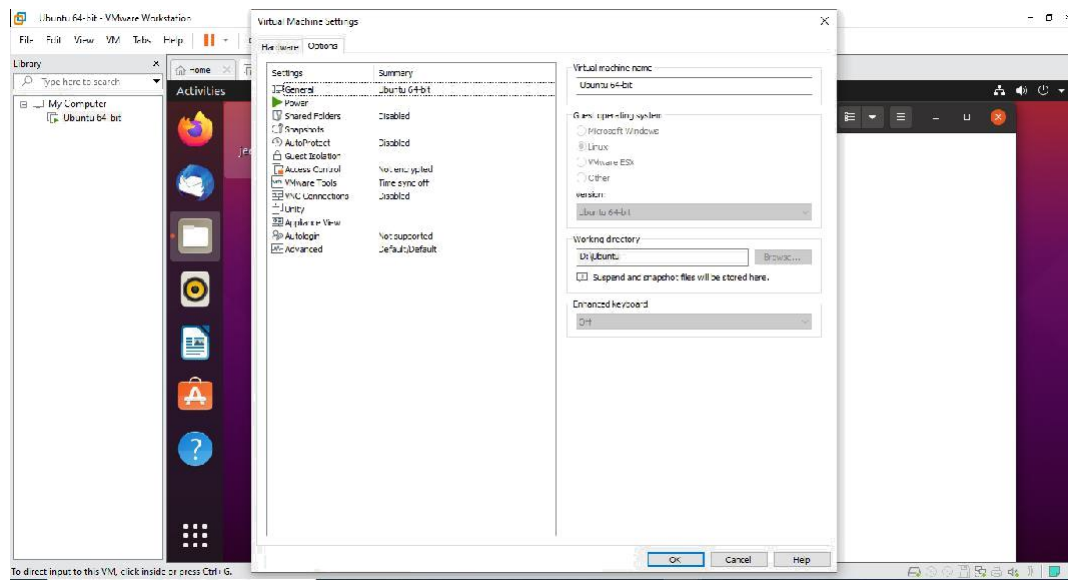
Step 4: Now open the “hgfs” folder



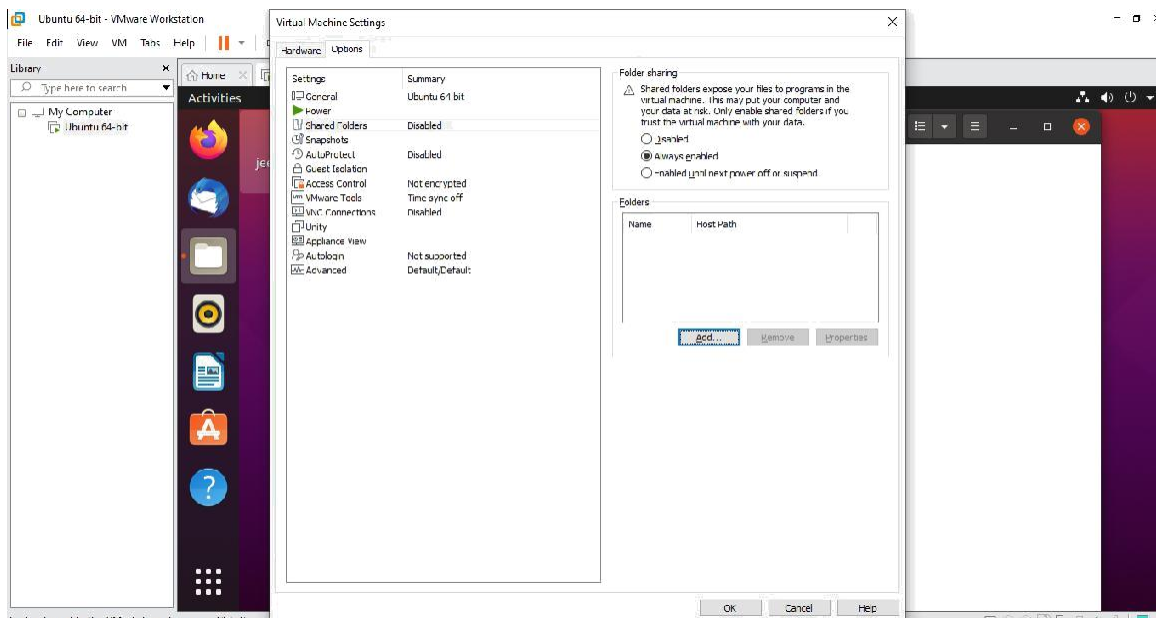
Initially the folder will be empty



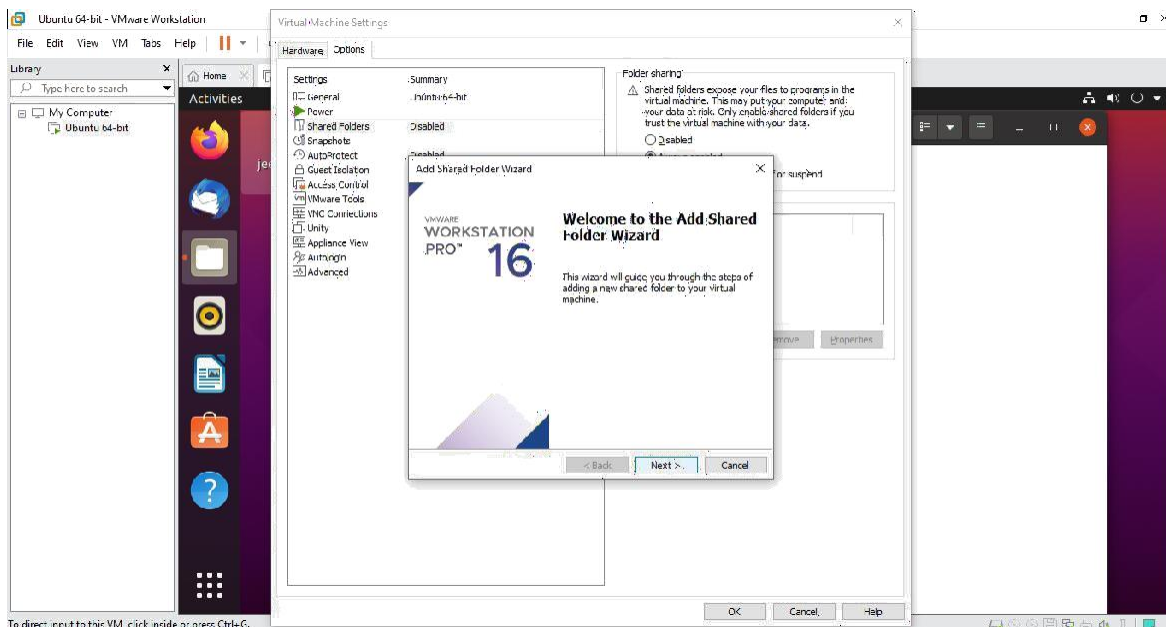
Step 5: Now right click ubuntu 64-bit (VM name) and select properties, then go to options tab



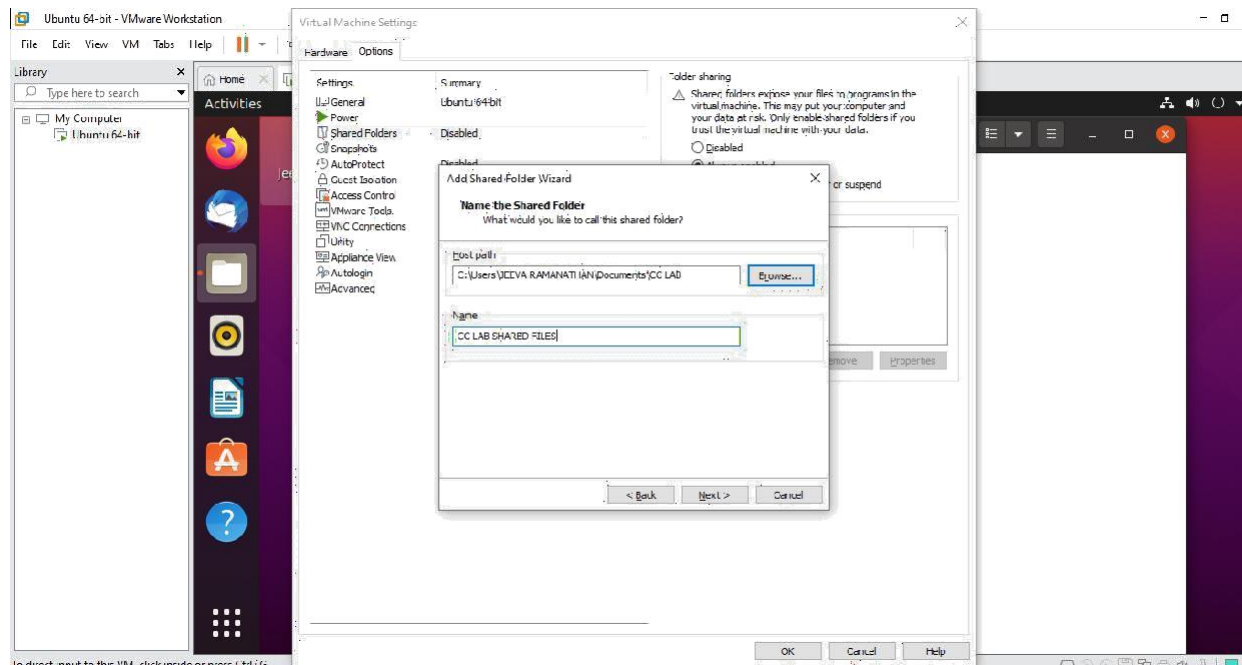
Step 6: Select the shared folders, change the radio button to “Always enabled” and click Add



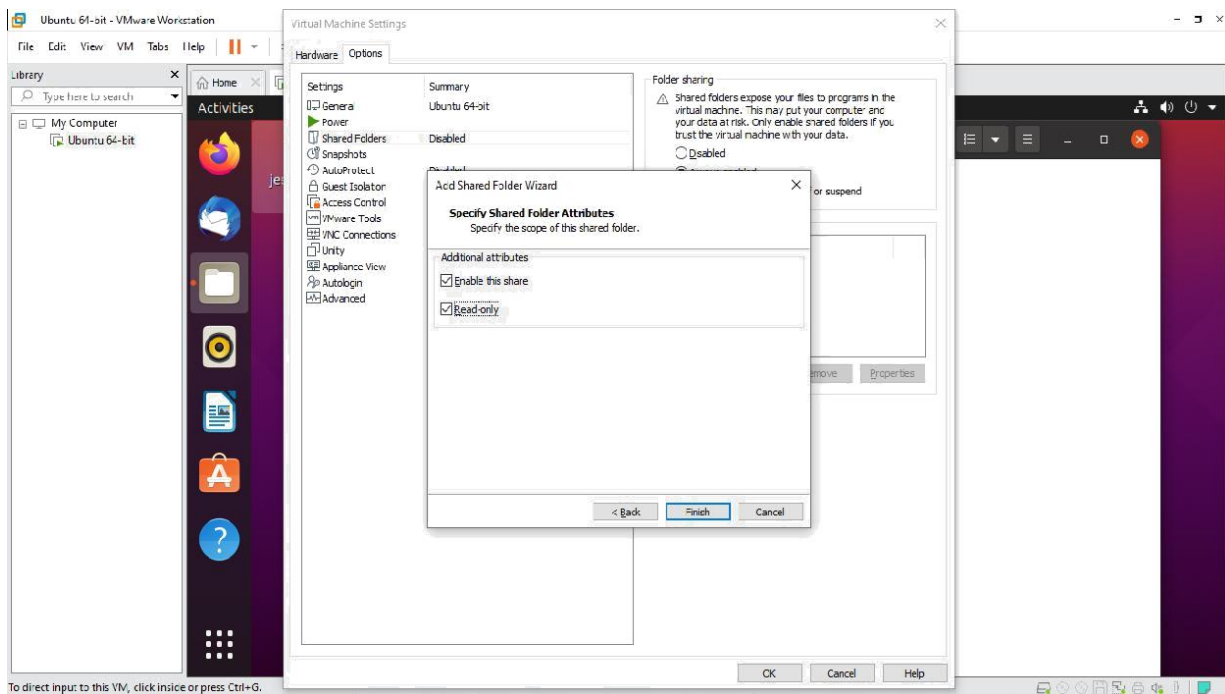
Step 7: Click Next



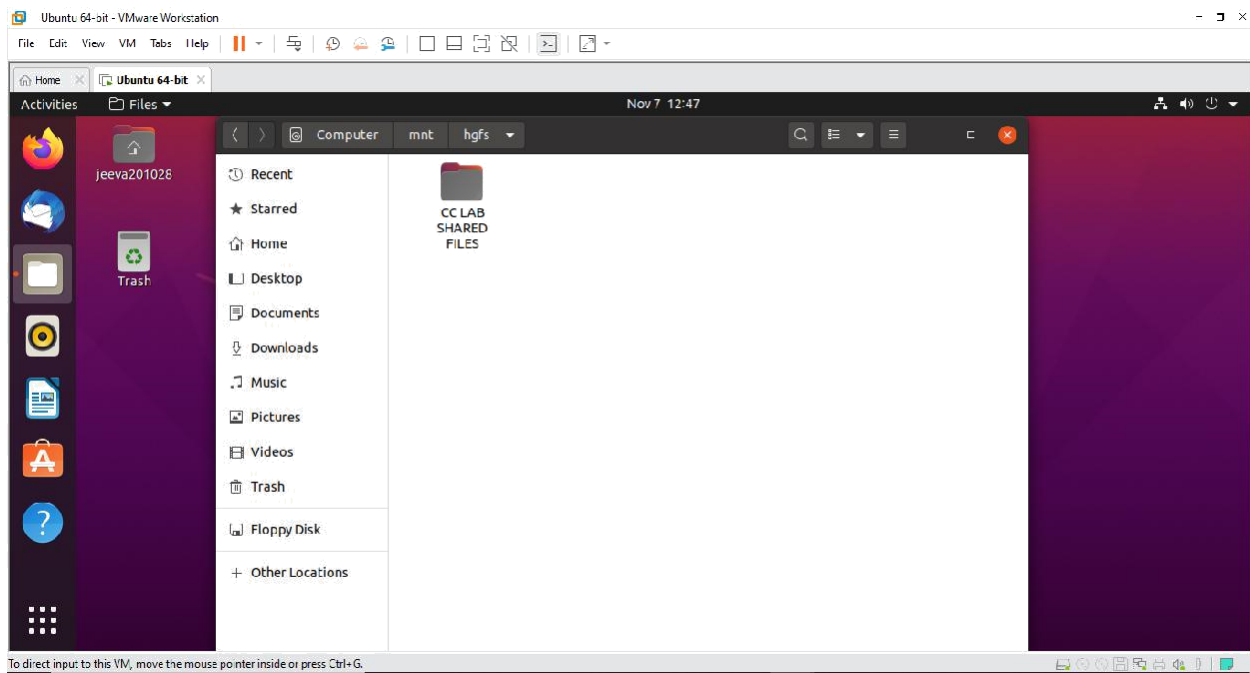
Step 8. Select the folder/file that has to be shared to VM in Host Path



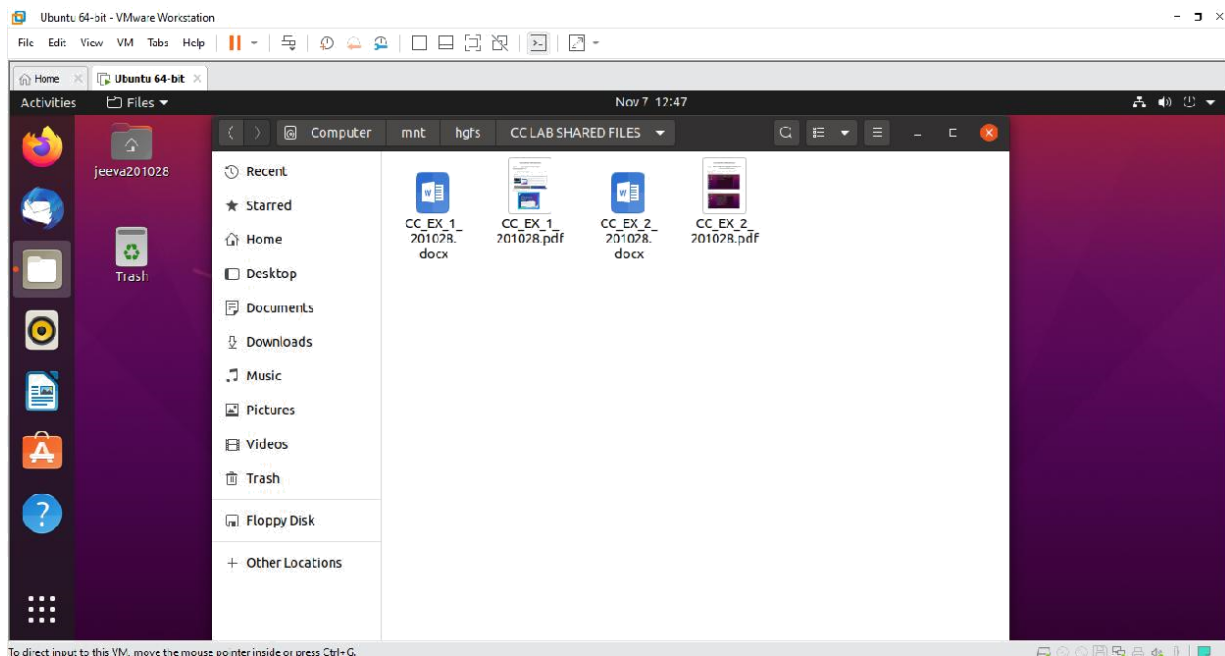
Step 9: Check Read Only and click Finish



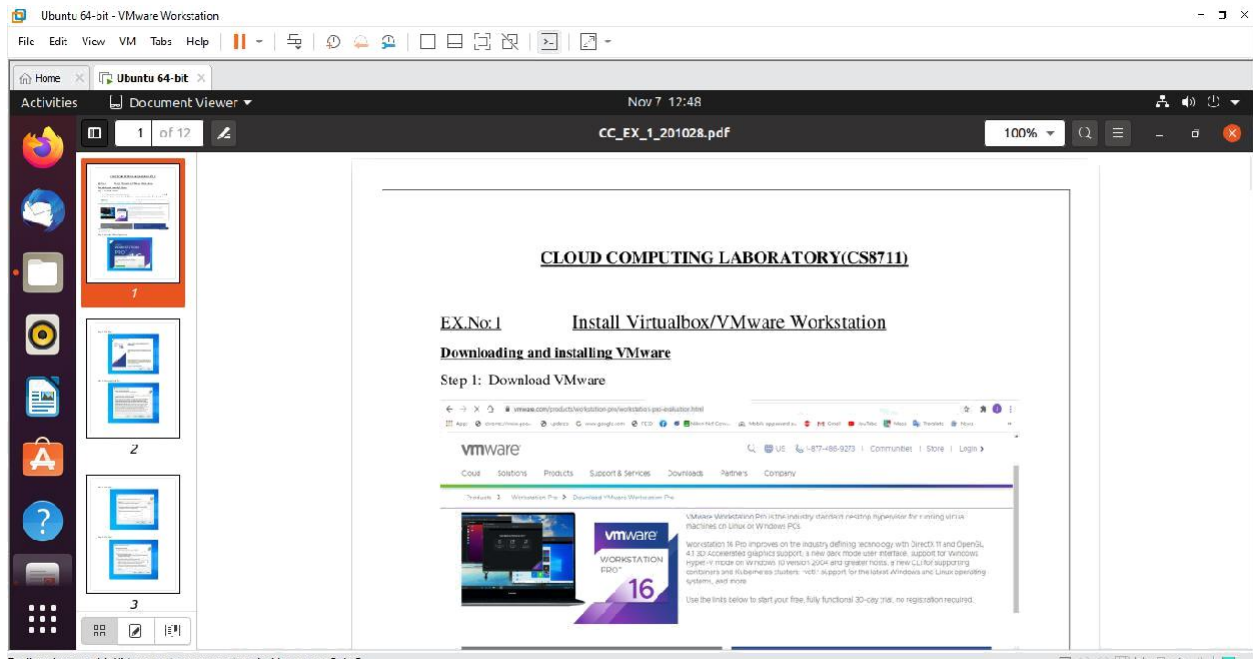
Step 10: Now in the same location “Computer>mnt>hgfs” the folder that are shared from host is visible



Step 11:



Step 12: The files can be accessed in Read Only Mode in the VM now



Result:

Thus file sharing between is done between hostmachine and the virtual machine successfully.

EX.No:4

Installation of a Google App Engine

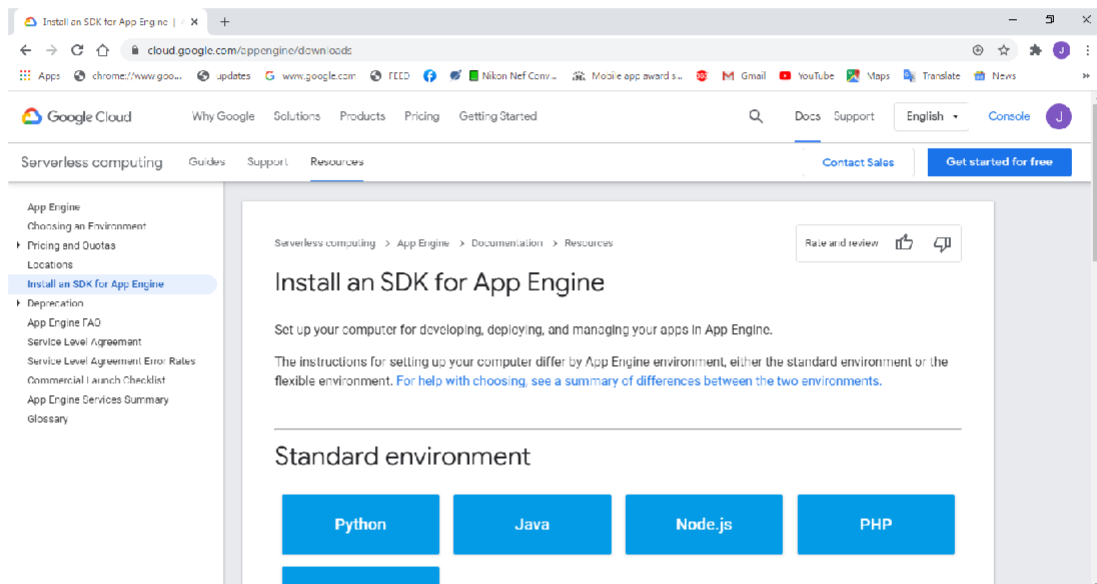
Date:

Aim:

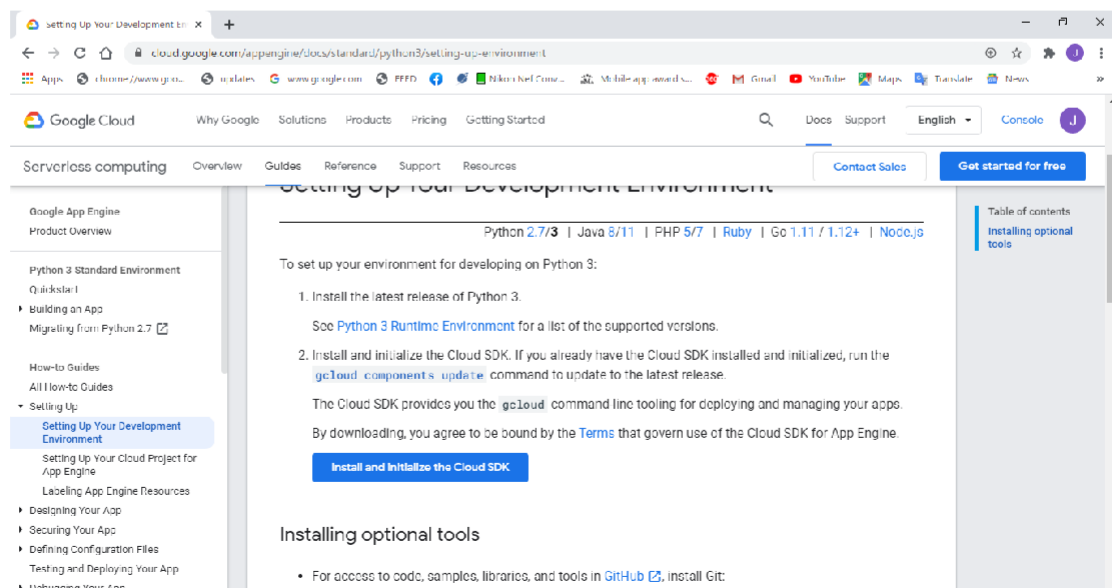
To install the google app engine in the system.

Procedure:

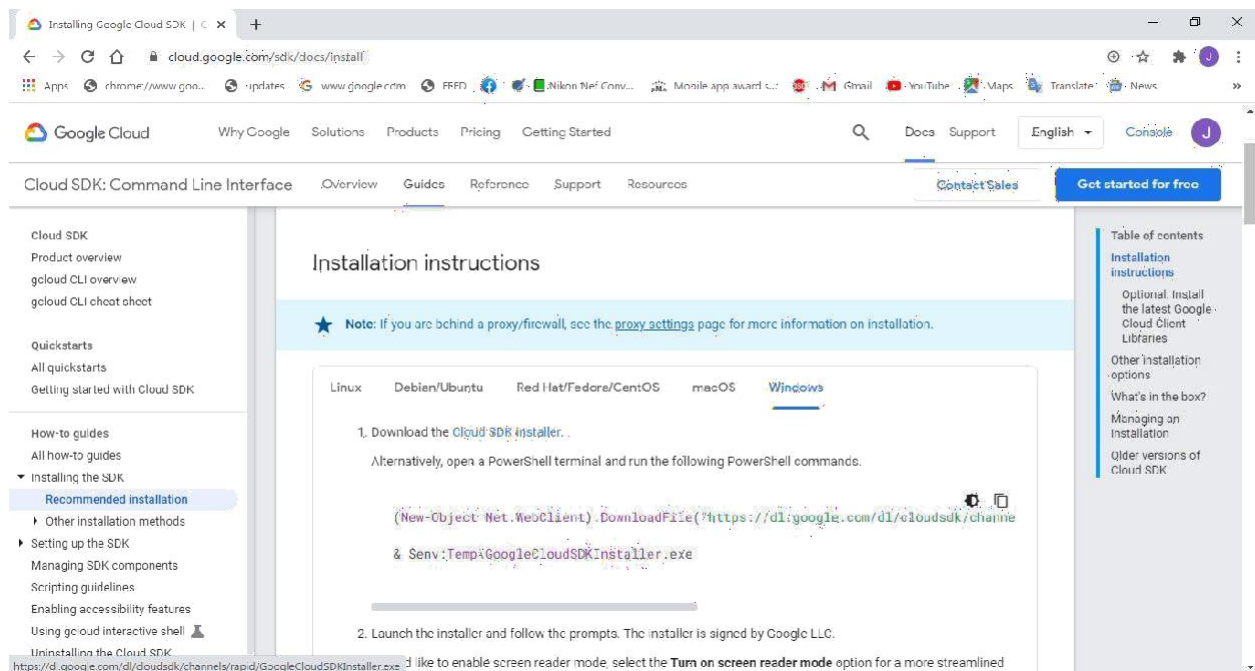
Step 1:Open the following link - <https://cloud.google.com/appengine/downloads> and click python.



Step 2:Select setting up your environment development and click on install the and initialize the cloud sdk.



Step 3: Download the sdk installer and install it.



The screenshot shows the Google Cloud SDK installation instructions page in a web browser. The browser's address bar shows the URL `cloud.google.com/sdk/docs/install`. The page has a navigation sidebar on the left with sections like 'Cloud SDK', 'Quickstarts', 'How-to guides', and 'Installing the SDK'. The main content area is titled 'Installation instructions' and includes a note about proxy/firewall settings. It lists two steps: 1. Download the Cloud SDK installer, and 2. Launch the installer and follow the prompts. Step 1 includes a PowerShell command to download the installer from a Google Cloud Storage bucket. Step 2 mentions that the installer is signed by Google LLC. A table of contents is visible on the right side of the page.

Installing Google Cloud SDK | C x +

cloud.google.com/sdk/docs/install

Google Cloud Why Google Solutions Products Pricing Getting Started Docs Support English Console

Cloud SDK: Command Line Interface Overview Guides Reference Support Resources Contact Sales Get started for free

Cloud SDK
Product overview
gcloud CLI overview
gcloud CLI cheat sheet

Quickstarts
All quickstarts
Getting started with Cloud SDK

How-to guides
All how-to guides

▼ Installing the SDK

Recommended installation

Other installation methods

Setting up the SDK

Managing SDK components

Scripting guidelines

Enabling accessibility features

Using gcloud interactive shell

Uninstalling the Cloud SDK

Installation instructions

★ Note: If you are behind a proxy/firewall, see the [proxy settings](#) page for more information on installation.

Linux Debian/Ubuntu Red Hat/Fedora/CentOS macOS Windows

1. Download the [Cloud SDK installer](#).

Alternatively, open a PowerShell terminal and run the following PowerShell commands.

```
(New-Object Net.WebClient).DownloadFile(https://dl.google.com/dl/cloudsdk/channel  
& .env\Temp\GoogleCloudSDKInstaller.exe
```

2. Launch the installer and follow the prompts. The installer is signed by Google LLC.

<https://dl.google.com/dl/cloudsdk/channel/rapid/GoogleCloudSDKInstaller.exe> I like to enable screen reader mode, select the **Turn on screen reader mode** option for a more streamlined

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Installation instructions

Optional: Install the latest Google Cloud Client Libraries

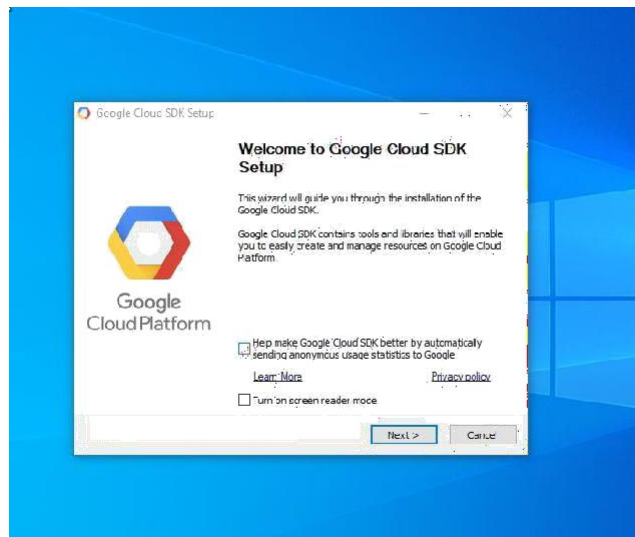
Other installation options

What's in the box?

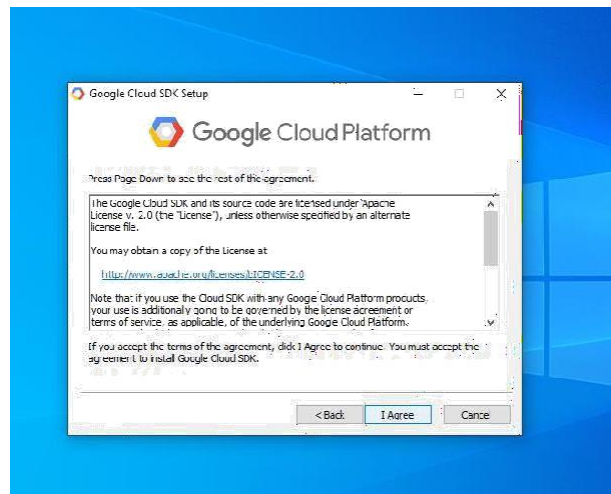
Managing an installation

Older versions of Cloud SDK

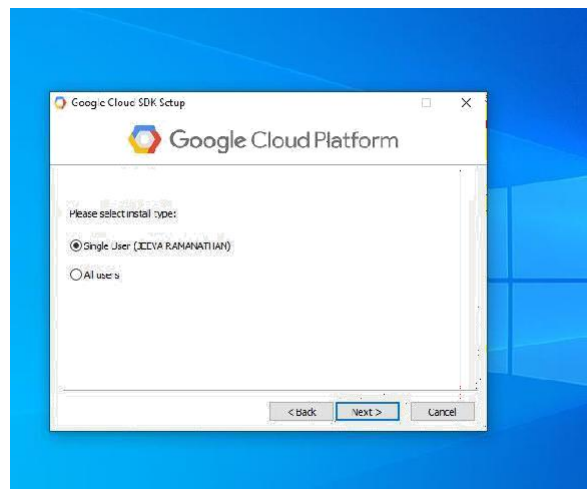
Step 4: Click Next.



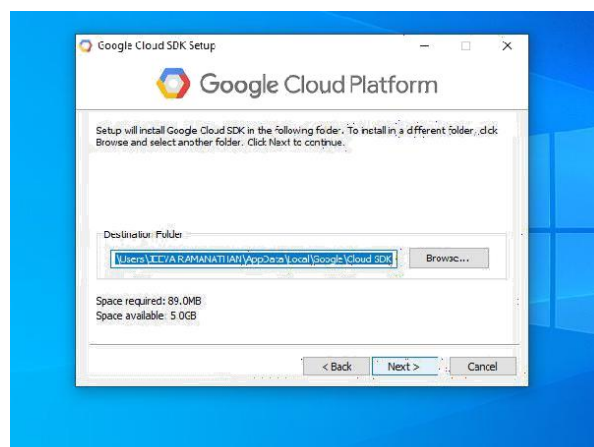
Step 5: Click I Agree.



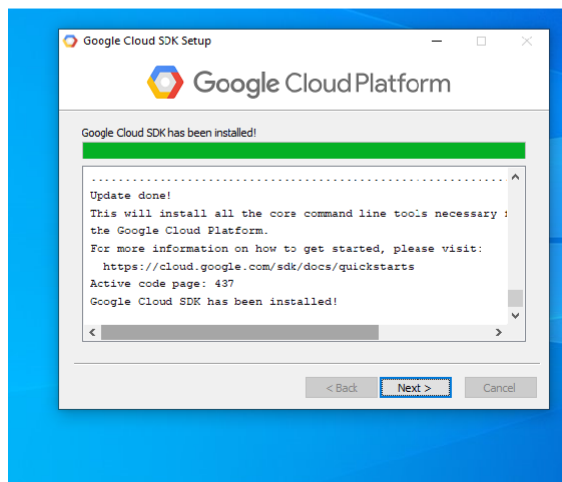
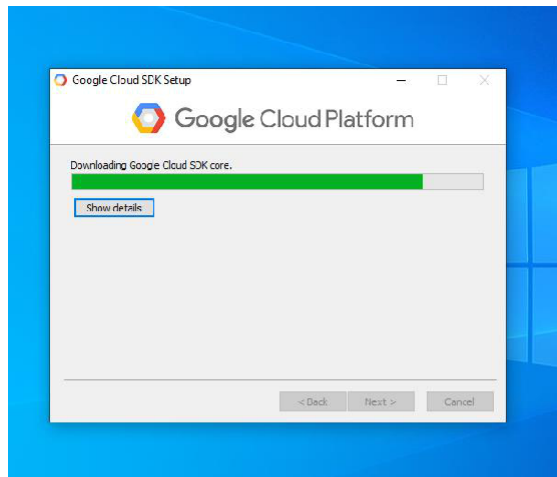
Step 6: Select single user and click Next.



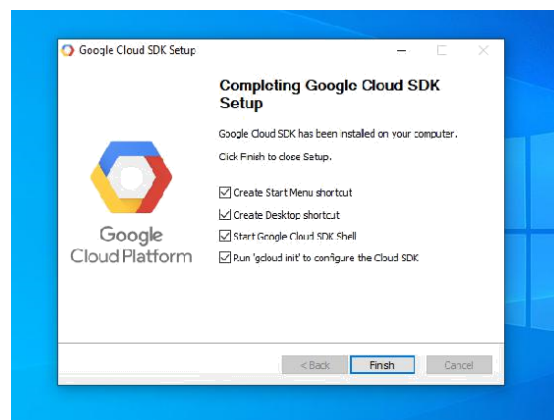
Step 7: Select the destination location and click Next.



Step 8: Downloading all the requirements and installing



Step 9: Click Finish.



Step 10: Once successfully installed cmd line in login with your google account.

```
C:\WINDOWS\SYSTEM32\cmd.exe - gcloud init
Welcome to the Google Cloud SDK! Run "gcloud -h" to get the list of available commands.
---
Welcome! This command will take you through the configuration of gcloud.

Your current configuration has been set to: [default]

You can skip diagnostics next time by using the following flag:
  gcloud init --skip-diagnostics

Network diagnostic detects and fixes local network connection issues.
Checking network connection...done.
Reachability Check passed.
Network diagnostic passed (1/1 checks passed).

You must log in to continue. Would you like to log in (Y/n)? Y
```

```
C:\Users\JEEVA RAMANATHAN\AppData\Local\Google\Cloud SDK>gcloud auth login
Your browser has been opened to visit:

https://accounts.google.com/o/oauth2/auth?client_id=32555940559.apps.googleusercontent.com&redirect_uri=http%3A%2F%2Flocalhost%3A8085%2F&scope=openid https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth&code_challenge=fr6eBPLwbenQ3uChkqT1l14Qf0WqjIiqD6dYj9Lae0&code_challenge_method=S256&access_type=offline&response_type=code&prompt=select_account

You are now logged in as [jeevaram2000@gmail.com].
Your current project is [None]. You can change this setting by running:
$ gcloud config set project PROJECT_ID

C:\Users\JEEVA RAMANATHAN\AppData\Local\Google\Cloud SDK>
```

Result:

Thus google app engine is installed successfully in the system.

EX.No:5

Launch a web application using GAE launcher

Date:

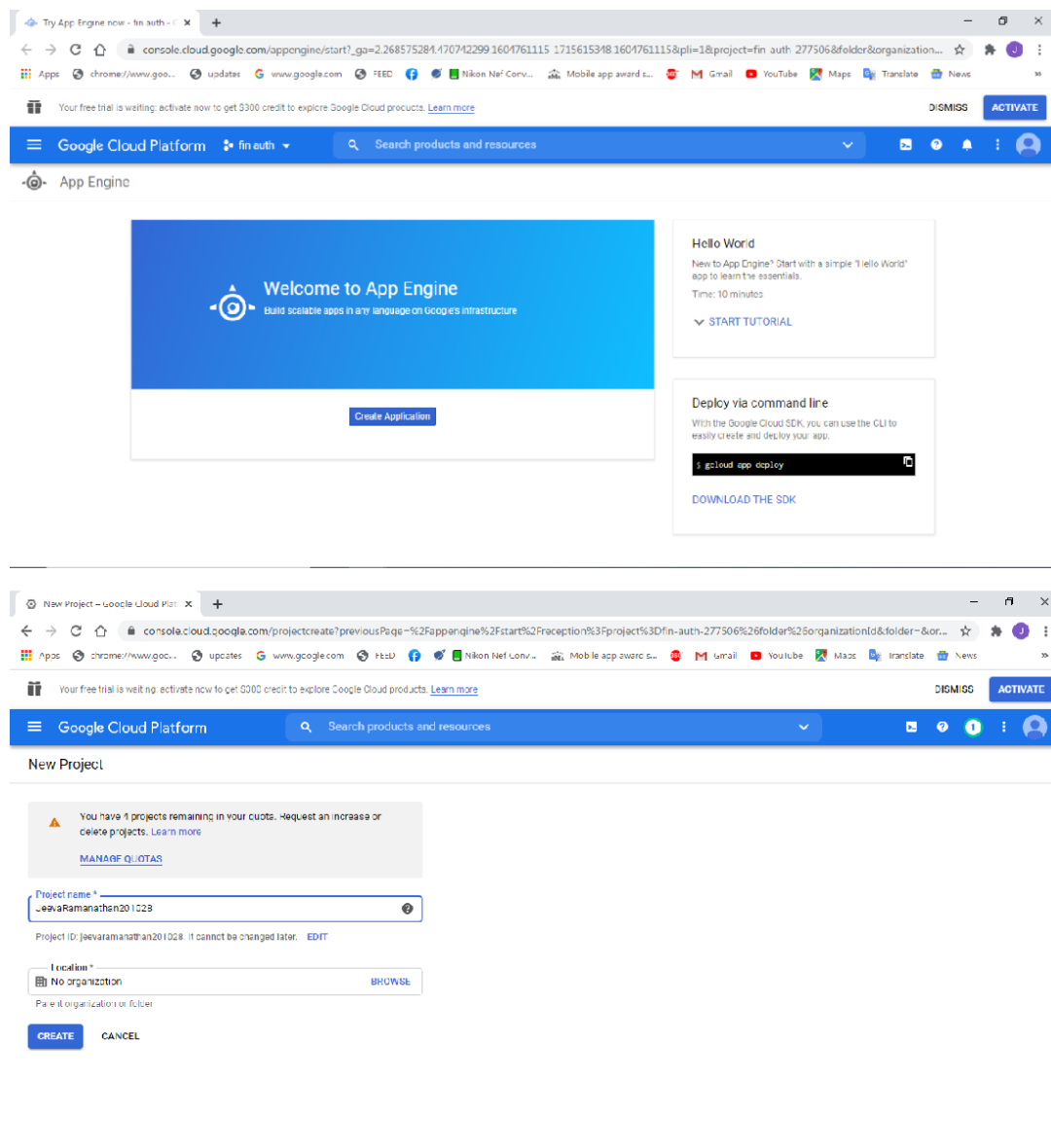
Aim:

To launch the web applications by using the GAE launcher.

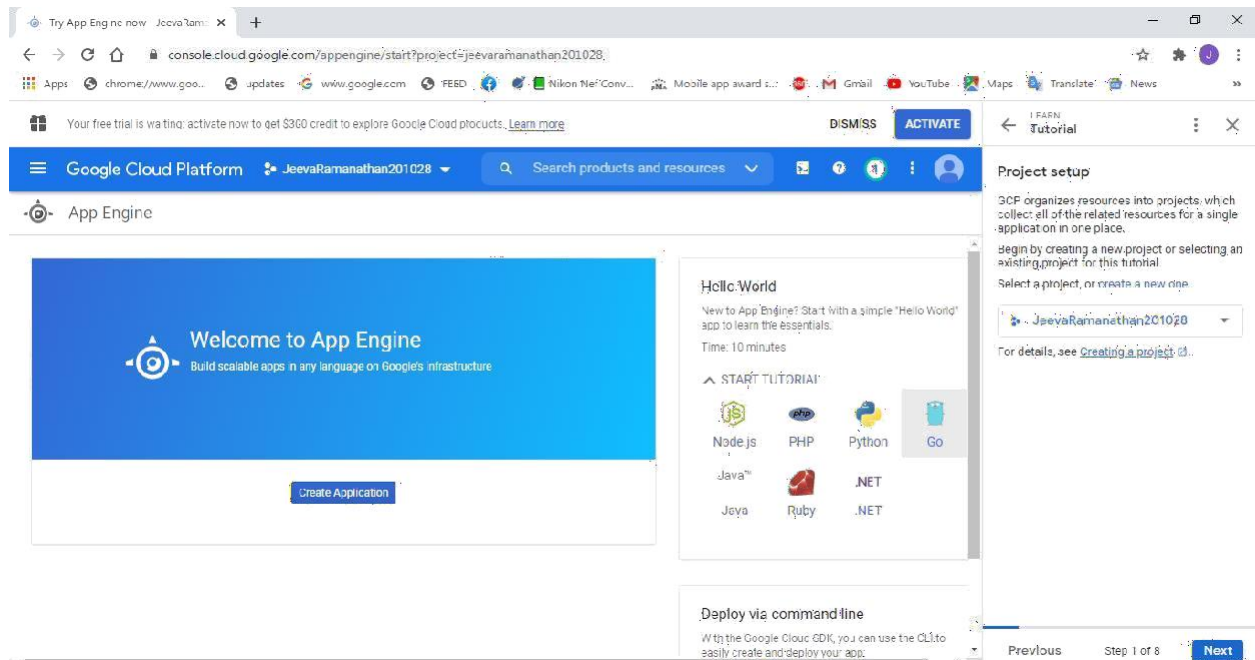
Procedure:

Step 1: Go to the following website

https://console.cloud.google.com/start/appengine?_ga=2.268575284.470742299.1604761115-1715615348.1604761115&pli=1 and create a new project.

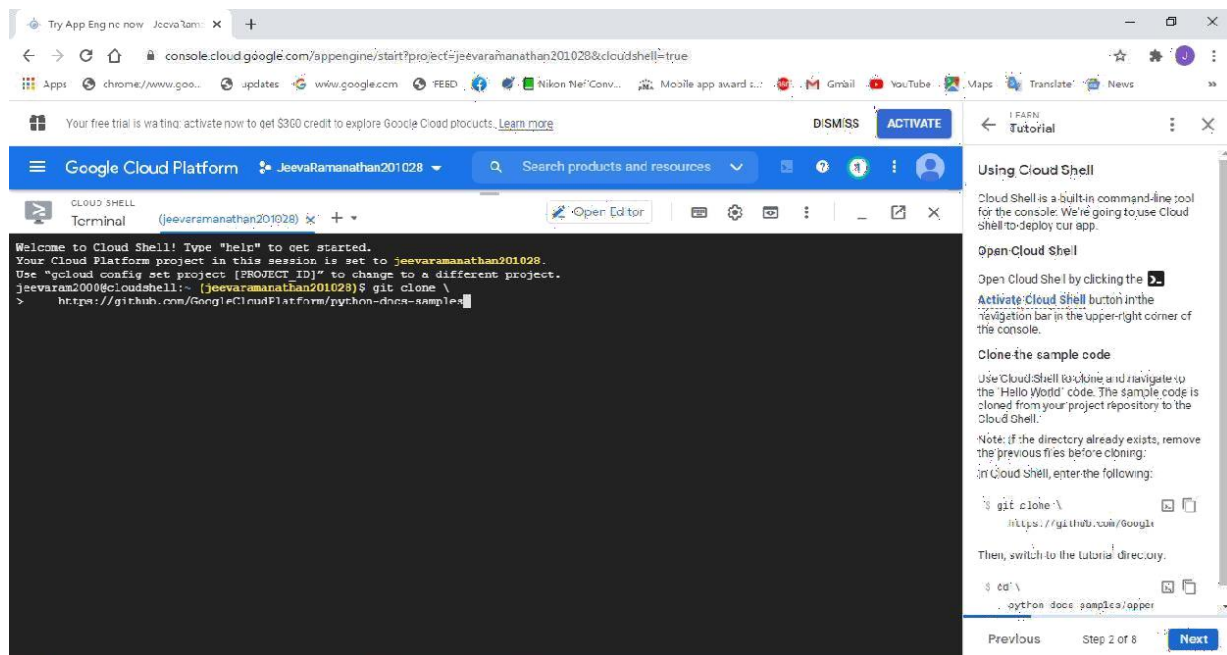


Step 2: Select python and click next.

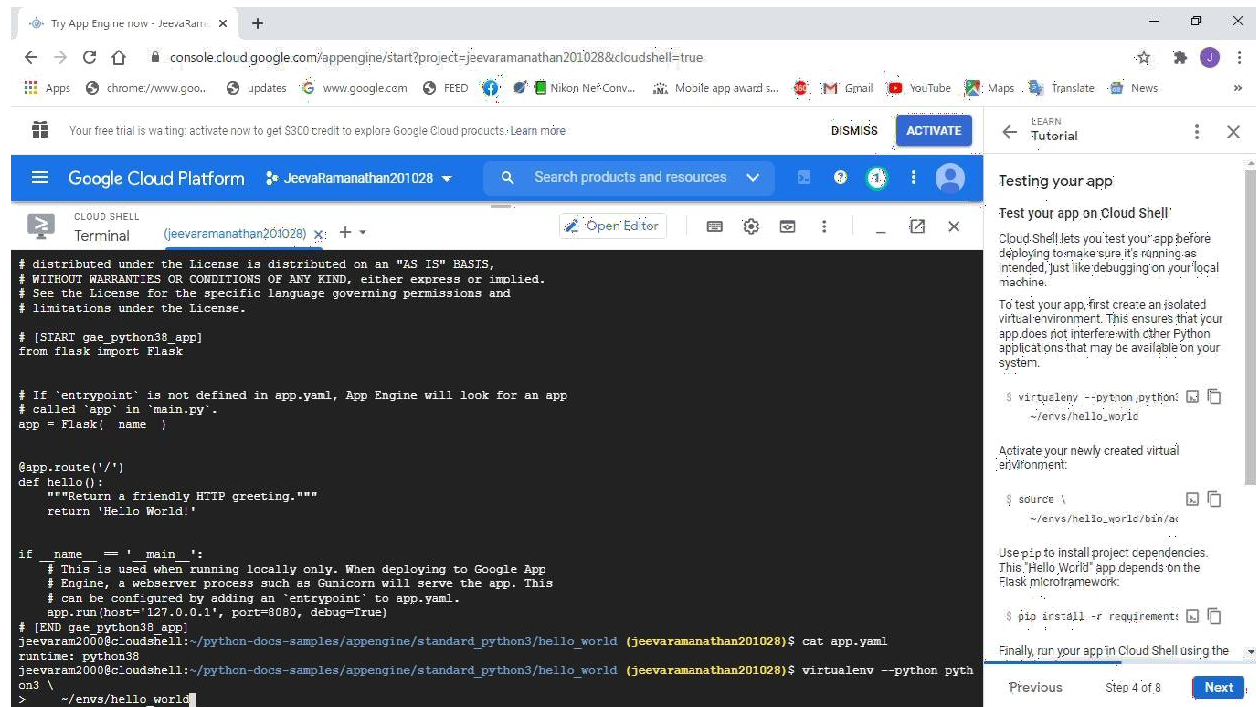


Step 3: Open the cloud shell and follow the steps in the tutorial.

Clone the repository by using the given command



Step 4: Create the virtual environment

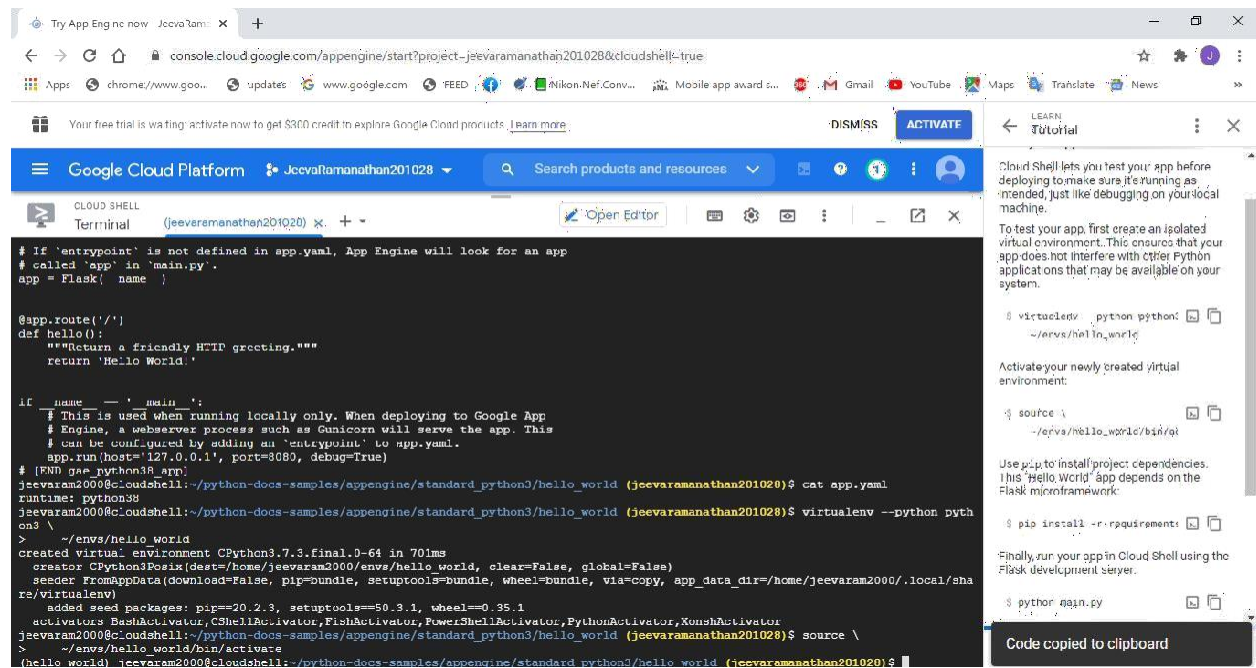


The screenshot shows the Google Cloud Platform console with a Cloud Shell terminal open. The terminal displays the contents of the `app.yaml` file, which defines a Flask application. The user has run the command `virtualenv --python python3` to create a virtual environment. The terminal output shows the creation of the virtual environment and the installation of dependencies.

```
# distributed under the License is distributed on an "AS IS" BASIS,  
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.  
# See the License for the specific language governing permissions and  
# limitations under the License.  
  
# [START gae_python38_app]  
from flask import Flask  
  
# If 'entrypoint' is not defined in app.yaml, App Engine will look for an app  
# called 'app' in 'main.py'.  
app = Flask(__name__)  
  
@app.route('/')  
def hello():  
    """Return a friendly HTTP greeting."""  
    return 'Hello World'  
  
if __name__ == '__main__':  
    # This is used when running locally only. When deploying to Google App  
    # Engine, a webserver process such as Gunicorn will serve the app. This  
    # can be configured by adding an 'entrypoint' to app.yaml.  
    app.run(host='127.0.0.1', port=8080, debug=True)  
# [END gae_python38_app]  
jeevaram2000@cloudshell:~/python-docs-samples/appengine/standard_python3/hello_world (jeevaramathan201028)$ cat app.yaml  
runtime: python38  
jeevaram2000@cloudshell:~/python-docs-samples/appengine/standard_python3/hello_world (jeevaramathan201028)$ virtualenv --python python3  
on3 \  
> ~/envs/hello_world  
>
```

The right sidebar shows the "Testing your app" section, which provides instructions on how to test the app on Cloud Shell. It includes steps for creating a virtual environment, installing dependencies, and running the app.

Step 5: Activate your virtual environment.



The screenshot shows the Google Cloud Platform console with a Cloud Shell terminal open. The terminal displays the output of the `virtualenv` command, showing the creation of the virtual environment and the installation of dependencies. The user has run the command `source ~/envs/hello_world/bin/activate` to activate the virtual environment. The terminal output shows the activation of the virtual environment and the installation of dependencies.

```
# If 'entrypoint' is not defined in app.yaml, App Engine will look for an app  
# called 'app' in 'main.py'.  
app = Flask(__name__)  
  
@app.route('/')  
def hello():  
    """Return a friendly HTTP greeting."""  
    return 'Hello World'  
  
if __name__ == '__main__':  
    # This is used when running locally only. When deploying to Google App  
    # Engine, a webserver process such as Gunicorn will serve the app. This  
    # can be configured by adding an 'entrypoint' to app.yaml.  
    app.run(host='127.0.0.1', port=8080, debug=True)  
# [END gae_python38_app]  
jeevaram2000@cloudshell:~/python-docs-samples/appengine/standard_python3/hello_world (jeevaramathan201028)$ cat app.yaml  
runtime: python38  
jeevaram2000@cloudshell:~/python-docs-samples/appengine/standard_python3/hello_world (jeevaramathan201028)$ virtualenv --python python3  
on3 \  
> ~/envs/hello_world  
> created virtual environment CPython3.7.3.final.0-64 in 701ms  
creator CPython3Posix(dest=/home/jeevaram2000/envs/hello_world, clear=False, global=False)  
scanner FromAppData(download=False, pip=bundle, setuptools=bundle, wheel=bundle, via=copy, app_data_dir=/home/jeevaram2000/.local/share/virtualenv)  
added seed packages: pip==20.2.3, setuptools==50.3.1, wheel==0.35.1  
activators BashActivator,CShellActivator,FishActivator,PowerShellActivator,PythonActivator,XonshActivator  
jeevaram2000@cloudshell:~/python-docs-samples/appengine/standard_python3/hello_world (jeevaramathan201028)$ source \  
> ~/envs/hello_world/bin/activate  
> (hello world) jeevaram2000@cloudshell:~/python-docs-samples/appengine/standard_python3/hello_world (jeevaramathan201028)$
```

The right sidebar shows the "Testing your app" section, which provides instructions on how to test the app on Cloud Shell. It includes steps for creating a virtual environment, installing dependencies, and running the app. A "Code copied to clipboard" notification is visible at the bottom right.

Step 6: Installing requirements and run the app.

The screenshot shows the Google Cloud Platform console with the Cloud Shell terminal open. The terminal displays the following commands and output:

```
created virtual environment CPython3.7.3.final.0-64 in 701ms
creator CPython3Posix(dest=/home/jeevaran2000/envs/hello_world, clear=False, global=False)
seeder FromAppData(download=False, pip=bundle, setuptools=bundle, wheel=bundle, via=copy, app_data_dir=/home/jeevaran2000/.local/share/virtualenv)
added seed packages: pip==20.2.3, setuptools==50.3.1, wheel==0.35.1
activators BashActivator,CShellActivator,FishActivator,PowerShellActivator,PythonActivator,KomodoActivator
jeevaran2000@cloudshell:~/python-docs-samples/appengine/standard_python3/hello_world (jeevaranathan201028)$ source \
~/envs/hello_world/bin/activate
(hello_world) jeevaran2000@cloudshell:~/python-docs-samples/appengine/standard_python3/hello_world (jeevaranathan201028)$ pip instal
l -r requirements.txt
Collecting Flask==1.1.2
  Downloading Flask-1.1.2-py2.py3-none-any.whl (94 kB)
    |#####| 94 kB 2.6 MB/s
Collecting click==5.1
  Downloading click-7.1.2-py2.py3-none-any.whl (82 kB)
    |#####| 82 kB 1.3 MB/s
Collecting itsdangerous==0.24
  Downloading itsdangerous-1.1.0-py2.py3-none-any.whl (16 kB)
Collecting Jinja2==2.10.1
  Downloading Jinja2-2.11.2-py2.py3-none-any.whl (125 kB)
    |#####| 125 kB 13.4 MB/s
Collecting Werkzeug==0.15
  Downloading Werkzeug-1.0.1-py2.py3-none-any.whl (298 kB)
    |#####| 298 kB 40.1 MB/s
Collecting MarkupSafe==0.23
  Downloading MarkupSafe-1.1.1-cp37-cp37m-manylinux1_x86_64.whl (27 kB)
Installing collected packages: click, itsdangerous, MarkupSafe, Jinja2, Werkzeug, Flask
Successfully installed Flask-1.1.2 Jinja2-2.11.2 MarkupSafe-1.1.1 Werkzeug-1.0.1 click-7.1.2 itsdangerous-1.1.0
WARNING: You are using pip version 20.2.3; however, version 20.2.4 is available.
You should consider upgrading via the '/home/jeevaran2000/envs/hello_world/bin/python -m pip install --upgrade pip' command.
(hello_world) jeevaran2000@cloudshell:~/python-docs-samples/appengine/standard_python3/hello_world (jeevaranathan201028)$
```

The screenshot shows the Google Cloud Platform console with the Cloud Shell terminal open. The terminal displays the following commands and output:

```
if name == 'main':
    # This is used when running locally only. When deploying to Google App
    # Engine, a webserver process such as Gunicorn will serve the app. This
    # can be configured by adding an 'entrypoint' to app.yaml.
    app.run(host='127.0.0.1', port=8080, debug=True)
# [END gae_python38_app]
jeevaran2000@cloudshell:~/python-docs-samples/appengine/standard_python3/hello_world (jeevaranathan201028)$ cat app.yaml
runtime: python38
jeevaran2000@cloudshell:~/python-docs-samples/appengine/standard_python3/hello_world (jeevaranathan201028)$ virtualenv --python pyth
on3 \
~/envs/hello_world
created virtual environment CPython3.7.3.final.0-64 in 701ms
creator CPython3Posix(dest=/home/jeevaran2000/envs/hello_world, clear=False, global=False)
seeder FromAppData(download=False, pip=bundle, setuptools=bundle, wheel=bundle, via=copy, app_data_dir=/home/jeevaran2000/.local/sha
re/virtualenv)
added seed packages: pip==20.2.3, setuptools==50.3.1, wheel==0.35.1
activators BashActivator,CShellActivator,FishActivator,PowerShellActivator,PythonActivator,KomodoActivator
jeevaran2000@cloudshell:~/python-docs-samples/appengine/standard_python3/hello_world (jeevaranathan201028)$ source \
~/envs/hello_world/bin/activate
(hello_world) jeevaran2000@cloudshell:~/python-docs-samples/appengine/standard_python3/hello_world (jeevaranathan201028)$ pip instal
l -r requirements.txt
Collecting Flask==1.1.2
  Downloading Flask-1.1.2-py2.py3-none-any.whl (94 kB)
    |#####| 94 kB 2.6 MB/s
Collecting click==5.1
  Downloading click-7.1.2-py2.py3-none-any.whl (82 kB)
    |#####| 82 kB 1.3 MB/s
Collecting itsdangerous==0.24
  Downloading itsdangerous-1.1.0-py2.py3-none-any.whl (16 kB)
```

Step 7: Create an application and deploy it in cloud shell.

The screenshot shows the Google Cloud Platform console with the 'Cloud Shell' terminal open. The terminal displays the process of downloading and installing various packages (Flask, Jinja2, Werkzeug, etc.) and then running the application. The application is a simple Flask app that returns 'hello world'. The terminal output shows the application is running on port 8080. The right sidebar shows the 'Deploying to App Engine' tutorial, which includes steps for creating an application, deploying it, and visiting it. The 'gcloud app create' command is entered in the terminal.

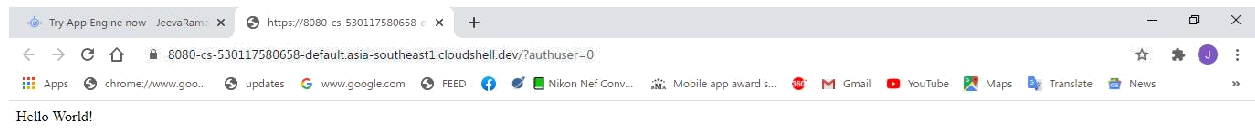
```
gcloud app create
```

Step 8: Click preview on port 8080 to see your deployed application

The screenshot shows the Google Cloud Platform console with the 'Cloud Shell' terminal open. The terminal displays the process of downloading and installing various packages (Flask, Jinja2, Werkzeug, etc.) and then running the application. The application is a simple Flask app that returns 'hello world'. The terminal output shows the application is running on port 8080. The right sidebar shows the 'View your app's status' section, which includes instructions on how to check the app's status and how to preview it. The 'Preview on port 8080' button is highlighted in the terminal output.

```
gcloud app create
gcloud app deploy app.yaml \
  --project jeevaramanathan201028
```

Step 9: Finally the application is deployed and the output is seen.



Result:

Thus a web application is launched by using the GAE launcher and the output is obtained successfully.

EX.No:6 Install Hadoop single mode cluster and run simple applications

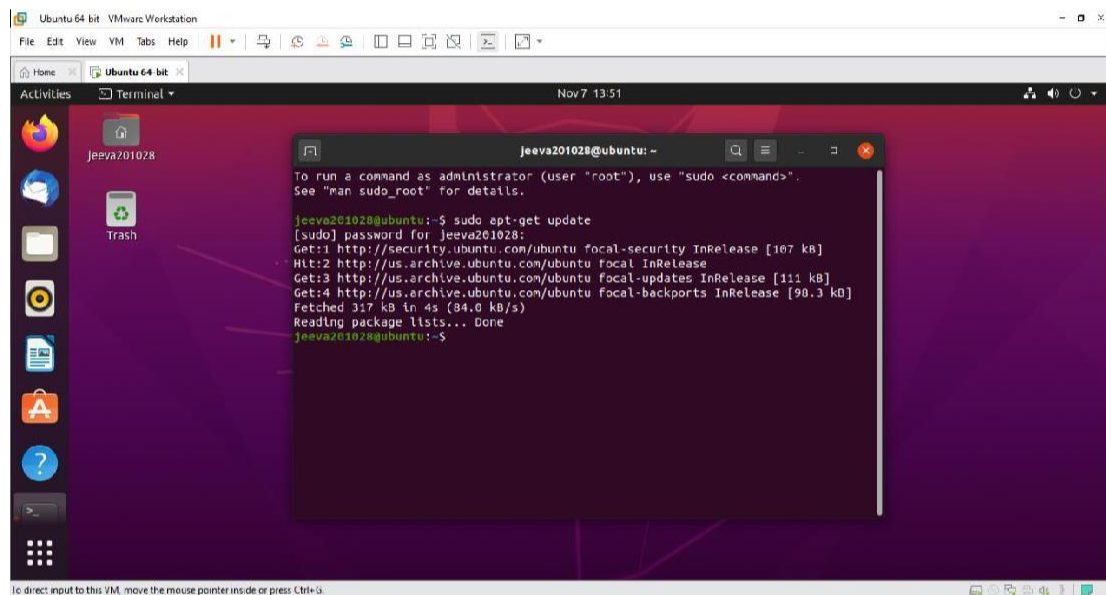
Date:

Aim:

To install hadoop single mode cluster and run a simple word count application.

Procedure:

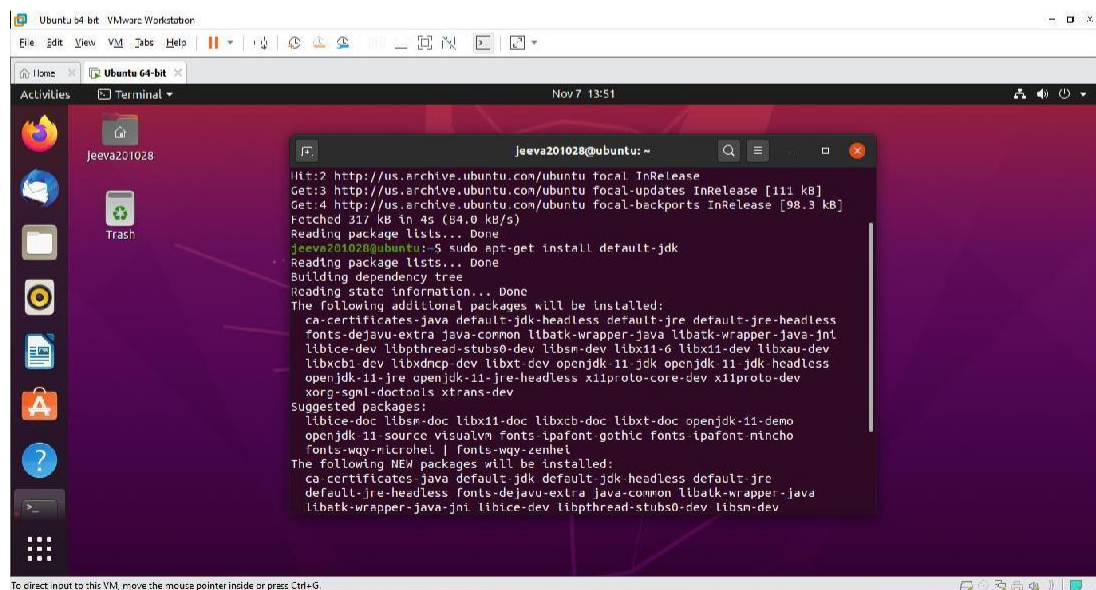
Update the packages



```
Ubuntu 64 bit VMware Workstation
File Edit View VM Tabs Help
Ubuntu 64 bit
Activities Terminal Nov 7 13:51
jeeva201028@ubuntu: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

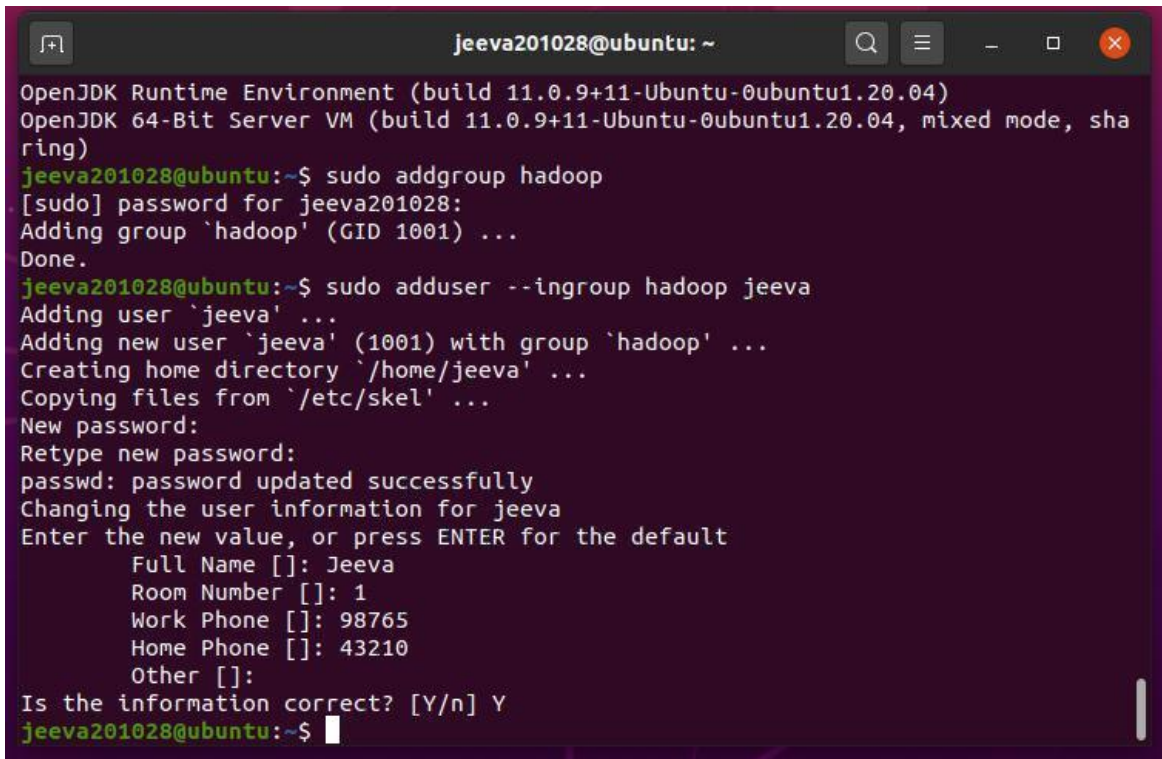
jeeva201028@ubuntu:~$ sudo apt-get update
[sudo] password for jeeva201028:
Get:1 http://security.ubuntu.com/ubuntu focal-security InRelease [107 kB]
Hit:2 http://us.archive.ubuntu.com/ubuntu focal InRelease
Get:3 http://us.archive.ubuntu.com/ubuntu focal-updates InRelease [111 kB]
Get:4 http://us.archive.ubuntu.com/ubuntu focal-backports InRelease [98.3 kB]
Fetched 317 kB in 4s (84.0 kB/s)
Reading package lists... Done
jeeva201028@ubuntu:~$
```

Installing java



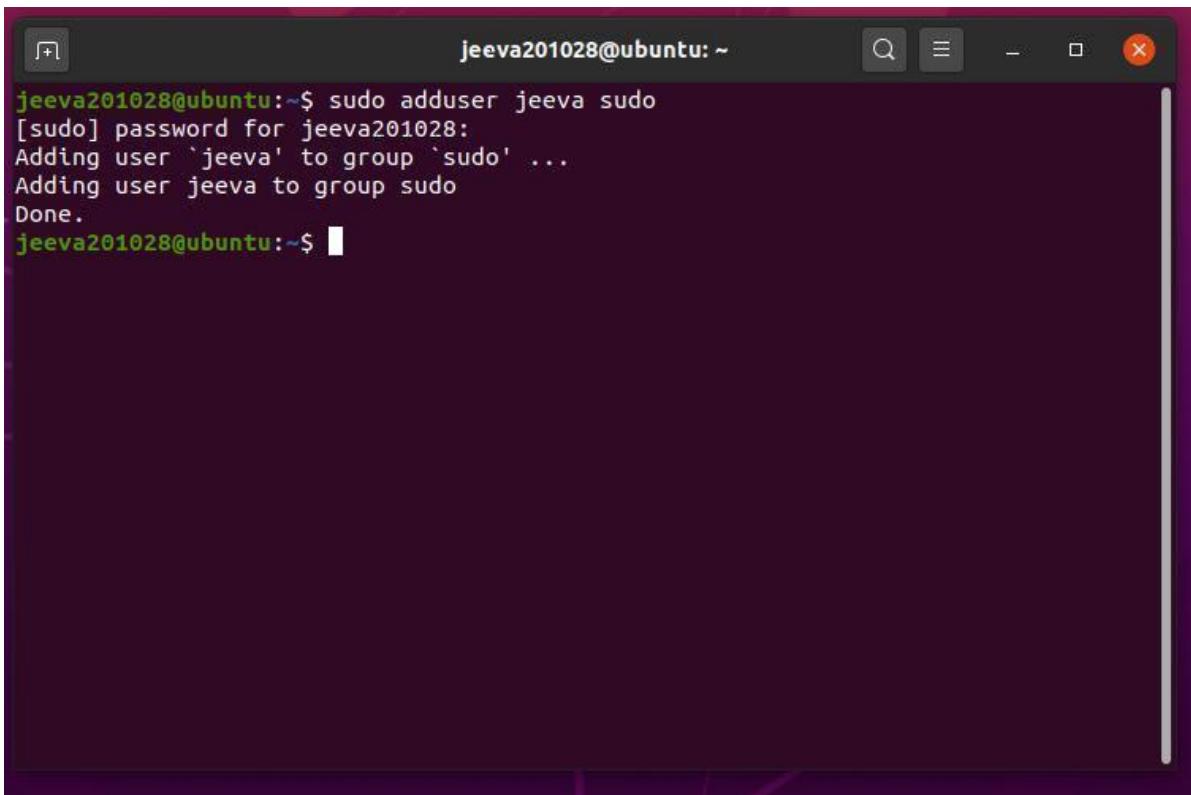
```
Ubuntu 64 bit VMware Workstation
File Edit View VM Tabs Help
Ubuntu 64 bit
Activities Terminal Nov 7 13:51
jeeva201028@ubuntu: ~
Hit:2 http://us.archive.ubuntu.com/ubuntu focal InRelease
Get:3 http://us.archive.ubuntu.com/ubuntu focal-updates InRelease [111 kB]
Get:4 http://us.archive.ubuntu.com/ubuntu focal-backports InRelease [98.3 kB]
Fetched 317 kB in 4s (84.0 kB/s)
Reading package lists... Done
jeeva201028@ubuntu:~$ sudo apt-get install default-jdk
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  ca-certificates-java default-jdk-headless default-jre default-jre-headless
  fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni
  libice-dev libpthread-stubs0-dev libsm-dev libx11-6 libx11-dev libxau-dev
  libxcb1-dev libxdmcp-dev libxt-dev openjdk-11-jdk openjdk-11-jdk-headless
  openjdk-11-jre openjdk-11-jre-headless x11proto-core-dev x11proto-dev
  xorg-sgml-doctools xtrans-dev
Suggested packages:
  libice-doc libsm-doc libx11-doc libxcb-doc libxt-doc openjdk-11-demo
  openjdk-11-source visualvm fonts-ipafont-gothic fonts-ipafont-mincho
  fonts-wqy-microhei fonts-wqy-zenhei
The following NEW packages will be installed:
  ca-certificates-java default-jdk default-jdk-headless default-jre
  default-jre-headless fonts-dejavu-extra java-common libatk-wrapper-java
  libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsm-dev
```

Assigning a dedicated user on hadoop to perform operations



```
jeeva201028@ubuntu: ~  
OpenJDK Runtime Environment (build 11.0.9+11-Ubuntu-0ubuntu1.20.04)  
OpenJDK 64-Bit Server VM (build 11.0.9+11-Ubuntu-0ubuntu1.20.04, mixed mode, sha  
ring)  
jeeva201028@ubuntu:~$ sudo addgroup hadoop  
[sudo] password for jeeva201028:  
Adding group `hadoop' (GID 1001) ...  
Done.  
jeeva201028@ubuntu:~$ sudo adduser --ingroup hadoop jeeva  
Adding user `jeeva' ...  
Adding new user `jeeva' (1001) with group `hadoop' ...  
Creating home directory `/home/jeeva' ...  
Copying files from `/etc/skel' ...  
New password:  
Retype new password:  
passwd: password updated successfully  
Changing the user information for jeeva  
Enter the new value, or press ENTER for the default  
    Full Name []: Jeeva  
    Room Number []: 1  
    Work Phone []: 98765  
    Home Phone []: 43210  
    Other []:  
Is the information correct? [Y/n] Y  
jeeva201028@ubuntu:~$
```

Adding user to sudo list



```
jeeva201028@ubuntu: ~  
jeeva201028@ubuntu:~$ sudo adduser jeeva sudo  
[sudo] password for jeeva201028:  
Adding user `jeeva' to group `sudo' ...  
Adding user jeeva to group sudo  
Done.  
jeeva201028@ubuntu:~$
```

Next install a package ssh(secured shell login)

```
jeeva201028@ubuntu: ~  
jeeva201028@ubuntu:~$ sudo adduser jeeva sudo  
[sudo] password for jeeva201028:  
Adding user `jeeva' to group `sudo' ...  
Adding user jeeva to group sudo  
Done.  
jeeva201028@ubuntu:~$ sudo apt-get install openssh-server  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following additional packages will be installed:  
  ncurses-term openssh-sftp-server ssh-import-id  
Suggested packages:  
  molly-guard monkeysphere ssh-askpass  
The following NEW packages will be installed:  
  ncurses-term openssh-server openssh-sftp-server ssh-import-id  
0 upgraded, 4 newly installed, 0 to remove and 237 not upgraded.  
Need to get 688 kB of archives.  
After this operation, 6,010 kB of additional disk space will be used.  
Do you want to continue? [Y/n]
```

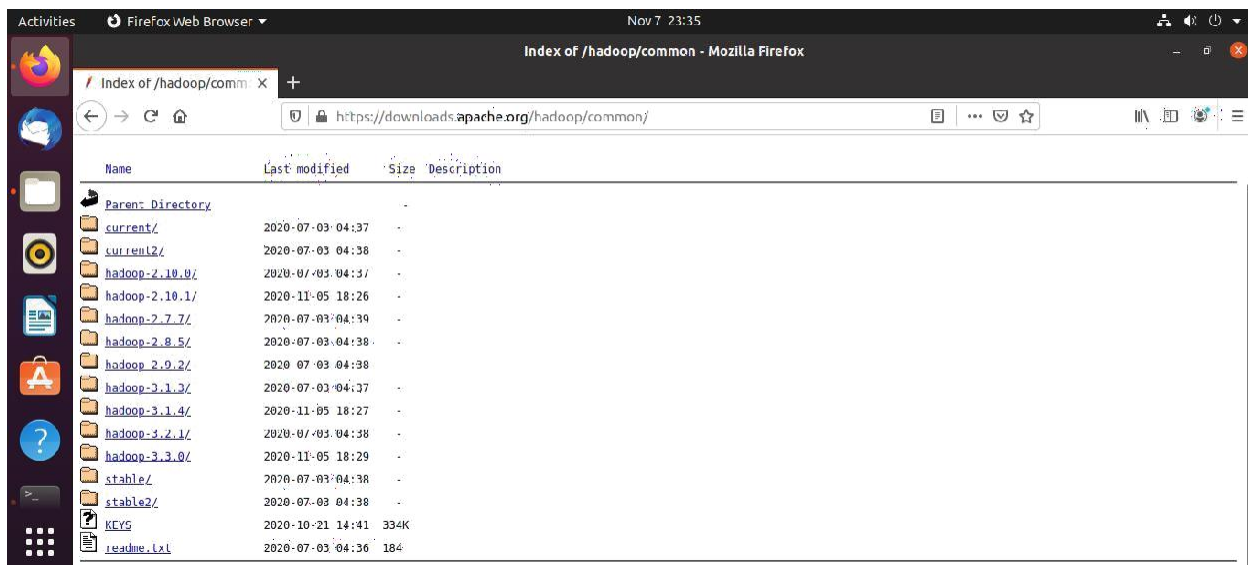
Next step is key generation and add a key to the file

```
jeeva@ubuntu: ~  
jeeva@ubuntu:~$ pwd  
/home/jeeva  
jeeva@ubuntu:~$ ssh-keygen -t rsa -P ""  
Generating public/private rsa key pair.  
Enter file in which to save the key (/home/jeeva/.ssh/id_rsa):  
Created directory '/home/jeeva/.ssh'.  
Your identification has been saved in /home/jeeva/.ssh/id_rsa  
Your public key has been saved in /home/jeeva/.ssh/id_rsa.pub  
The key fingerprint is:  
SHA256:kdjFKbAxpSwyYzTHae7PiBjotPzG+CK741F5xPc9yDs jeeva@ubuntu  
The key's randomart image is:  
+---[RSA 3072]---+  
|  o...+o....  |  
| . o=. B.oo   |  
| =o.o=.+.    |  
| . ++.. o.o   |  
| . + . S+ o   |  
|o.. o        |  
|+++ . + E     |  
|+*oo. o      |  
|===o         |  
+----[SHA256]-----+  
jeeva@ubuntu:~$
```

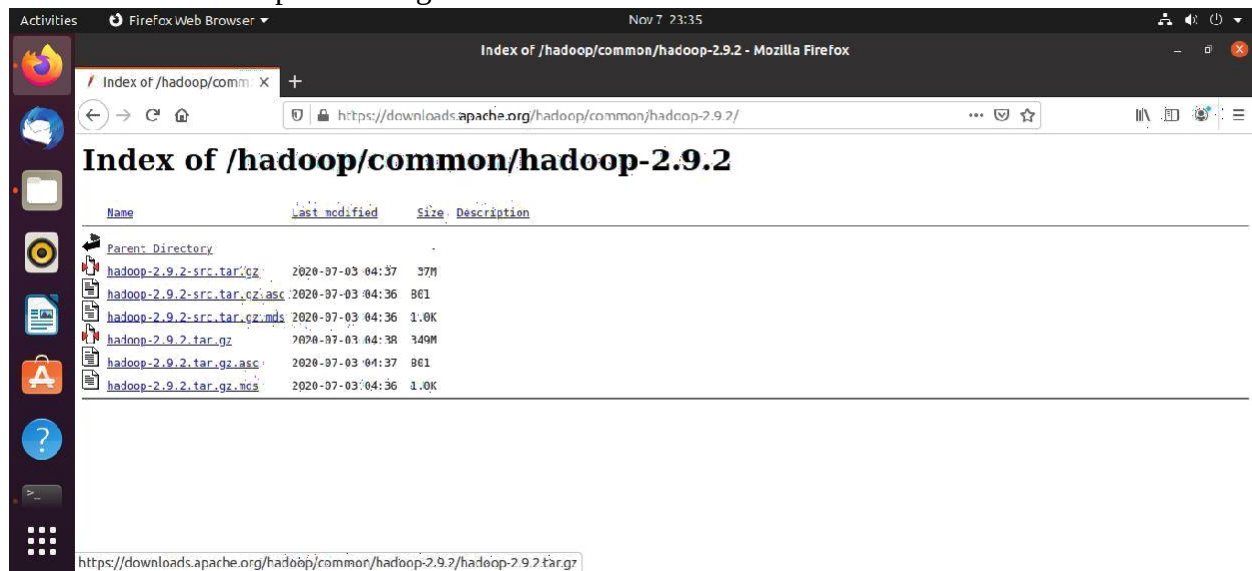
To check whether ssh is installed properly by logging in and after that exit from it

```
jeeva201028@ubuntu: ~  
jeeva201028@ubuntu:~$ ssh localhost  
The authenticity of host 'localhost (127.0.0.1)' can't be established.  
ECDSA key fingerprint is SHA256:3w08Zvw2Q8dtjtAaSLWahjenZ7zywgGfYjr0Im8eYM.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'localhost' (ECDSA) to the list of known hosts.  
jeeva201028@localhost's password:  
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-52-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/advantage  
  
237 updates can be installed immediately.  
100 of these updates are security updates.  
To see these additional updates run: apt list --upgradable  
  
Your Hardware Enablement Stack (HWE) is supported until April 2025.  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
jeeva201028@ubuntu:~$ exit  
logout  
Connection to localhost closed.
```

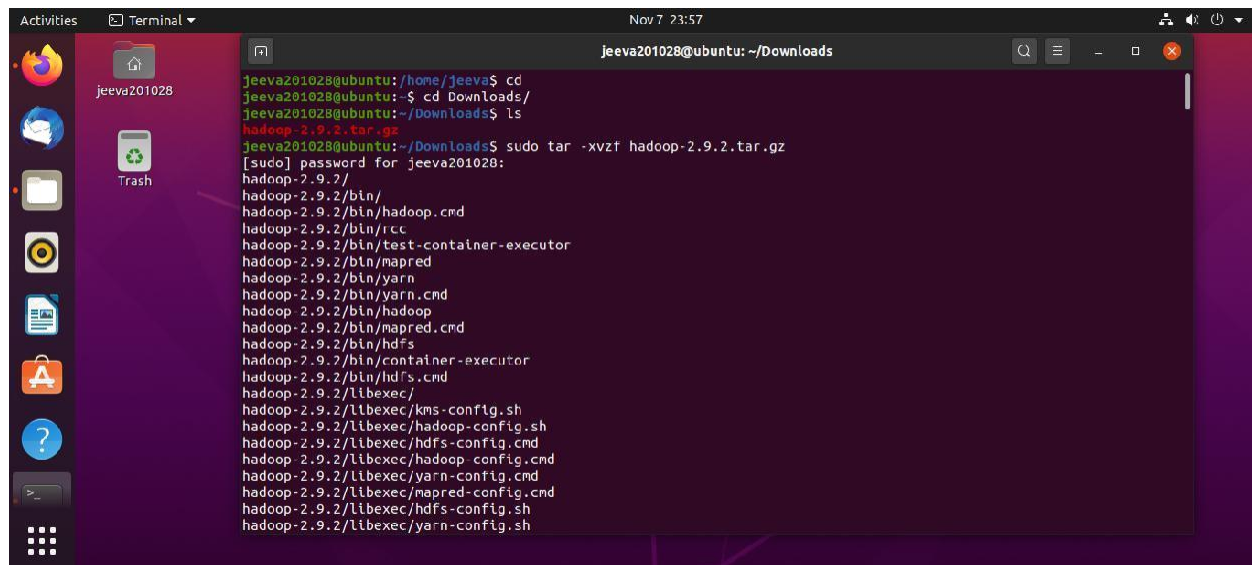
Download Hadoop using this link : <https://downloads.apache.org/hadoop/common/>



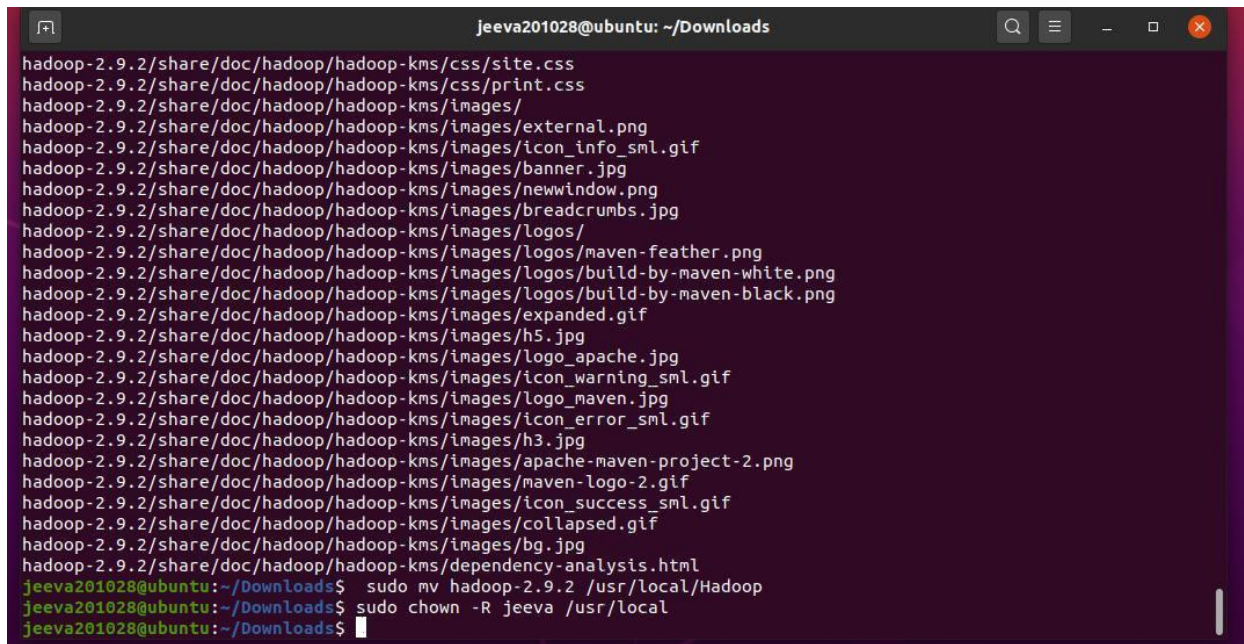
Download the hadoop-2.9.2.tar.gz file



Now extract the tar file

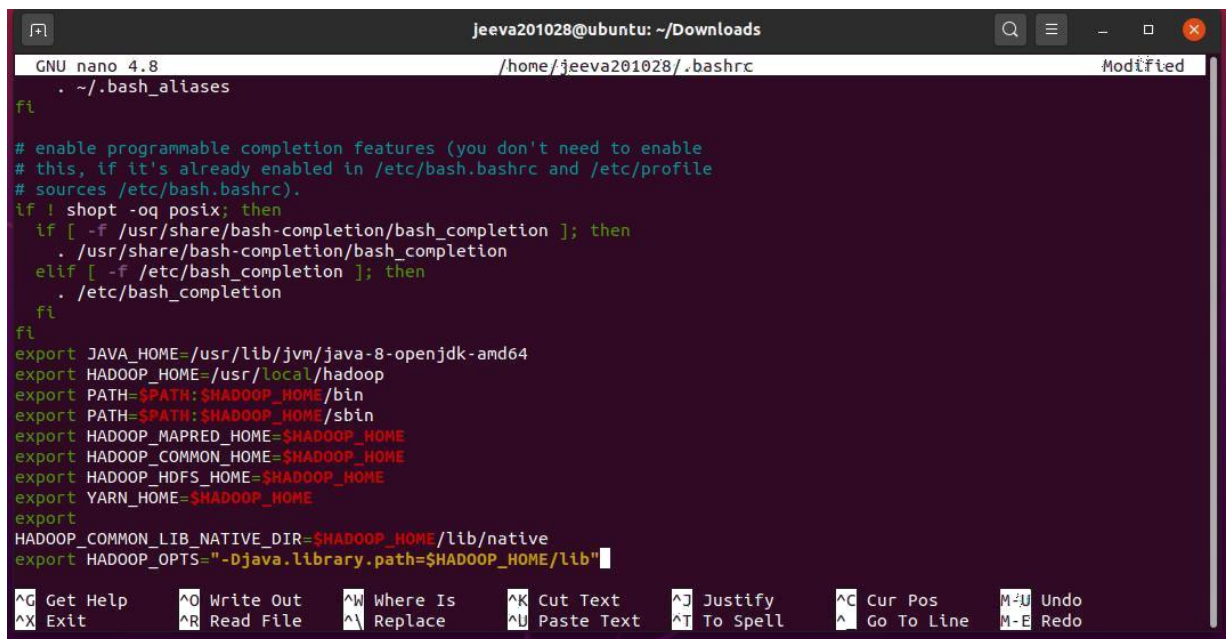


Move the file to local and Change the ownership of hadoop folder



```
jeeva201028@ubuntu: ~/Downloads
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/css/site.css
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/css/print.css
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/external.png
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/icon_info_sml.gif
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/banner.jpg
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/newwindow.png
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/breadcrumbs.jpg
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/logos/
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/logos/maven-feather.png
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/logos/build-by-maven-white.png
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/logos/build-by-maven-black.png
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/expanded.gif
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/h5.jpg
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/logo_apache.jpg
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/icon_warning_sml.gif
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/logo_maven.jpg
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/icon_error_sml.gif
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/h3.jpg
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/apache-maven-project-2.png
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/maven-logo-2.gif
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/icon_success_sml.gif
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/collapsed.gif
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/bg.jpg
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/dependency-analysis.html
jeeva201028@ubuntu:~/Downloads$ sudo mv hadoop-2.9.2 /usr/local/Hadoop
jeeva201028@ubuntu:~/Downloads$ sudo chown -R jeeva /usr/local
jeeva201028@ubuntu:~/Downloads$
```

In .bashrc file add the following and make it at source



```
GNU nano 4.8 /home/jeeva201028/.bashrc Modified
. ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi

export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export HADOOP_HOME=/usr/local/hadoop
export PATH=$PATH:$HADOOP_HOME/bin
export PATH=$PATH:$HADOOP_HOME/sbin
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export
HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib"
```

```
jeeva201028@ubuntu: ~/Downloads
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/collapsed.gif
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/images/bg.jpg
hadoop-2.9.2/share/doc/hadoop/hadoop-kms/dependency-analysis.html
jeeva201028@ubuntu:~/Downloads$ sudo mv hadoop-2.9.2 /usr/local/Hadoop
jeeva201028@ubuntu:~/Downloads$ sudo chown -R jeeva /usr/local
jeeva201028@ubuntu:~/Downloads$ sudo nano ~/.bashrc
jeeva201028@ubuntu:~/Downloads$ source ~/.bashrc
declare -x CLUTTER_IM_MODULE="ibus"
declare -x COLORTERM="truecolor"
declare -x DBUS_SESSION_BUS_ADDRESS="unix:path=/run/user/1000/bus"
declare -x DESKTOP_SESSION="ubuntu"
declare -x DISPLAY=":0"
declare -x GDMSESSION="ubuntu"
declare -x GJS_DEBUG_OUTPUT="stderr"
declare -x GJS_DEBUG_TOPICS="JS ERROR;JS LOG"
declare -x GNOME_DESKTOP_SESSION_ID="this-is-deprecated"
declare -x GNOME_SHELL_SESSION_MODE="ubuntu"
declare -x GNOME_TERMINAL_SCREEN="/org/gnome/terminal/screen/c98c3178_c4bb_40a4_9647_3c0ea63ea95a"
declare -x GNOME_TERMINAL_SERVICE=:1.128"
declare -x GPG_AGENT_INFO="/run/user/1000/gnupg/S.gpg-agent:0:1"
declare -x GTK_IM_MODULE="ibus"
declare -x GTK_MODULES="gail:atk-bridge"
declare -x HADOOP_COMMON_HOME="/usr/local/hadoop"
declare -x HADOOP_HDFS_HOME="/usr/local/hadoop"
declare -x HADOOP_HOME="/usr/local/hadoop"
declare -x HADOOP_MAPRED_HOME="/usr/local/hadoop"
declare -x HOME="/home/jeeva201028"
declare -x IM_CONFIG_PHASE="1"
```

Edit the `hadoop-env.sh` file as following

```
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop
GNU nano 4.8 hadoop-env.sh
# limitations under the License.

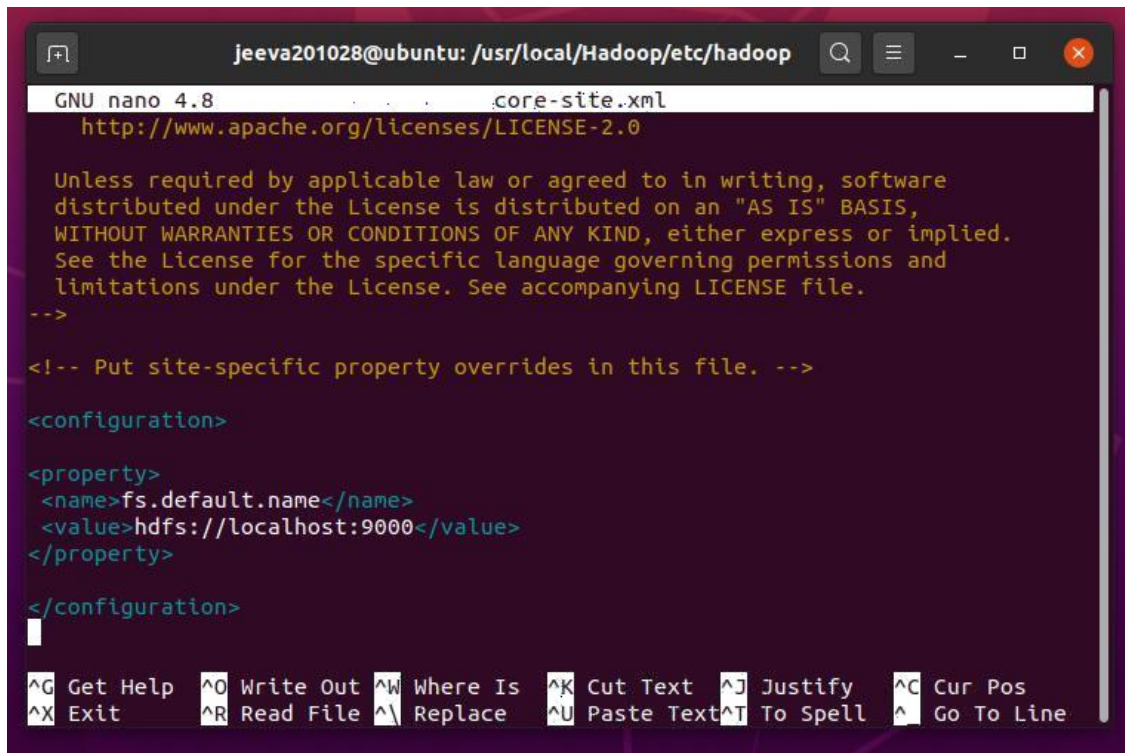
# Set Hadoop-specific environment variables here.

# The only required environment variable is JAVA_HOME. All others are
# optional. When running a distributed configuration it is best to
# set JAVA_HOME in this file, so that it is correctly defined on
# remote nodes.

# The java implementation to use.
#export JAVA_HOME=${JAVA_HOME}
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64

# The jsvc implementation to use. Jsvc is required to run secure datanodes
# that bind to privileged ports to provide authentication of data transfer
# protocol. Jsvc is not required if SASL is configured for authentication of
# data transfer protocol using non-privileged ports.
#export JSVC_HOME=${JSVC_HOME}
[ line 28/121 (23%), col 52/52 (100%), char 1175/5024 (23%) ]
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Line
```

Edit the core-site.xml



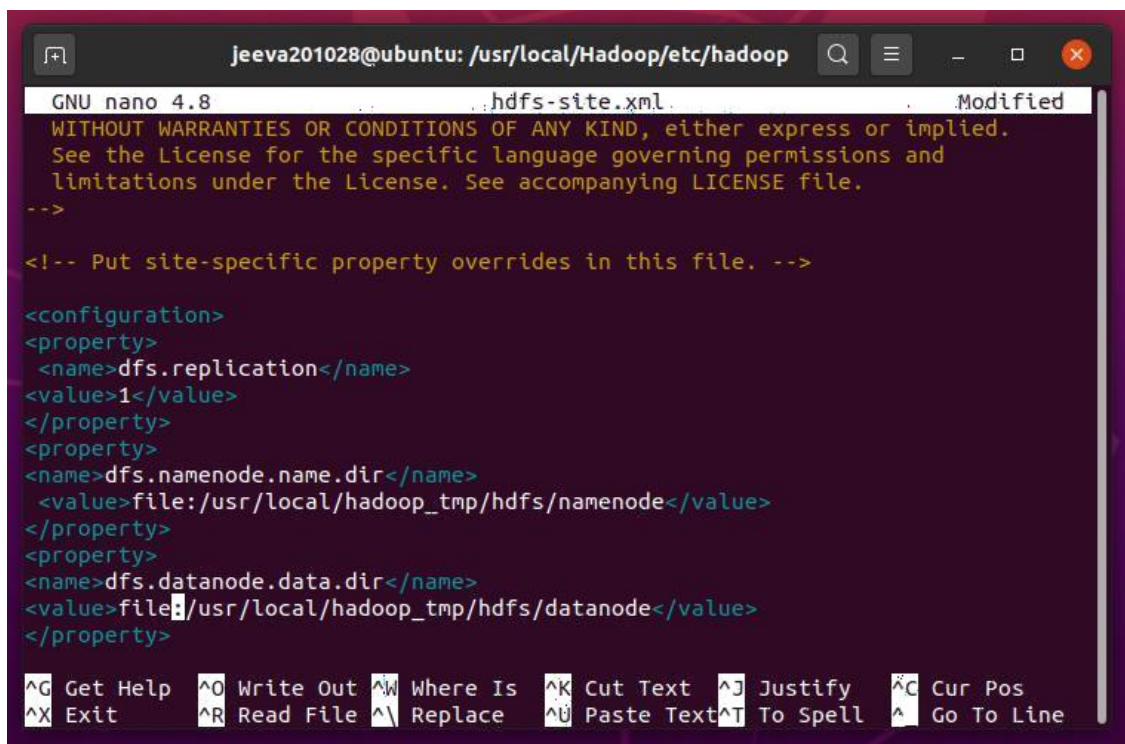
```
GNU nano 4.8 /usr/local/Hadoop/etc/hadoop/core-site.xml
http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>fs.default.name</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text  ^J Justify   ^C Cur Pos
^X Exit      ^R Read File  ^\ Replace   ^U Paste Text ^T To Spell  ^_ Go To Line
```

Edit the hdfs-site.xml



```
GNU nano 4.8 /usr/local/Hadoop/etc/hadoop/hdfs-site.xml Modified
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:/usr/local/hadoop_tmp/hdfs/namenode</value>
  </property>
  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:/usr/local/hadoop_tmp/hdfs/datanode</value>
  </property>

```

Edit the yarn-site.xml

```

jееva201028@ubuntu: /usr/local/Hadoop/etc/hadoop
GNU nano 4.8                                yarn-site.xml                                Modified

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distributed under the License is distributed on an "AS IS" BASIS,
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limitations under the License. See accompanying LICENSE file.

-->
<configuration>

<!-- Site specific YARN configuration properties -->
<property>
  <name>yarn.nodemanager.aux-services</name>
  <value>mapreduce_shuffle</value>
</property>
<property>
  <name>yarn.nodemanager.auxservices.mapreduce.shuffle.class</name>
  <value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>

```

^G Get Help
^X Exit

^O Write Out
^R Read File

^W Where Is
^_ Replace

^K Cut Text
^U Paste Text

^J Justify
^T To Spell

^C Cur Pos
^_ Go To Line

Edit the `mapred-site.xml` by remaning the `mapred-site.xml.template`

```
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$ sudo nano mapred-site.xml  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$ sudo cp mapred-site.xml.templat  
e mapred-site.xml  
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$
```

```
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop
GNU nano 4.8      mapred-site.xml      Modified
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you may not use this file except in compliance with the License.
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    http://www.apache.org/licenses/LICENSE-2.0

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WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>

    <name>mapreduce.framework.name</name>
    <value>yarn</value>
</property>

^G Get Help  ^O Write Out  ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos
^X Exit      ^R Read File  ^\ Replace    ^U Paste Text ^T To Spell   ^_ Go To Line
```

Create the following directory

```
jeeva201028@ubuntu: ~/Desktop
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$ sudo nano mapred-site.xml
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$ sudo cp mapred-site.xml.templat
e mapred-site.xml
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$ sudo nano mapred-site.xml
jeeva201028@ubuntu: /usr/local/Hadoop/etc/hadoop$ cd
jeeva201028@ubuntu: ~$ cd Desktop/
jeeva201028@ubuntu: ~/Desktop$ sudo mkdir -p /usr/local/hadoop_tmp
jeeva201028@ubuntu: ~/Desktop$ sudo mkdir -p /usr/local/hadoop_tmp/hdfs/namenode
jeeva201028@ubuntu: ~/Desktop$ sudo mkdir -p /usr/local/hadoop_tmp/hdfs/datanode
jeeva201028@ubuntu: ~/Desktop$ sudo chown -R jeeva /usr/local/hadoop_tmp
jeeva201028@ubuntu: ~/Desktop$
```

Format hdfs namenode

```
jeeva201028@ubuntu: ~$ hdfs
Usage: hdfs [--config confdir] [--loglevel loglevel] COMMAND
       where COMMAND is one of:
       dfs                run a filesystem command on the file systems supported in Hadoop.
       classpath          prints the classpath
       namenode -format    format the DFS filesystem
       secondarynamenode  run the DFS secondary namenode
       namenode            run the DFS namenode
       journalnode         run the DFS journalnode
       zkfc                run the ZK Failover Controller daemon
       datanode            run a DFS datanode
       debug              run a Debug Admin to execute HDFS debug commands
       dfsadmin           run a DFS admin client
       dfsrouter           run the DFS router
       dfsrouteradmin      manage Router-based federation
       haadmin            run a DFS HA admin client
       fsck               run a DFS filesystem checking utility
       balancer            run a cluster balancing utility
       jmxget             get JMX exported values from NameNode or DataNode.
       mover              run a utility to move block replicas across
                           storage types
       oiv                apply the offline fsimage viewer to an fsimage
       oiv_legacy         apply the offline fsimage viewer to an legacy fsimage
       oev                apply the offline edits viewer to an edits file
       fetchdt            fetch a delegation token from the NameNode
       getconf            get config values from configuration
```

```
jeeva201028@ubuntu: ~$ hdfs
       balancer            run a cluster balancing utility
       jmxget             get JMX exported values from NameNode or DataNode.
       mover              run a utility to move block replicas across
                           storage types
       oiv                apply the offline fsimage viewer to an fsimage
       oiv_legacy         apply the offline fsimage viewer to an legacy fsimage
       oev                apply the offline edits viewer to an edits file
       fetchdt            fetch a delegation token from the NameNode
       getconf            get config values from configuration
       groups             get the groups which users belong to
       snapshotdiff       diff two snapshots of a directory or diff the
                           current directory contents with a snapshot
       lsSnapshottableDir list all snapshottable dirs owned by the current user
                           Use -help to see options
       portmap            run a portmap service
       nfs3               run an NFS version 3 gateway
       cacheadmin         configure the HDFS cache
       crypto             configure HDFS encryption zones
       storagepolicies     list/get/set block storage policies
       version            print the version

Most commands print help when invoked w/o parameters.
jeeva201028@ubuntu:~$ hdfs namenode -format
```

```
jeeva201028@ubuntu: ~$ hdfs namenode -format
20/11/08 04:19:35 INFO namenode.FSNamesystem: Retry cache will use 0.03 of total
heap and retry cache entry expiry time is 600000 millis
20/11/08 04:19:35 INFO util.GSet: Computing capacity for map NameNodeRetryCache
20/11/08 04:19:35 INFO util.GSet: VM type = 64-bit
20/11/08 04:19:35 INFO util.GSet: 0.029999999329447746% max memory 1000 MB = 307
.2 KB
20/11/08 04:19:35 INFO util.GSet: capacity = 2^15 = 32768 entries
20/11/08 04:19:35 INFO namenode.FSImage: Allocated new BlockPoolId: BP-725375456
-127.0.1.1-1604837975223
20/11/08 04:19:35 INFO common.Storage: Storage directory /usr/local/hadoop_tmp/h
dfs/namenode has been successfully formatted.
20/11/08 04:19:35 INFO namenode.FSImageFormatProtobuf: Saving image file /usr/lo
cal/hadoop_tmp/hdfs/namenode/current/fsimage.ckpt_000000000000000000 using no c
ompression
20/11/08 04:19:35 INFO namenode.FSImageFormatProtobuf: Image file /usr/local/had
oop_tmp/hdfs/namenode/current/fsimage.ckpt_000000000000000000 of size 329 bytes
saved in 0 seconds .
20/11/08 04:19:35 INFO namenode.NNStorageRetentionManager: Going to retain 1 ima
ges with txid >= 0
20/11/08 04:19:35 INFO namenode.NameNode: SHUTDOWN_MSG:
/*****
SHUTDOWN_MSG: Shutting down NameNode at ubuntu/127.0.1.1
*****/
jeeva201028@ubuntu:~$
```

\$ start-dfs.sh

\$ start-yarn.sh

To check whether hadoop is correctly installed or not: \$ jps

```
jeeva201028@ubuntu: ~  
doop_tmp/hdfs/namenode/current/fsimage.ckpt_000000000000000000 of size 330 bytes saved in 0 seconds .  
20/11/08 12:04:47 INFO namenode.NNStorageRetentionManager: Going to retain 1 images with txid >= 0  
20/11/08 12:04:47 INFO namenode.NameNode: SHUTDOWN_MSG:  
/*****  
SHUTDOWN_MSG: Shutting down NameNode at ubuntu/127.0.1.1  
*****/  
jeeva201028@ubuntu: /usr/local/hadoop$ cd  
jeeva201028@ubuntu: ~$ start-dfs.sh  
WARNING: An illegal reflective access operation has occurred  
WARNING: Illegal reflective access by org.apache.hadoop.security.authentication.util.KerberosUtil (file:/usr/local/hadoop/share/hadoop/common/lib/hadoop-auth-2.9.2.jar) to method sun.security.krb5.Config.getInstance()  
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.security.authentication.util.KerberosUtil  
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations  
WARNING: All illegal access operations will be denied in a future release  
20/11/08 12:06:10 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable  
Starting namenodes on [localhost]  
jeeva201028@localhost's password: █
```

```
jeeva201028@ubuntu: ~  
028-datanode-ubuntu.out  
Starting secondary namenodes [0.0.0.0]  
jeeva201028@0.0.0.0's password:  
0.0.0.0: secondarynamenode running as process 20058. Stop it first.  
WARNING: An illegal reflective access operation has occurred  
WARNING: Illegal reflective access by org.apache.hadoop.security.authentication.util.KerberosUtil (file:/usr/local/hadoop/share/hadoop/common/lib/hadoop-auth-2.9.2.jar) to method sun.security.krb5.Config.getInstance()  
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.security.authentication.util.KerberosUtil  
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations  
WARNING: All illegal access operations will be denied in a future release  
20/11/08 12:09:32 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable  
jeeva201028@ubuntu: ~$ start-yarn.sh  
starting yarn daemons  
starting resourcemanager, logging to /usr/local/hadoop/logs/yarn-jeeva201028-resourcemanager-ubuntu.out  
jeeva201028@localhost's password:  
localhost: starting nodemanager, logging to /usr/local/hadoop/logs/yarn-jeeva201028-nodemanager-ubuntu.out  
jeeva201028@ubuntu: ~$ █
```

```
jeeva201028@ubuntu: ~  
  
^Cjeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$ jps  
20898 NodeManager  
19363 NameNode  
20756 ResourceManager  
21146 Jps  
20058 SecondaryNameNode  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$
```

To execute word count program:

Create a directory and add a text file in it named “sample.txt”

```
jeeva201028@ubuntu: ~/Desktop/data1  
  
^Cjeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$ jps  
20898 NodeManager  
19363 NameNode  
20756 ResourceManager  
21146 Jps  
20058 SecondaryNameNode  
jeeva201028@ubuntu:~$  
jeeva201028@ubuntu:~$ cd /home/jeeva201028/Desktop/  
jeeva201028@ubuntu:~/Desktop$ mkdir data1  
jeeva201028@ubuntu:~/Desktop$ cd data1  
jeeva201028@ubuntu:~/Desktop/data1$ sudo nano sample.txt  
jeeva201028@ubuntu:~/Desktop/data1$
```

Input: 'sample.txt' file

```
hello
how are you
communication
components
computational
computer
computing
coordinate
compiler
distributed file system
hadoop
single node cluster
word count
```

AG Get Help AO Write Out AW Where Is AK Cut Text AJ Justify AC Cur Pos MU Undo MA Mark Text MT To Bracket
AX Exit AR Read File AL Replace AU Paste Text AT To Spell GC Go To Line ME Redo MC Copy Text MQ Where Was

Now run bin/hdfs dfs -mkdir /user

```
jeeva201028@ubuntu: /usr/local/hadoop
20058 SecondaryNameNode
jeeva201028@ubuntu:~$
jeeva201028@ubuntu:~$ cd /home/jeeva201028/Desktop/
jeeva201028@ubuntu:~/Desktop$ mkdir data1
jeeva201028@ubuntu:~/Desktop$ cd data1
jeeva201028@ubuntu:~/Desktop/data1$ sudo nano sample.txt
jeeva201028@ubuntu:~/Desktop/data1$ cd
jeeva201028@ubuntu:~$ cd /usr/local/hadoop
jeeva201028@ubuntu:/usr/local/hadoop$ bin/hdfs dfs -mkdir /user
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.hadoop.security.authentication.util.KerberosUtil (file:/usr/local/hadoop/share/hadoop/common/lib/hadoop-auth-2.9.2.jar) to method sun.security.krb5.Config.getInstance()
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.security.authentication.util.KerberosUtil
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
20/11/08 12:14:30 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
```

Then run `bin/hdfs dfs -mkdir /ypm`

```
file:/usr/local/hadoop/share/hadoop/common/lib/hadoop-auth-2.9.2.jar) to method sun.security.krb5.Con
fig.getInstance()
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.security.authenticat
ion.util.KerberosUtil
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operation
s
WARNING: All illegal access operations will be denied in a future release
20/11/08 12:14:30 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform
... using builtin-java classes where applicable
jeeva201028@ubuntu: /usr/local/hadoop$ bin/hdfs dfs -mkdir /ypm
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.hadoop.security.authentication.util.KerberosUtil (f
ile:/usr/local/hadoop/share/hadoop/common/lib/hadoop-auth-2.9.2.jar) to method sun.security.krb5.Con
fig.getInstance()
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.security.authenticat
ion.util.KerberosUtil
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operation
s
WARNING: All illegal access operations will be denied in a future release
20/11/08 12:15:49 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform
... using builtin-java classes where applicable
jeeva201028@ubuntu: /usr/local/hadoop$
```

Run the program //word count prgm will be in the jar file by default which we are using now.

`$ bin/hadoop jar share/hadoop/mapreduce/hadoop-mapreduce-examples-2.9.2.jar wordcount /user/input output`

```
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$ bin/hadoop jar share/hadoop/mapreduce/hadoop-mapreduce-exampl
e-2.9.2.jar wordcount /user/input output
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.hadoop.security.authentication.util.KerberosUtil (f
ile:/usr/local/hadoop/share/hadoop/common/lib/hadoop-auth-2.9.2.jar) to method sun.security.krb5.Con
fig.getInstance()
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.security.authenticat
ion.util.KerberosUtil
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operation
s
WARNING: All illegal access operations will be denied in a future release
20/11/08 12:24:53 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform
... using builtin-java classes where applicable
20/11/08 12:24:54 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
```

```
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$ bin/hdfs dfs -put /home/jeeva/Desktop/data/user/input
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.hadoop.security.authentication.util.KerberosUtil (f
ile:/usr/local/hadoop/share/hadoop/common/lib/hadoop-auth-2.9.2.jar) to method sun.security.krb5.Con
fig.getInstance()
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.security.authenticat
ion.util.KerberosUtil
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operation
s
WARNING: All illegal access operations will be denied in a future release
20/11/08 12:37:51 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform
... using builtin-java classes where applicable
```

```
jeeva201028@ubuntu: ~  
Total megabyte-millisecond taken by all reduce tasks=2942976  
Map-Reduce Framework  
  Map input records=16  
  Map output records=23  
  Map output bytes=274  
  Map output materialized bytes=285  
  Input split bytes=108  
  Combine input records=23  
  Combine output records=20  
  Reduce input groups=20  
  Reduce shuffle bytes=285  
  Reduce input records=20  
  Reduce output records=20  
  Spilled Records=40  
  Shuffled Maps =1  
  Failed Shuffles=0  
  Merged Map outputs=1  
  GC time elapsed (ms)=62  
  CPU time spent (ms)=1510  
  Physical memory (bytes) snapshot=455553024  
  Virtual memory (bytes) snapshot=4140194752  
  Total committed heap usage (bytes)=273678336  
Shuffle Errors  
  BAD_ID=0  
  CONNECTION=0  
  IO_ERROR=0  
  WRONG_LENGTH=0  
  WRONG_MAP=0  
  WRONG_REDUCE=0  
File Input Format Counters  
  Bytes Read=186  
File Output Format Counters  
  Bytes Written=192
```

Now run the following command to see the output

\$ bin/hdfs dfs -cat output/*

```
jeeva201028@ubuntu: /usr/local/hadoop$  
jeeva201028@ubuntu: /usr/local/hadoop$  
jeeva201028@ubuntu: /usr/local/hadoop$  
jeeva201028@ubuntu: /usr/local/hadoop$  
jeeva201028@ubuntu: /usr/local/hadoop$ bin/hdfs dfs -cat output/*  
WARNING: An illegal reflective access operation has occurred  
WARNING: Illegal reflective access by org.apache.hadoop.security.authentication.util.KerberosUtil (file:/usr/local/hadoop/share/hadoop/common/lib/hadoop-auth-2.9.2.jar) to method sun.security.krb5.Config.getInstance()  
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.security.authentication.util.KerberosUtil  
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations  
WARNING: All illegal access operations will be denied in a future release  
26/11/07 17:25:31 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable  
are 1  
cluster 1  
communication 1  
compiler 1  
components 1  
computational 1  
computer 2  
computing 1  
coordinate 1  
count 1  
distributed 1  
file 1  
hadoop 3  
hello 1  
how 1  
node 1  
single 1  
system 1  
word 1  
you 1
```

Using gui: In browser open port 50700, <http://localhost:50070> (Hadoop must be running)



Overview 'localhost:9000' (active)

Started:	Sat Nov 07 15:45:51 +0530 2020
Version:	2.9.2, 1926afbea931ca687bc2f8471dc841b66ed2c6704
Compiled:	Tue Nov 13 18:12:00 +0530 2018 by ajsaka from branch-2.9.2
Cluster ID:	C10-412455a2-d8d7-4c6a-898b-e279108927c1
Block Pool ID:	BP-434539900-127.0.0.1-1604743359075

Summary

Security is off.
Safemode is off.
20 files and directories, 5 blocks = 25 total filesystem object(s).
Heap Memory used 73.32 MB of 111 MB Heap Memory. Max Heap Memory is 1000 MB.
Non Heap Memory used 50.57 MB of 54.81 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

Configured Capacity:	97.63 GB
DFS Used:	296 KB (0%)
Non DFS Used:	15.43 GB
DFS Remaining:	77.2 GB (79.07%)



Datanode Information

✓ In service ⚠ Down ⚠ Decommissioned ⚠ Decommissioned & dead ⚠ Maintenance & dead

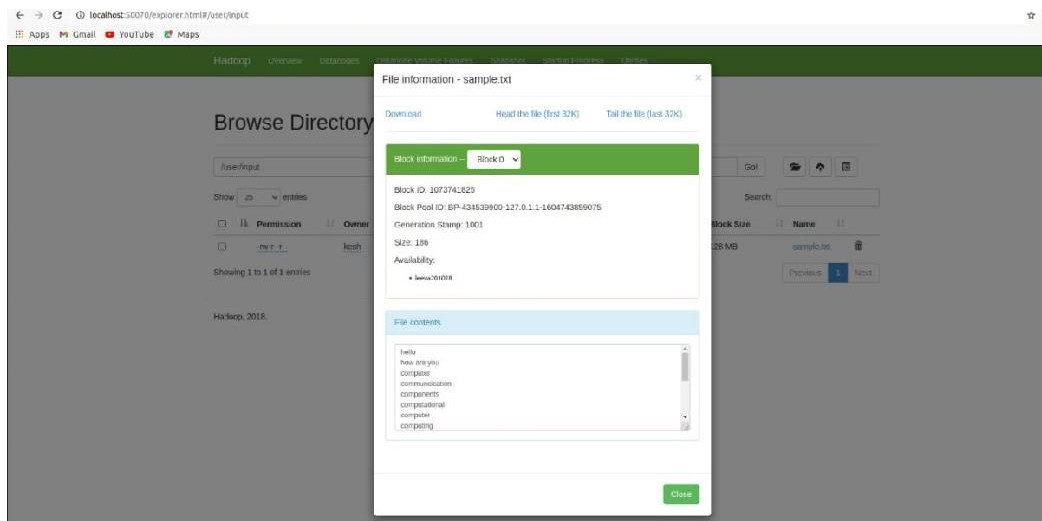
Datanode usage histogram

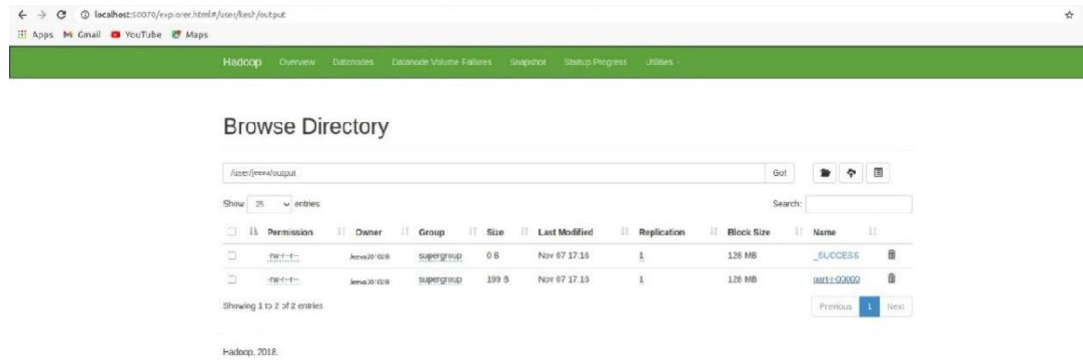


In operation

Show	25	▼ entries	Search:				
Node	Http Address	Last contact	Last Block Report	Capacity	Blocks	Block pool used	Version
✓ jeeva50019 (127.0.0.1:50010)	http://eeva:50075	0s	114m	97.63 GB	5	296 KB (0%)	2.9.2
Showing 1 to 1 of 1 entries							
				Peeking	1	Next	

localhost:50070/explorer.html





After completing stop hadoop \$ stop-all.sh

```
jeeva201028@ubuntu: /usr/local/hadoop
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$
jeeva201028@ubuntu: /usr/local/hadoop$ stop-all.sh
This script is Deprecated. Instead use stop-dfs.sh and stop-yarn.sh
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.hadoop.security.authentication.
util.KerberosUtil (file:/usr/local/hadoop/share/hadoop/common/lib/hadoop-auth-2.
9.2.jar) to method sun.security.krb5.Config.getInstance()
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.
security.authentication.util.KerberosUtil
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflect
ive access operations
WARNING: All illegal access operations will be denied in a future release
20/11/09 00:21:44 WARN util.NativeCodeLoader: Unable to load native-hadoop libra
ry for your platform... using builtin-java classes where applicable
Stopping namenodes on [localhost]
jeeva201028@localhost's password: █
```

Result:

Thus the installation of hadoop single mode cluster and execution of word count program is done and the output is obtained successfully.

EX.No:7

Date:06/11/2020

Simulate a cloud scenario using CloudSim and run a scheduling algorithm

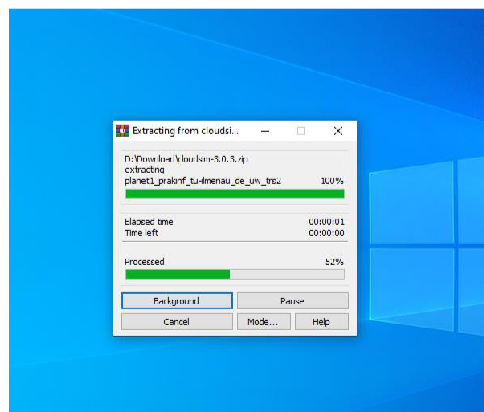
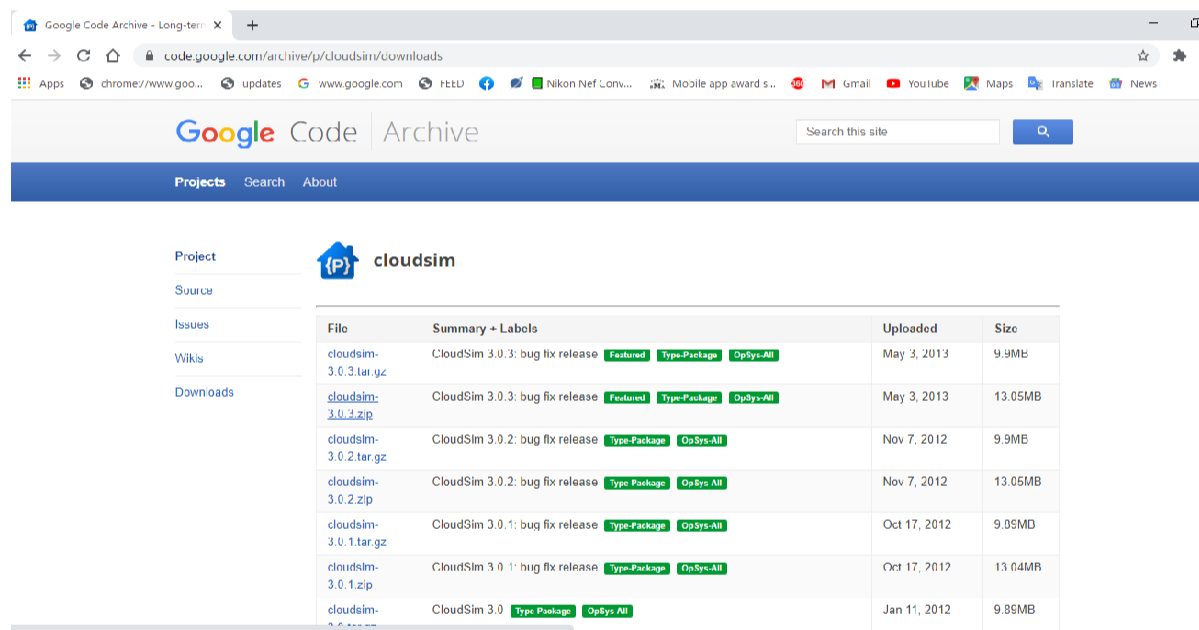
Aim:

To simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim.

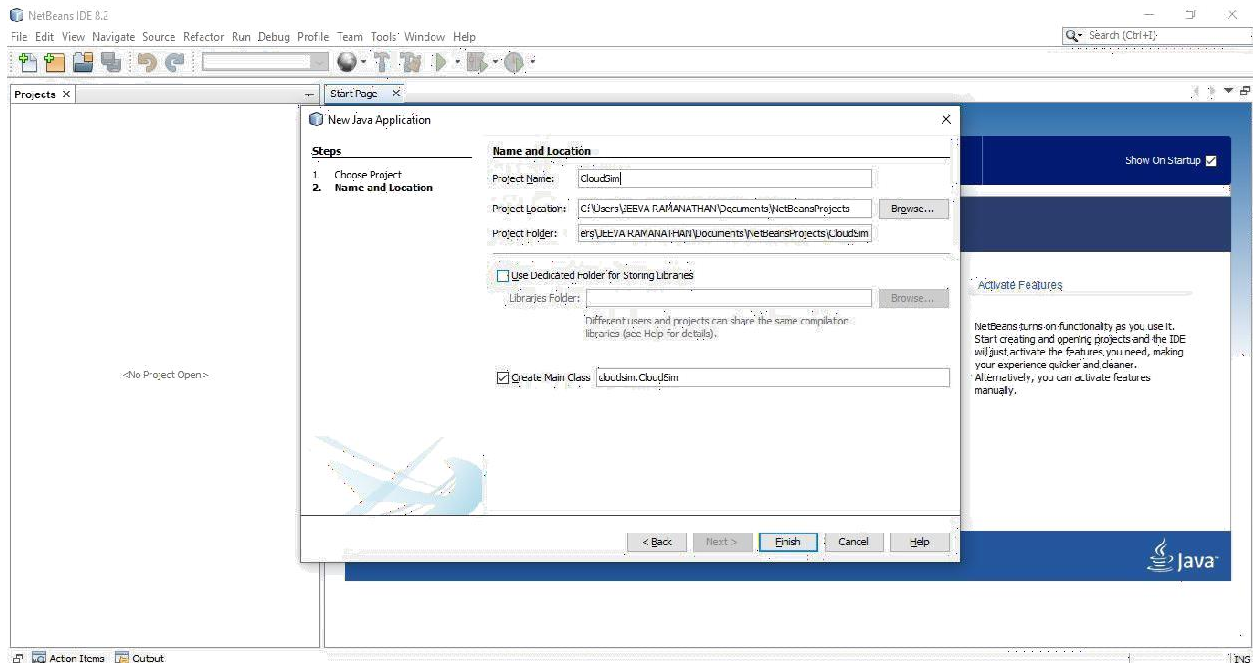
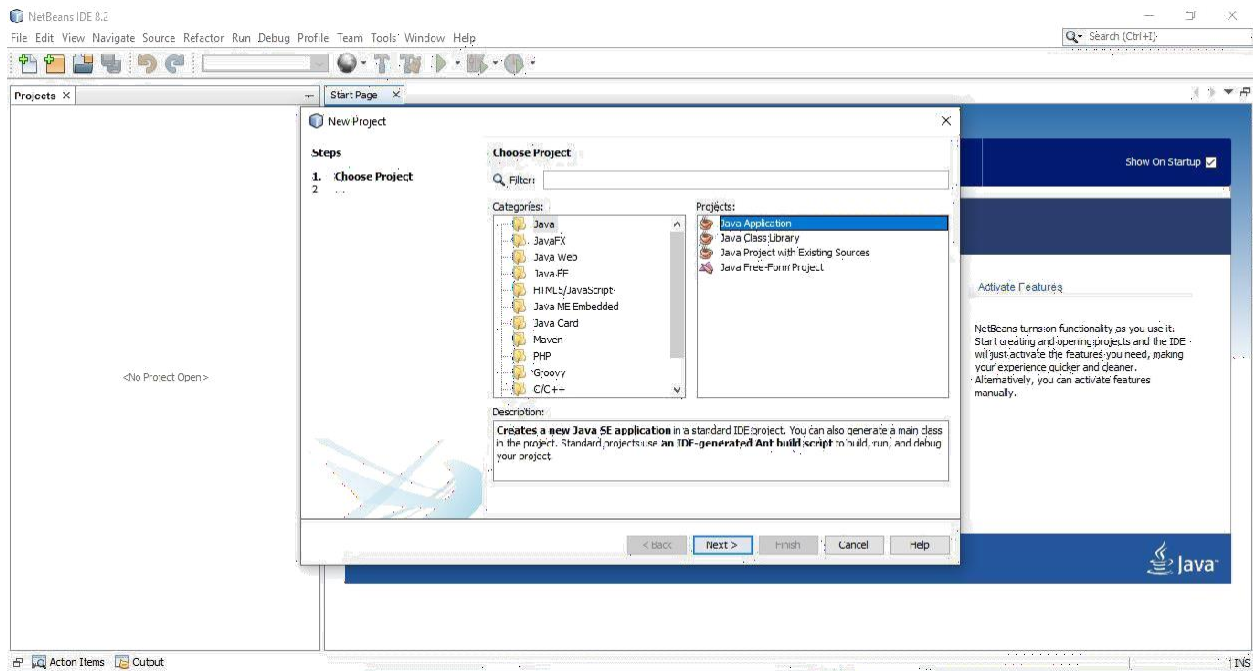
Procedure:

Step 1: Download CloudSim installable files from:

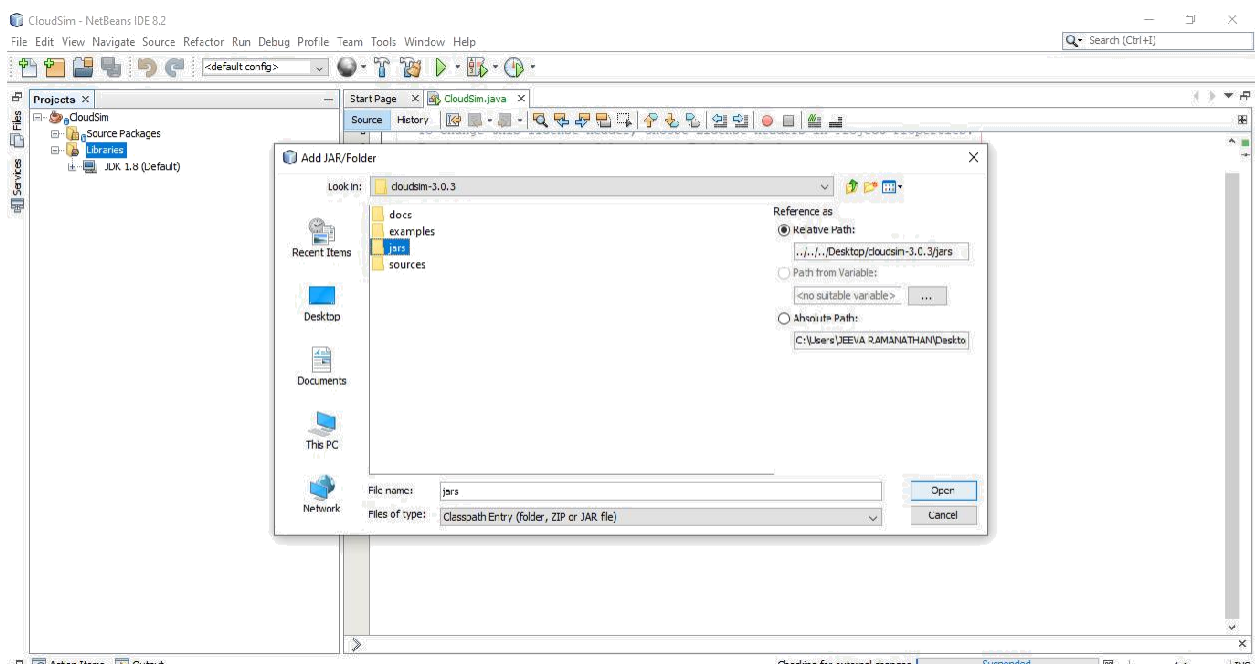
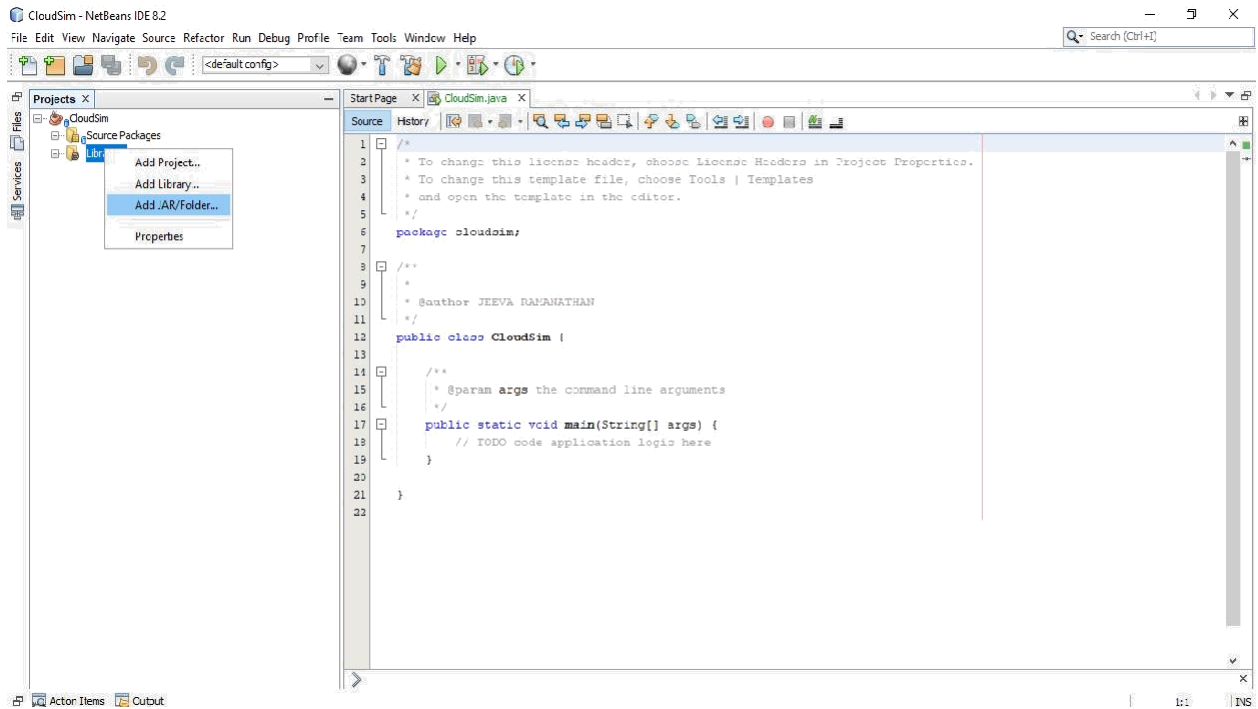
<https://code.google.com/p/cloudsim/downloads/list> and unzip the download.

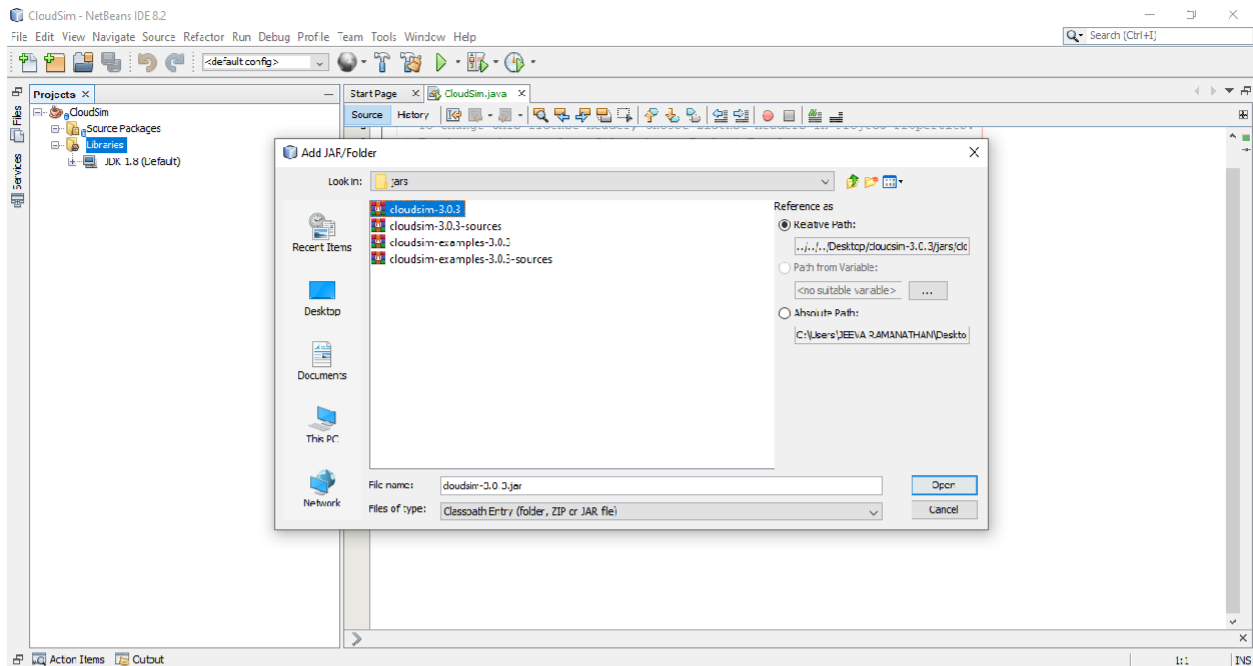


Step 2: Open Netbeans and create a new project named “Clousim”

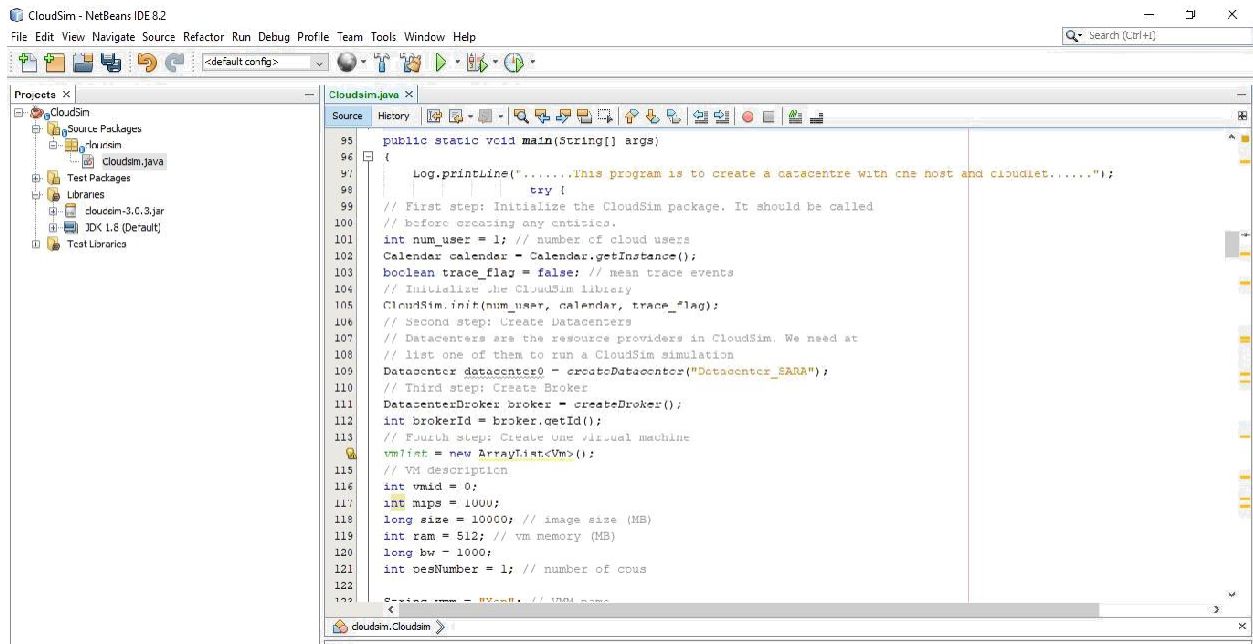


Step 3: Type the code and add the jar file to the libraries. The jar file will be in the extracted cloudsimsim.

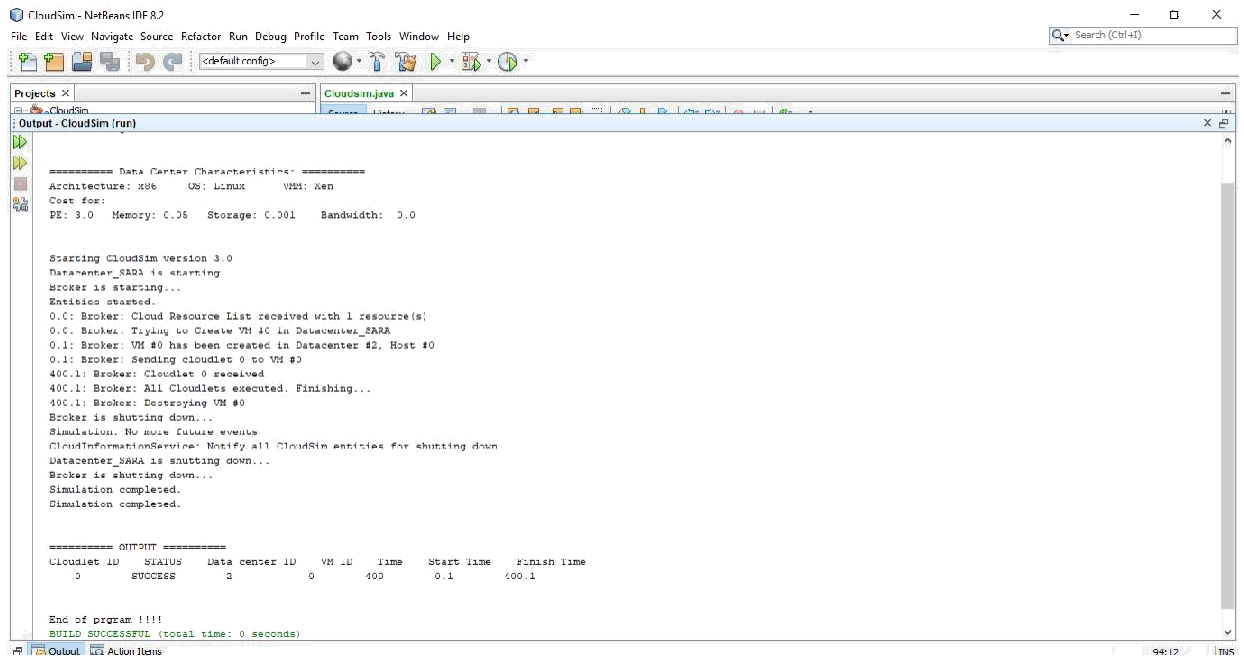




Step 4: Run the application



OUTPUT:



The screenshot shows the NetBeans IDE with the 'Output - CloudSim (run)' window open. The output text is as follows:

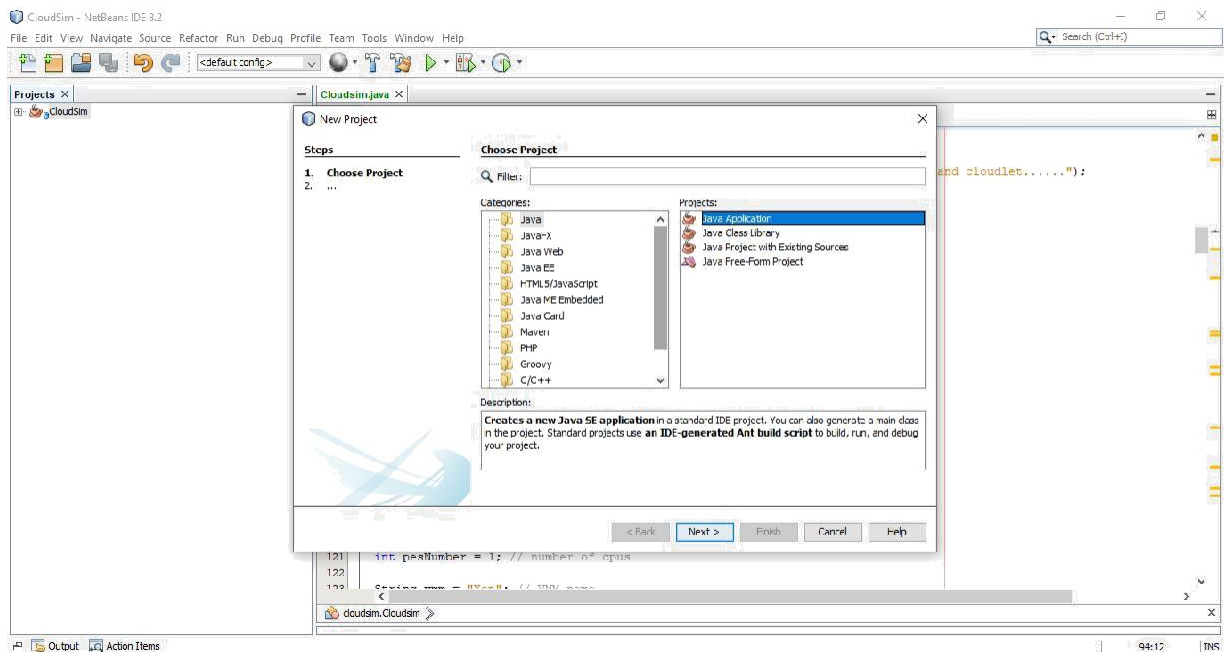
```
===== Data Center Characteristics: =====
Architecture: x86  OS: LINUX  VM: Xen
Cost for:
PE: 3.0  Memory: 0.05  Storage: 0.001  Bandwidth: 0.0

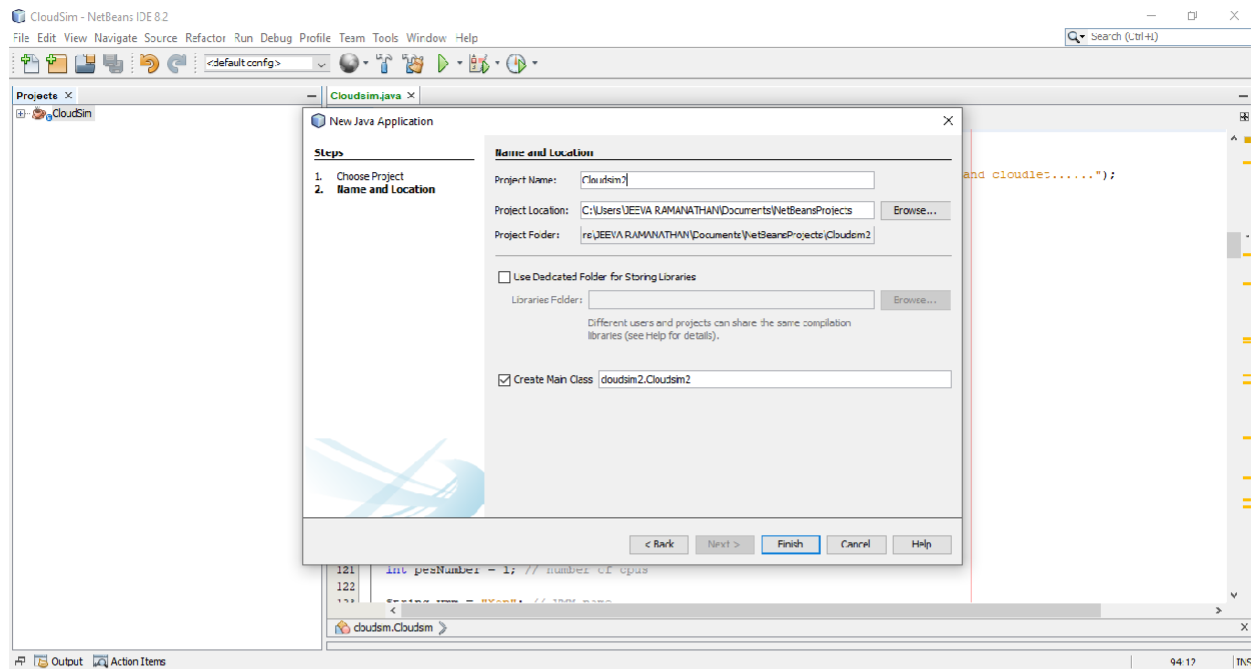
Starting CloudSim version 3.0
Datacenter_SARA is starting
Broker is starting...
Entities started.
0.0: Broker: Cloud Resource List received with 1 resource(s)
0.0: Broker: Trying to create VM 10 in Datacenter_SARA
0.1: Broker: VM #0 has been created in Datacenter #2, Host #0
0.1: Broker: Sending cloudlet 0 to VM #0
400.1: Broker: Cloudlet 0 received
400.1: Broker: All Cloudlets executed. Finishing...
400.1: Broker: Destroying VM #0
Broker is shutting down...
Simulation: No more future events
CloudInformationReceiver: Notify all CloudSim entities for shutting down
Datacenter_SARA is shutting down...
Broker is shutting down...
Simulation completed.
Simulation completed.

===== OUTPUT =====
Cloudlet ID  STATUS  Data center ID  VM ID  Time  Start time  Finish time
0           SUCCESS  2              0      400  0.1         400.1

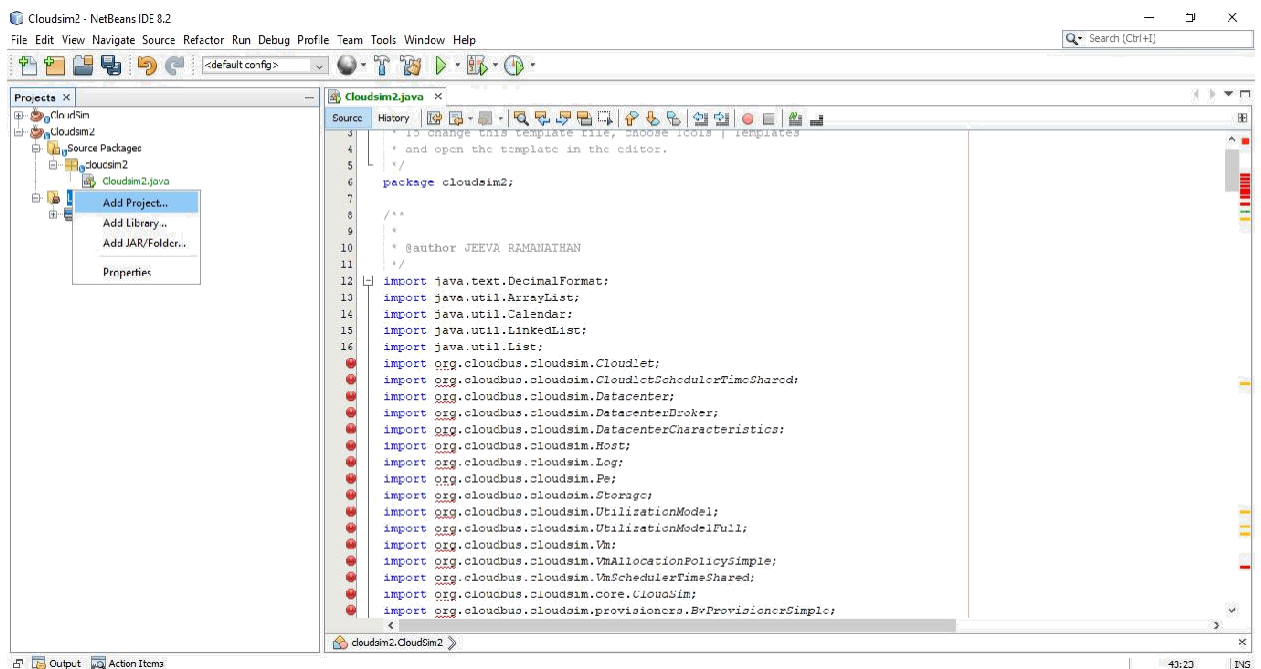
End of program !!!!
BUILD SUCCESSFUL (total time: 0 seconds)
```

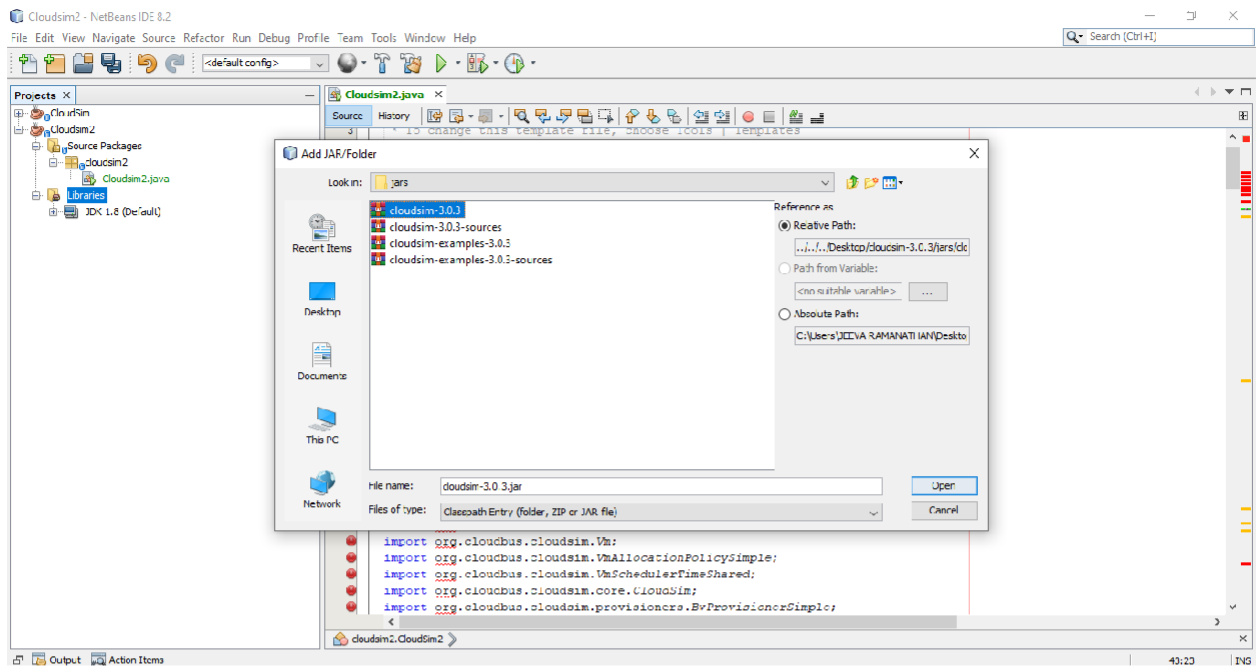
Step 5: Again create a new project and name it as “Cloudsim2”



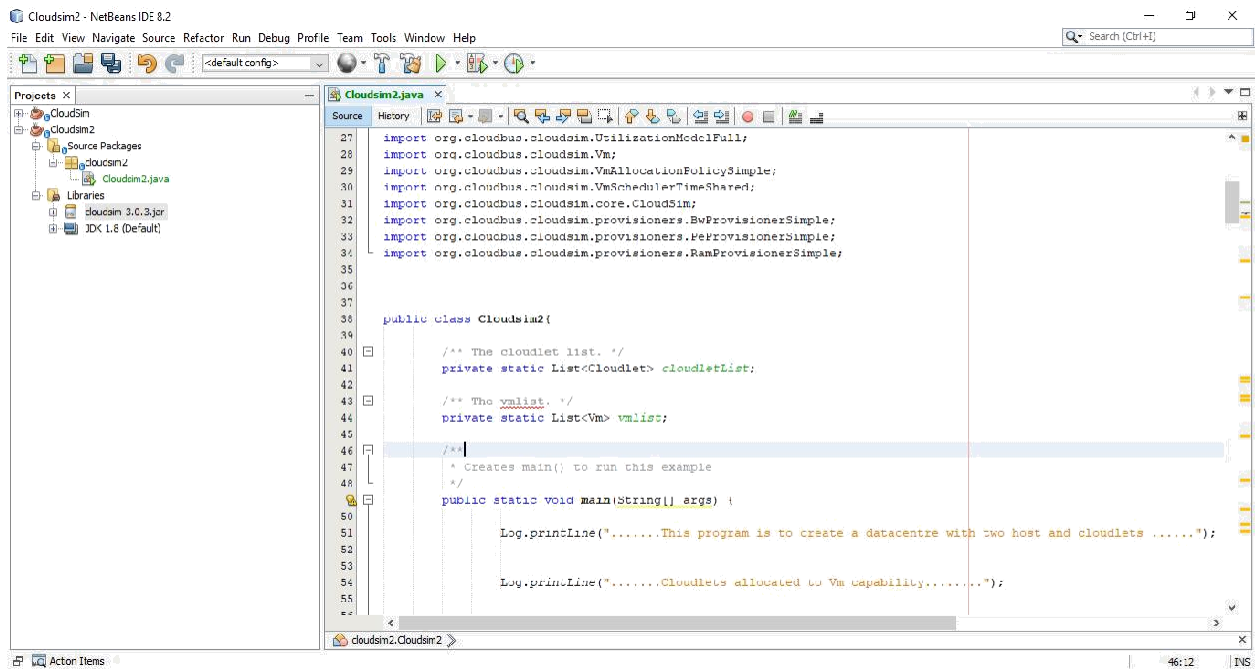


Step 6: Type the code and include the jar file

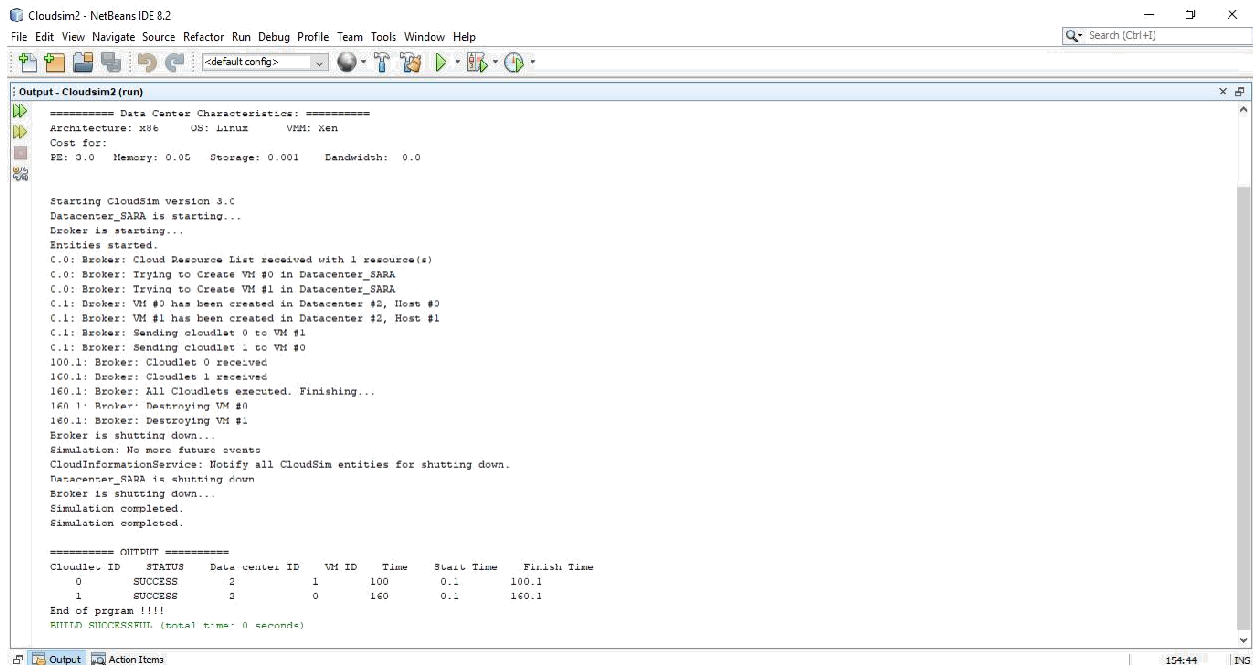




Step 7: Run the application



OUTPUT:



```
Cloudsim2 - NetBeans IDE 8.2
File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help
<default config>
Search (Ctrl+F)

Output - Cloudsim2 (run)

===== Data Center Characteristics: =====
Architecture: X86    OS: Linux    VM: Xen
Cost for:
FE: 0.0    Memory: 0.05    Storage: 0.001    Bandwidth: 0.0

Starting CloudSim version 3.0
Datacenter_SARA is starting...
Broker is starting...
Entities started.
C.0: Broker: Cloud Resource List received with 1 resource(s)
C.0: Broker: Trying to Create VM #0 in Datacenter_SARA
C.0: Broker: Trying to Create VM #1 in Datacenter_SARA
C.1: Broker: VM #0 has been created in Datacenter #2, Host #0
C.1: Broker: VM #1 has been created in Datacenter #2, Host #1
C.1: Broker: Sending cloudlet 0 to VM #1
C.1: Broker: Sending cloudlet 1 to VM #0
100.1: Broker: Cloudlet 0 received
100.1: Broker: Cloudlet 1 received
160.1: Broker: All Cloudlets executed. Finishing...
160.1: Broker: Destroying VM #0
160.1: Broker: Destroying VM #1
Broker is shutting down...
Simulation: No more future events
CloudInformationService: Notify all CloudSim entities for shutting down.
Datacenter_SARA is shutting down
Broker is shutting down...
Simulation completed.
Simulation completed.

===== OUTPUT =====
Cloudlet ID   STATUS   Data center ID   VM ID   Time   Start Time   Finish Time
0            SUCCESS   2                1       100    0.1          100.1
1            SUCCESS   2                0       160    0.1          160.1

End of program !!!!
SIMULATED SUCCESSFULLY (total time: 0 seconds)
```

Result:

Thus the simulation a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim is done and the output is obtained successfully.

EX.No:8

Find a procedure to launch virtual machine using

Date:

trystack (Online Openstack Demo Version)

OpenStack is an open-source software cloud computing platform. OpenStack is primarily used for deploying an infrastructure as a service (IaaS) solution like Amazon Web Service (AWS). In other words, you can make your own AWS by using OpenStack. If you want to try out OpenStack, TryStack is the easiest and free way to do it.

Overview: What we will do? In this , I will show you how to run an OpenStack instance. The instance will be accessible through the internet (have a public IP address).

Step 1: Create Network

Yes, the network in here is our own local network. So, your instances will be not mixed up with the others. You can imagine this as your own LAN (Local Area Network) in the cloud.

1. Go to Network > Networks and then click Create Network.
2. In Network tab, fill Network Name for example internal and then click Next.
3. In Subnet tab,
 1. Fill Network Address with appropriate CIDR, for example 192.168.1.0/24.
Use private network CIDR block as the best practice.
 2. Select IP Version with appropriate IP version, in this case IPv4.
 3. Click Next.
4. In Subnet Details tab, fill DNS Name Servers with 8.8.8.8 (Google DNS) and then click Create.

Step 2: Create Instance

Now, we will create an instance. The instance is a virtual machine in the cloud, like AWS EC2. You need the instance to connect to the network that we just created in the previous step.

1. Go to Compute > Instances and then click Launch Instance.
2. In Details tab,

1. Fill Instance Name, for example Ubuntu 1.
2. Select Flavor, for example m1.medium.
3. Fill Instance Count with 1.
4. Select Instance Boot Source with Boot from Image.
5. Select Image Name with Ubuntu 14.04 amd64 (243.7 MB) if you want install Ubuntu 14.04 in your virtual machine.

3. In Access & Security tab,

1. Click [+] button of Key Pair to import key pair. This key pair is a public and private key that we will use to connect to the instance from our machine.

2. In Import Key Pair dialog,

1. Fill Key Pair Name with your machine name (for example Edward-Key).
2. Fill Public Key with your SSH public key (usually is in `~/.ssh/id_rsa.pub`). See description in Import Key Pair dialog box for more information. If you are using Windows, you can use Puttygen to generate key pair.
3. Click Import key pair.

3. In Security Groups, mark/check default.

4. In Networking tab,

1. In Selected Networks, select network that have been created in Step 1, for example internal.

5. Click Launch.

6. If you want to create multiple instances, you can repeat step 1-5. I created one more instance with instance name Ubuntu 2.

Step 3: Create Router

I guess you already know what router is. In the step 1, we created our network, but it is isolated. It doesn't connect to the internet. To make our network has an

internet connection, we need a router that running as the gateway to the internet.

1. Go to Network > Routers and then click Create Router.
2. Fill Router Name for example router1 and then click Create router.
3. Click on your router name link, for example router1, Router Details page.
4. Click Set Gateway button in upper right: 1. Select External networks with external.
2. Then OK.
5. Click Add Interface button.

1. Select Subnet with the network that you have been created in Step 1.
2. Click Add interface.

6. Go to Network > Network Topology. You will see the network topology. In the example, there are two network, i.e. external and internal, those are bridged by a router. There are instances those are joined to internal network.

Step 4: Configure Floating IP Address

Floating IP address is public IP address. It makes your instance is accessible from the internet. When you launch your instance, the instance will have a private network IP, but no public IP. In OpenStack, the public IPs is collected in a pool and managed by admin (in our case is TryStack). You need to request a public (floating) IP address to be assigned to your instance.

1. Go to Compute > Instance.
2. In one of your instances, click More > Associate Floating IP.
3. In IP Address, click Plus [+].
4. Select Pool to external and then click Allocate IP.
5. Click Associate.
6. Now you will get a public IP, e.g. 8.21.28.120, for your instance.

Step 5: Configure Access & Security

OpenStack has a feature like a firewall. It can whitelist/blacklist your in/out connection. It is called Security Group.

1. Go to Compute > Access & Security and then open Security Groups tab.

2. In default row, click Manage Rules.
3. Click Add Rule, choose ALL ICMP rule to enable ping into your instance, and then click Add.
4. Click Add Rule, choose HTTP rule to open HTTP port (port 80), and then click Add.
5. Click Add Rule, choose SSH rule to open SSH port (port 22), and then click Add.
6. You can open other ports by creating new rules.

Step 6: SSH to Your Instance

Now, you can SSH your instances to the floating IP address that you got in the step 4. If you are using Ubuntu image, the SSH user will be ubuntu.